Engineering Mathematics-1

Basic Formulae of Integration



By Gulshan sir

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① $\int x^n dx = \frac{x^{n+1}}{n+1} + c \qquad n \neq -1$

② $\int \frac{1}{x} dx = \log x + c$

③ $\int e^x dx = e^x + c$

④ $\int a^x dx = \frac{a^x}{\log a} + c$

⑤ $\int \sin x \, dx = -\cos x + c$

⑥ $\int \cos x \, dx = \sin x + c$

⑦ $\int \tan x \, dx = -\log|\cos x| + c$

⑧ $\int \cot x \, dx = \log|\sin x| + c$

⑨ $\int \sec x \, dx = \log|\sec x + \tan x| + c$

⑩ $\int \operatorname{cosec} x \, dx = \log|\operatorname{cosec} x - \cot x| + c$

(11) $\int \sec^2 x \, dx = \tan x + C$

(12) $\int \operatorname{cosec}^2 x \, dx = -\cot x + C$

(13) $\int \sec x \tan x \, dx = \sec x + C$

(14) $\int \operatorname{cosec} x \cot x \, dx = -\operatorname{cosec} x + C$

(15) $\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1}\frac{x}{a} + C$

(16) $\int \frac{-1}{\sqrt{a^2 - x^2}} dx = \cos^{-1}\frac{x}{a} + C$

(17) $\int \frac{1}{a^2 + x^2} dx = \frac{1}{a}\tan^{-1}\frac{x}{a} + C$

(18) $\int \frac{-1}{a^2 + x^2} dx = \frac{1}{a}\cot^{-1}\frac{x}{a} + C$

(19) $\int \frac{1}{x\sqrt{x^2 - a^2}} dx = \frac{1}{a}\sec^{-1}\frac{x}{a} + C$

(20) $\int \frac{-1}{x\sqrt{x^2 - a^2}} dx = \frac{1}{a}\operatorname{cosec}^{-1}\frac{x}{a} + C$

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(21) $\int \frac{1}{x^2-a^2}\,dx = \frac{1}{2a}\log\left|\frac{x-a}{x+a}\right| + c$

(22) $\int \frac{1}{a^2-x^2}\,dx = \frac{1}{2a}\log\left|\frac{a+x}{a-x}\right| + c$

(23) $\int \frac{1}{\sqrt{x^2-a^2}}\,dx = \log\left|x+\sqrt{x^2-a^2}\right| + c$

(24) $\int \frac{1}{\sqrt{x^2+a^2}}\,dx = \log\left|x+\sqrt{x^2+a^2}\right| + c$

(25) $\int \sqrt{x^2-a^2}\,dx = \frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2}\log\left|x+\sqrt{x^2-a^2}\right| + c$

(26) $\int \sqrt{x^2+a^2}\,dx = \frac{x}{2}\sqrt{x^2+a^2} + \frac{a^2}{2}\log\left|x+\sqrt{x^2+a^2}\right| + c$

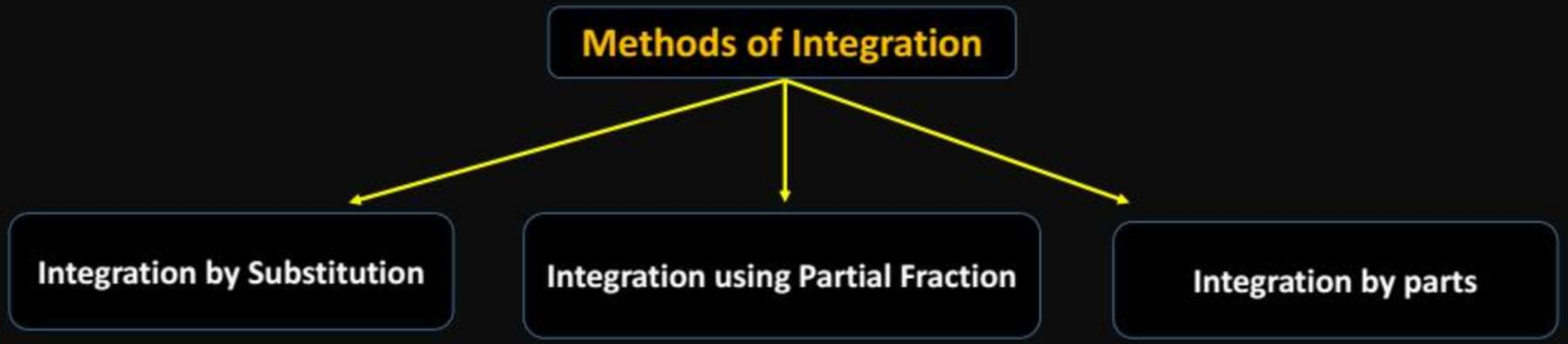
(27) $\int \sqrt{a^2-x^2}\,dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + c$

① $\int \sin(ax+b)\,dx = -\frac{\cos(ax+b)}{a} + C$

② $\int \frac{1}{ax+b}\,dx = \frac{1}{a}\log(ax+b) + C$

$\int \frac{1}{x}\,dx = \log x$

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Integration by parts

$$\int \underset{\text{I}}{u}\,\underset{\text{II}}{v}\,dx = u\int v\,dx - \int\left(\frac{d}{dx}(u)\int v\,dx\right)$$

I L A T E

I- Inverse Trigonometry Function

L- Logarithm Function

A- Algebraic Function

T- Trigonometry Function

E- Exponential Function

Properties of Definite Integral

① $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$

② $\int_a^b f(x) = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$

③ $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$

④ $\int_0^{2a} f(x)\,dx = 2\int_0^a f(x)\,dx$ If $f(2a-x)=f(x)$

$= 0$ If $f(2a-x)=-f(x)$

⑤ $\int_{-a}^{a} f(x)\,dx = 2\int_0^a f(x)\,dx$ If $f(x)$ is an even function i.e $f(-x)=f(x)$

$= 0$ If $f(x)$ is an odd function i.e $f(-x)=-f(x)$

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Cartesian System and some important graphs

I. Equation of x-axis $y = 0$

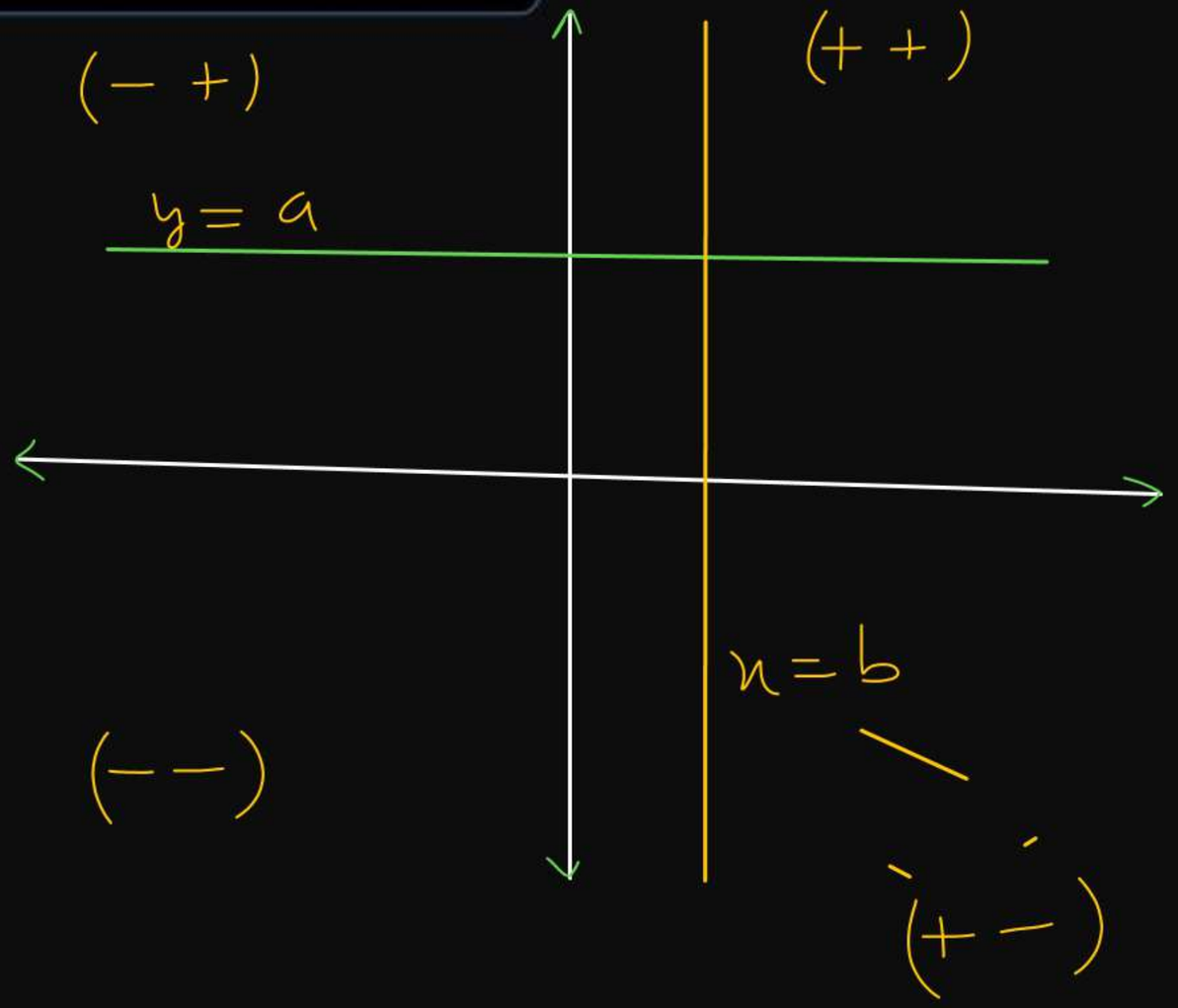
II. Equation of y-axis $x = 0$

III. Equation of a line parallel to x-axis $y = a$

IV. Equation of a line parallel to y-axis $x = b$

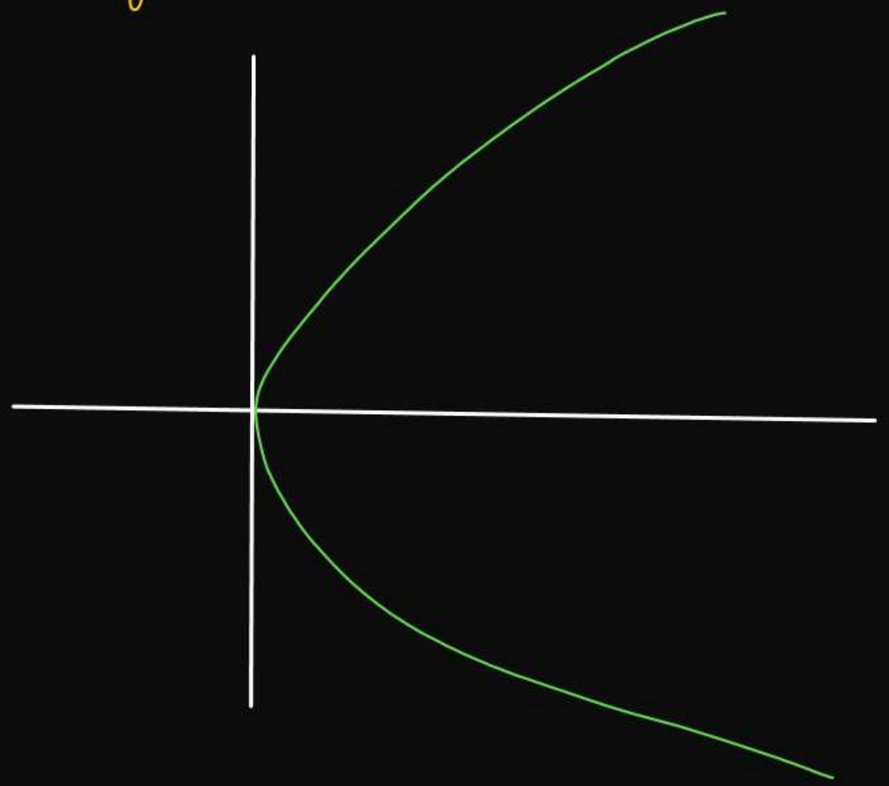
V. Equation of a line inclined to both the axes

$$ax + by + c = 0$$

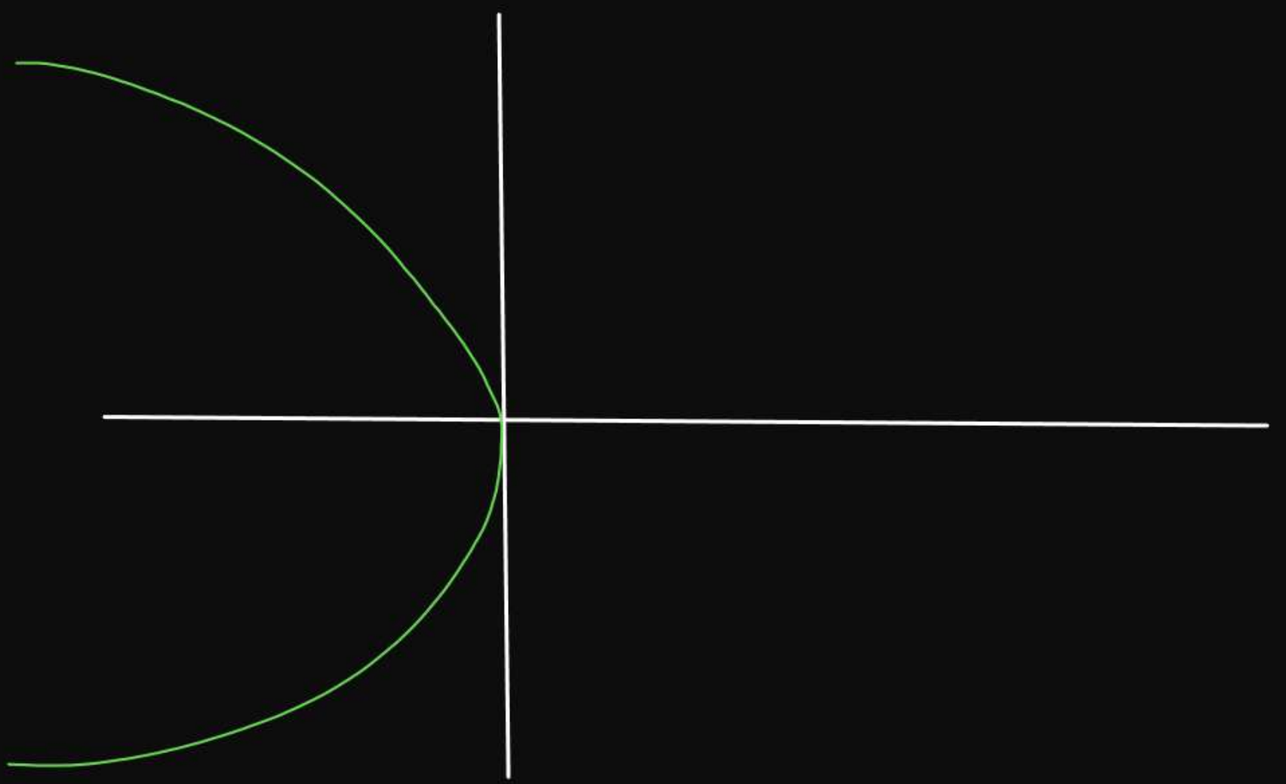


Equation of parabola

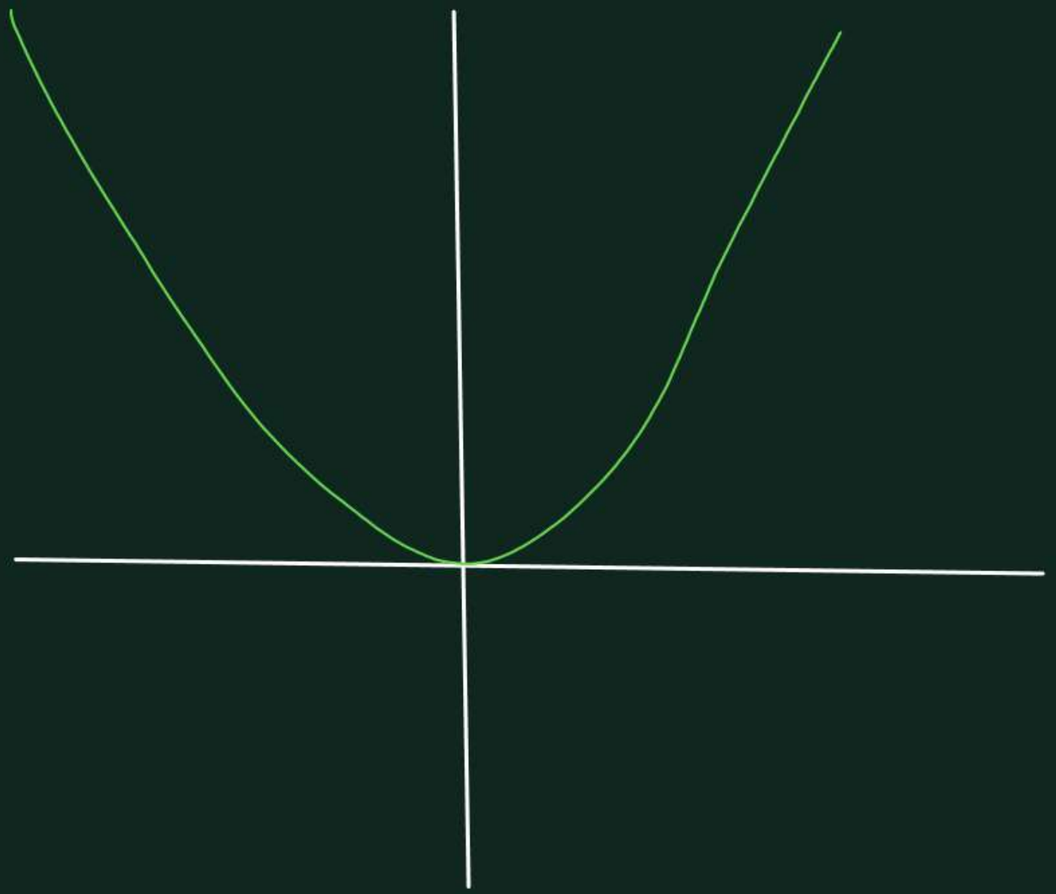
① $y^2 = 4ax$



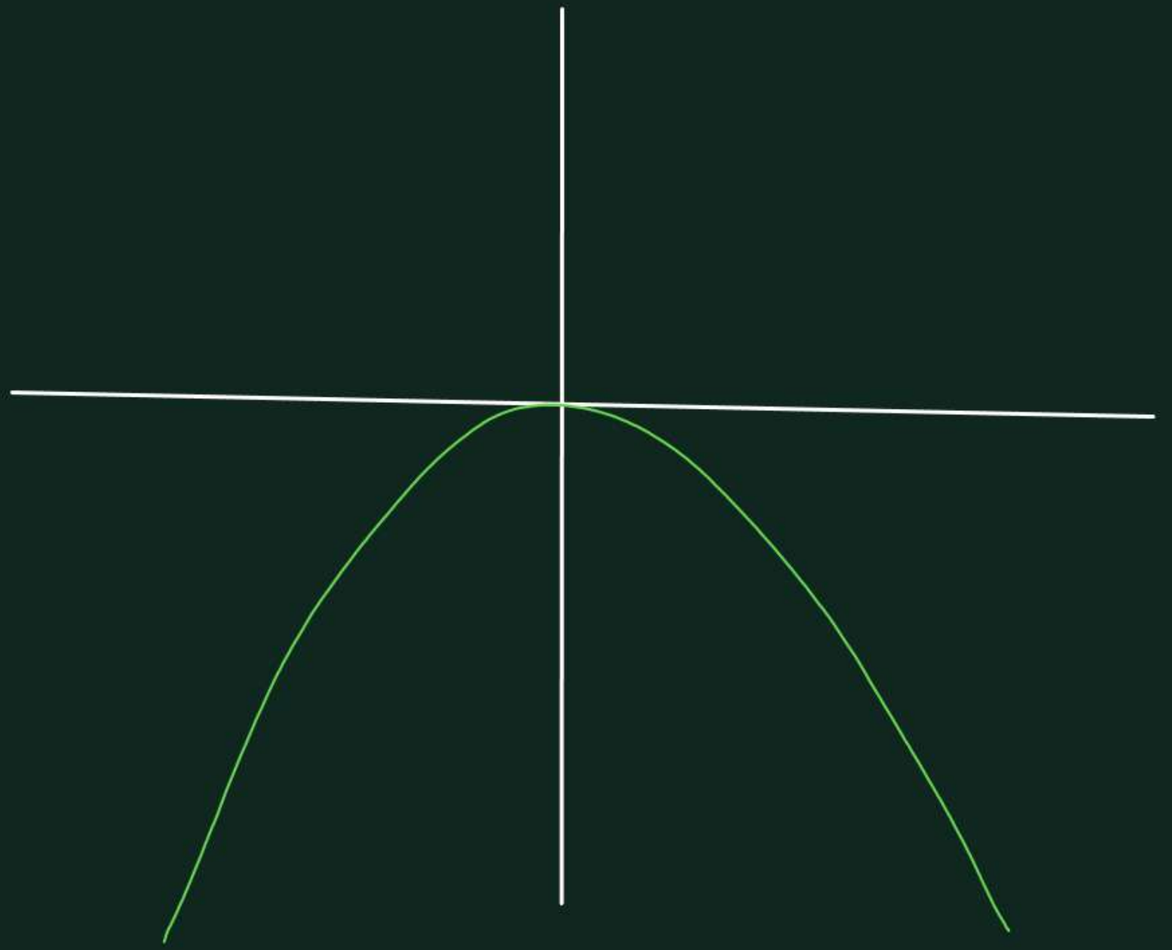
② $y^2 = -4ax$



③ $x^2 = 4ay$

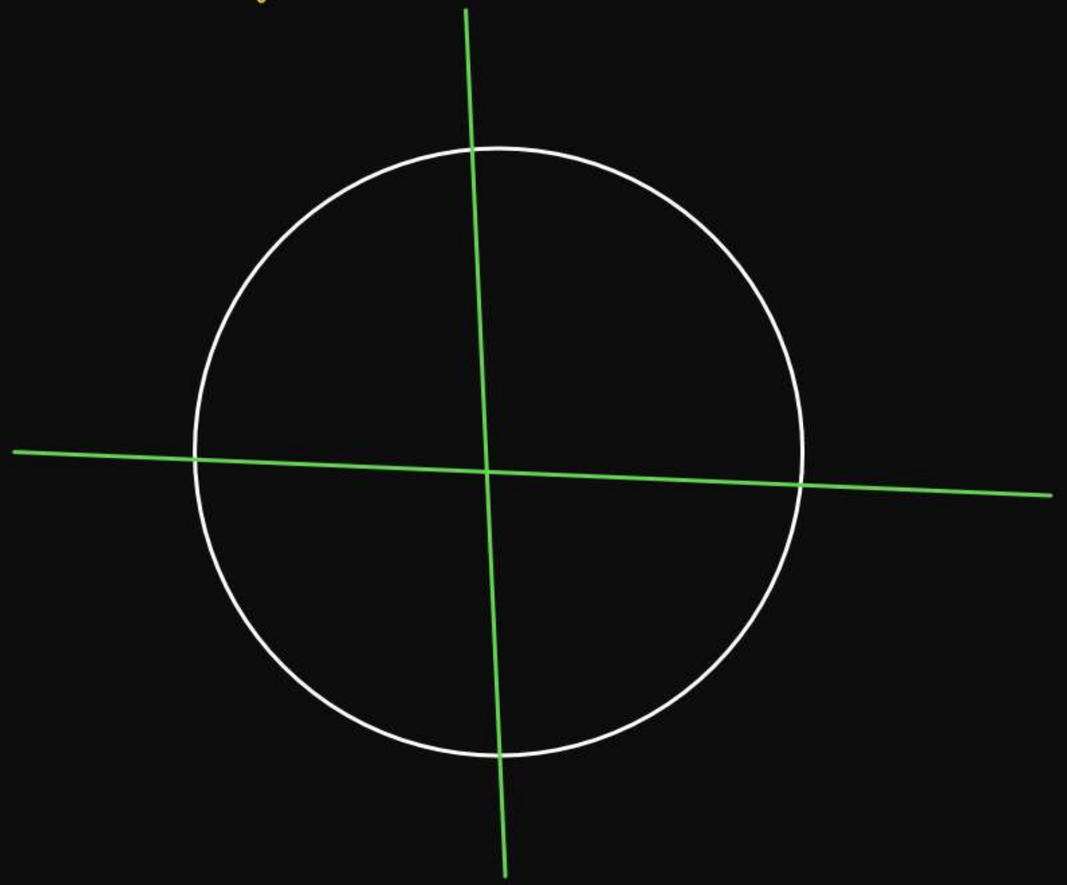


④ $x^2 = -4ay$



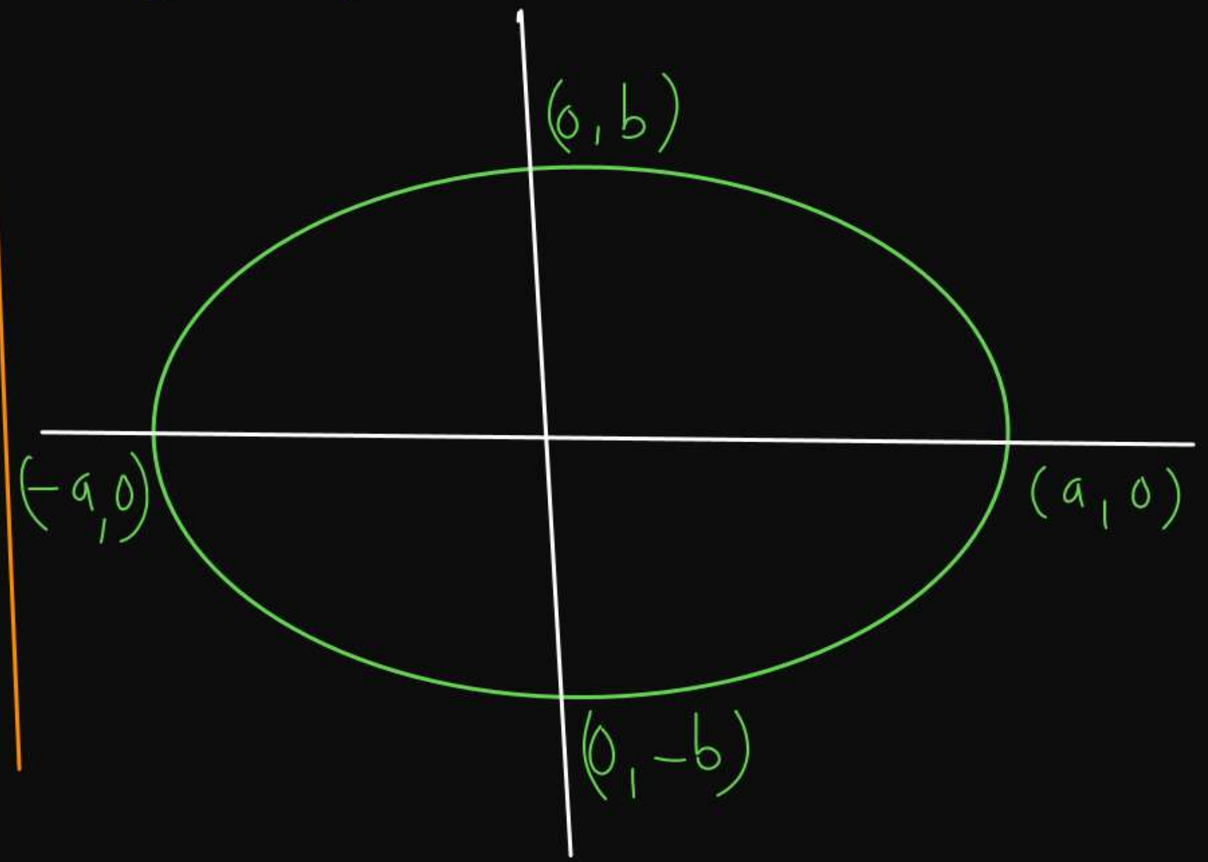
Equation of Circle

$$x^2 + y^2 = r^2$$

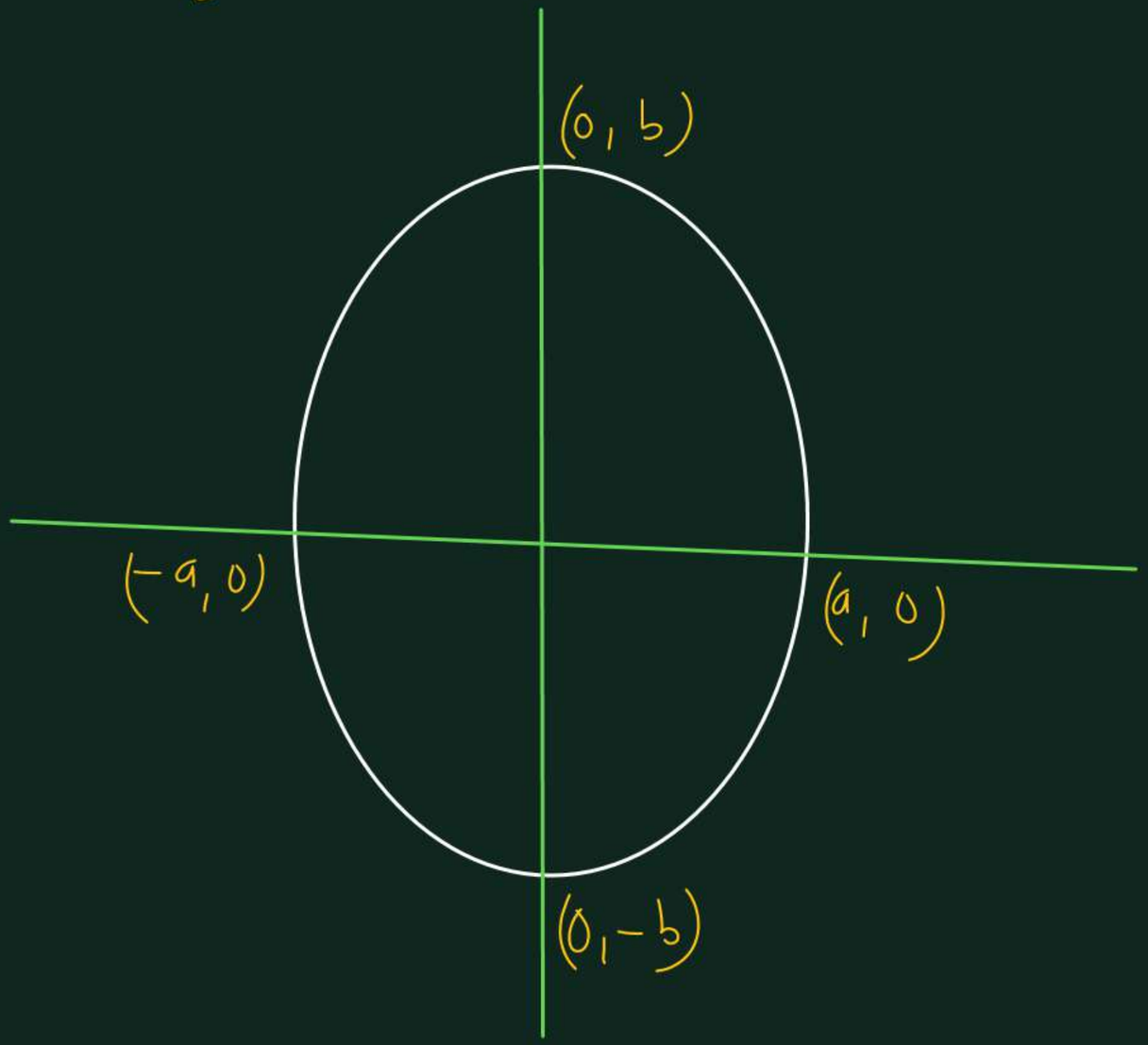


Equation of ellipse

① $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ If $a > b$



② $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ If $b > a$



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Multiple Integration : Double Integral

Lec-1

Today's Target

- ✓ Evaluation of Double Integral
- ✓ AKTU Questions
- ✓ Practice Questions

Double Integral

The integral $\int\int_R f(x,y)dx\,dy$ is called Double Integral of $f(x,y)$ over the region R .

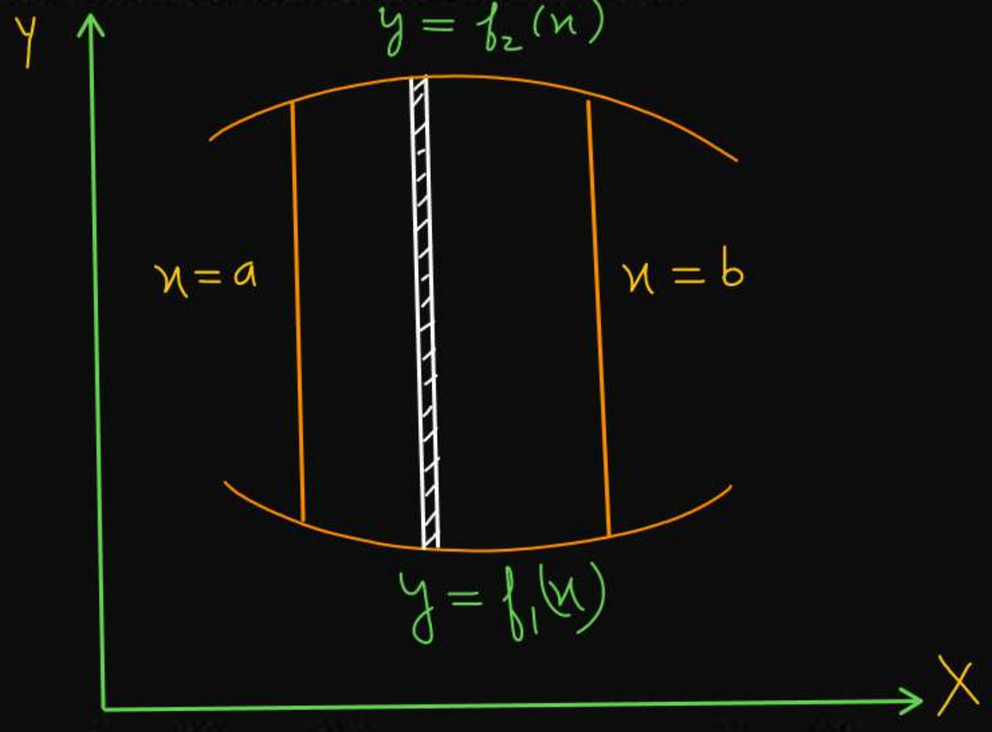
Evaluation of Double Integral

The methods of evaluating of double integral depend on the nature of the curves bounding the region R .

Case-I If the region R is defined as

(i) $a \le x \le b$ (ii) $f_1(x) \le y \le f_2(x)$

$$\int\int_R f(x,y)\,dx\,dy = \int_a^b \int_{f_1(x)}^{f_2(x)} f(x,y)\,dy\,dx$$

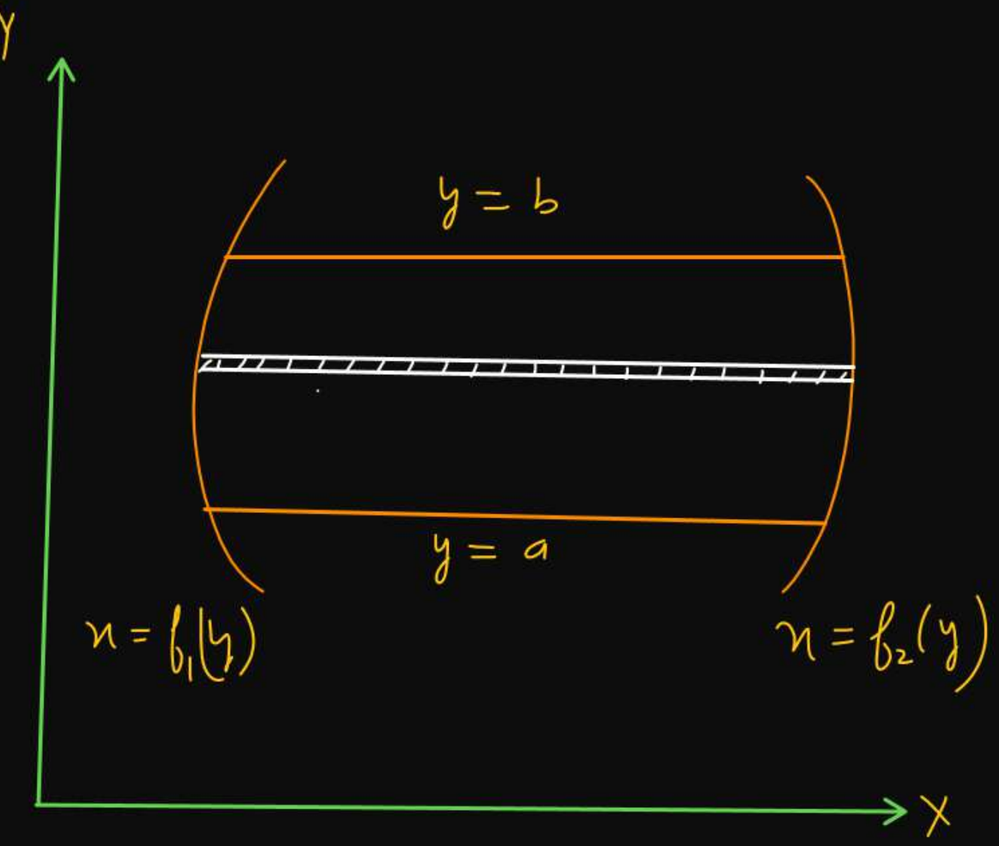


Note : Integral with variable limits calculated first i.e. w.r.t. y (treating x as a constant).

Case-II If the region R is defined as

(i) $a \le y \le b$ (ii) $f_1(y) \le x \le f_2(y)$

$$\iint_R f(x,y)\,dx\,dy = \int_a^b \int_{f_1(y)}^{f_2(y)} f(x,y)\,dx\,dy$$

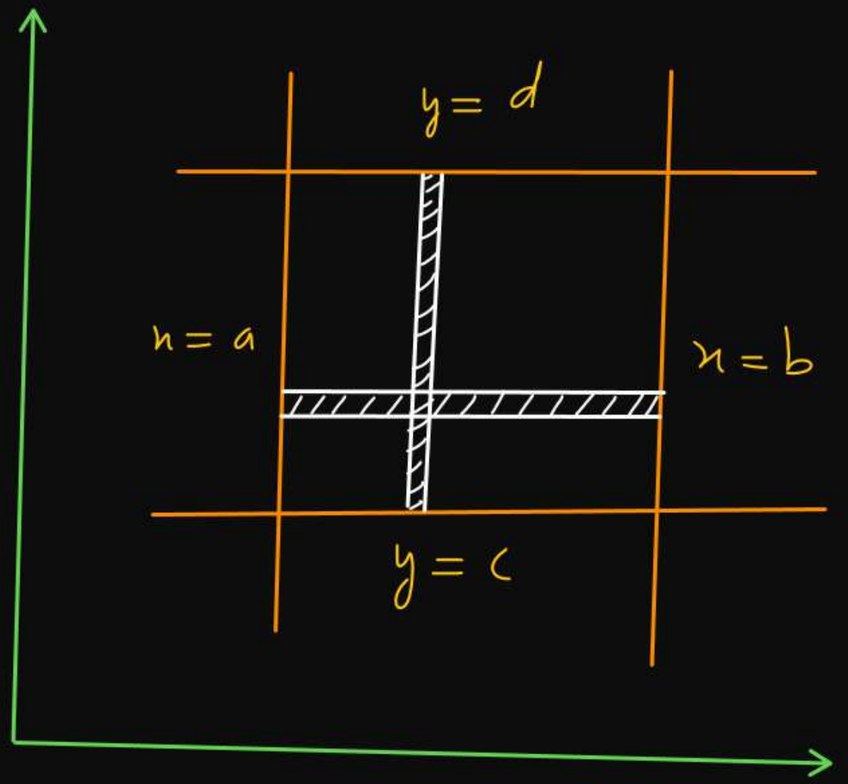


Note : Integral with variable limits calculated first i.e. w.r.t. x (treating y as a constant).

Case-III If the region R is defined as

(i) $a \le x \le b$ (ii) $c \le y \le d$

$$\int\int_R f(x,y)\,dx\,dy = \int_a^b \int_c^d f(x,y)\,dx\,dy$$



Q.1 Evaluate $\int_0^1 \int_0^1 \frac{dxdy}{\sqrt{(1-x^2)(1-y^2)}}$.

$$\int_0^1 \int_0^1 \frac{dx\,dy}{\sqrt{(1-x^2)(1-y^2)}}$$

$$= \int_0^1 \left[\int_0^1 \frac{dx}{\sqrt{(1-x^2)}\sqrt{(1-y^2)}} \right] dy$$

$$= \int_0^1 \frac{1}{\sqrt{1-y^2}} \left[\int_0^1 \frac{dx}{\sqrt{1-x^2}} \right] dy$$

$$= \int_0^1 \frac{1}{\sqrt{1-y^2}} \left(\sin^{-1}x \right)_0^1 dy$$

$$= \int_0^1 \frac{1}{\sqrt{1-y^2}} \left(\sin^{-1}1 - \sin^{-1}0 \right) dy$$

$$= \int_0^1 \frac{1}{\sqrt{1-y^2}} \left(\frac{\pi}{2} - 0 \right) dy$$

$$= \int_0^1 \frac{1}{\sqrt{1-y^2}} \times \frac{\pi}{2}\, dy$$

$$= \frac{\pi}{2} \int_0^1 \frac{dy}{\sqrt{1-y^2}}$$

$$= \frac{\pi}{2} \left(\sin^{-1}y \right)_0^1$$

$$= \frac{\pi}{2} \left(\sin^{-1}1 - \sin^{-1}0 \right)$$

$$= \frac{\pi}{2} \left(\frac{\pi}{2} - 0 \right)$$

$$= \frac{\pi^2}{4}$$

$$\int_0^1 \int_0^1 \frac{dx\,dy}{\sqrt{(1-x^2)(1-y^2)}} = \frac{\pi^2}{4}$$

Q.2 Evaluate $\int_0^1 \int_0^{\sqrt{1+x^2}} \frac{dydx}{1+x^2+y^2}$.

$$\int_0^1 \int_0^{\sqrt{1+x^2}} \frac{dy\,dx}{1+x^2+y^2}$$

$$= \int_0^1 \left[\int_0^{\sqrt{1+x^2}} \frac{dy}{1+x^2+y^2} \right] dx$$

$$= \int_0^1 \left[\int_0^{\sqrt{1+x^2}} \frac{dy}{\left(\sqrt{1+x^2}\right)^2+y^2} \right] dx$$

$$= \int_0^1 \frac{1}{\sqrt{1+x^2}} \left(\tan^{-1}\frac{y}{\sqrt{1+x^2}} \right)_0^{\sqrt{1+x^2}} dx$$

$$= \int_0^1 \frac{1}{\sqrt{1+x^2}} \left[\tan^{-1}\frac{\sqrt{1+x^2}}{\sqrt{1+x^2}} - \tan^{-1} 0 \right] dx$$

$$= \int_0^1 \frac{1}{\sqrt{1+x^2}} \left(\tan^{-1} 1 - 0 \right) dx$$

$$= \frac{\pi}{4} \int_0^1 \frac{dx}{\sqrt{1+x^2}}$$

$$= \frac{\pi}{4} \left[\log\left(x + \sqrt{1+x^2}\right) \right]_0^1$$

$$= \frac{\pi}{4} \left[\log(1+\sqrt{2}) - \log(0+1) \right]$$

$$= \frac{\pi}{4} \log(\sqrt{2}+1)$$

$$\int_0^1 \int_0^{\sqrt{1+x^2}} \frac{dy\,dx}{1+x^2+y^2} = \frac{\pi}{4} \log(\sqrt{2}+1)$$

Q.3 Evaluate $\iint_R y\,dx\,dy$, where R is the region bounded by the parabolas $x^2 = 4y$ and $y^2 = 4x$.

$y^2 = 4x$ — ①

$x^2 = 4y$ — ②

Solve ① and ②

$y = \frac{x^2}{4}$

$\left(\frac{x^2}{4}\right)^2 = 4x$

$\frac{x^4}{16} = 4x$

$x^4 = 64x$

$x^4 - 64x = 0$

$x(x^3 - 64) = 0$

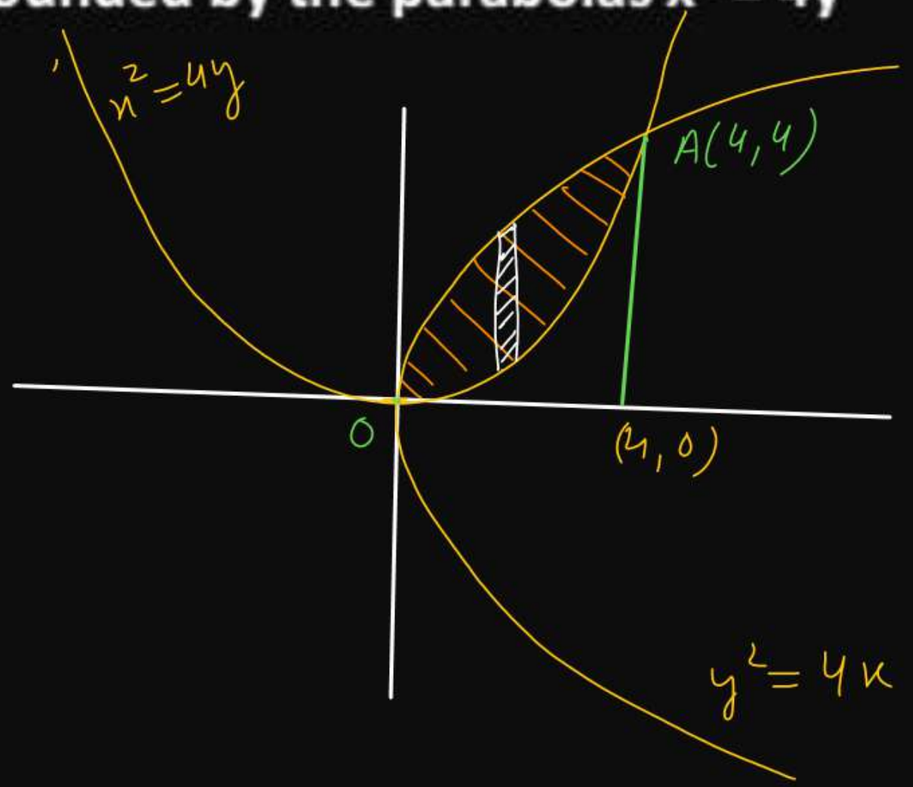
$x(x^3 - 4^3) = 0$

$x = 0$

$x = 4$

when $x = 0$, $y = 0$

when $x = 4$, $y = 4$



$$\iint_R y\,dx\,dy = \int_0^4 \int_{\frac{x^2}{4}}^{2\sqrt{x}} y\,dy\,dx = \int_0^4 \left(\frac{y^2}{2}\right)_{\frac{x^2}{4}}^{2\sqrt{x}} dx = \frac{1}{2}\int_0^4 \left[(2\sqrt{x})^2 - \left(\frac{x^2}{4}\right)^2\right] dx$$

$$= \frac{1}{2}\int_0^4 \left(4x - \frac{x^4}{16}\right) dx$$

$$= \frac{1}{2}\left[\frac{4x^2}{2} - \frac{x^5}{5\times 16}\right]_0^4$$

$$= \frac{1}{2}\left[\frac{4\times 4^2}{2} - \frac{4^5}{5\times 16}\right]$$

$$= \frac{1}{2}\left[\frac{4\times 4\times 4}{2} - \frac{4\times 4\times 4\times \cancel{4}\times \cancel{4}}{5\times \cancel{16}_{\cancel{4}}}\right]$$

$$= \frac{\cancel{4}^{2}\times 4\times 4}{\cancel{2}}\left(\frac{1}{2} - \frac{1}{5}\right)$$

$$= 32\left(\frac{5-2}{10}\right)$$

$$= \cancel{32}^{16} \times \frac{3}{\cancel{10}_{5}}$$

$$= \frac{48}{5} \quad \text{Ans}$$

Q.4 Evaluate $\int\int x\,y\,dx\,dy$ over the positive quadrant of the circle $x^2 + y^2 = a^2$.

$$x^2+y^2=a^2$$

$$y^2 = a^2-x^2$$

$$y = \pm\sqrt{a^2-x^2}$$

$$y = \sqrt{a^2-x^2}$$

$$\int\int xy\,dx\,dy = \int_0^a \int_0^{\sqrt{a^2-x^2}} xy\,dy\,dx$$

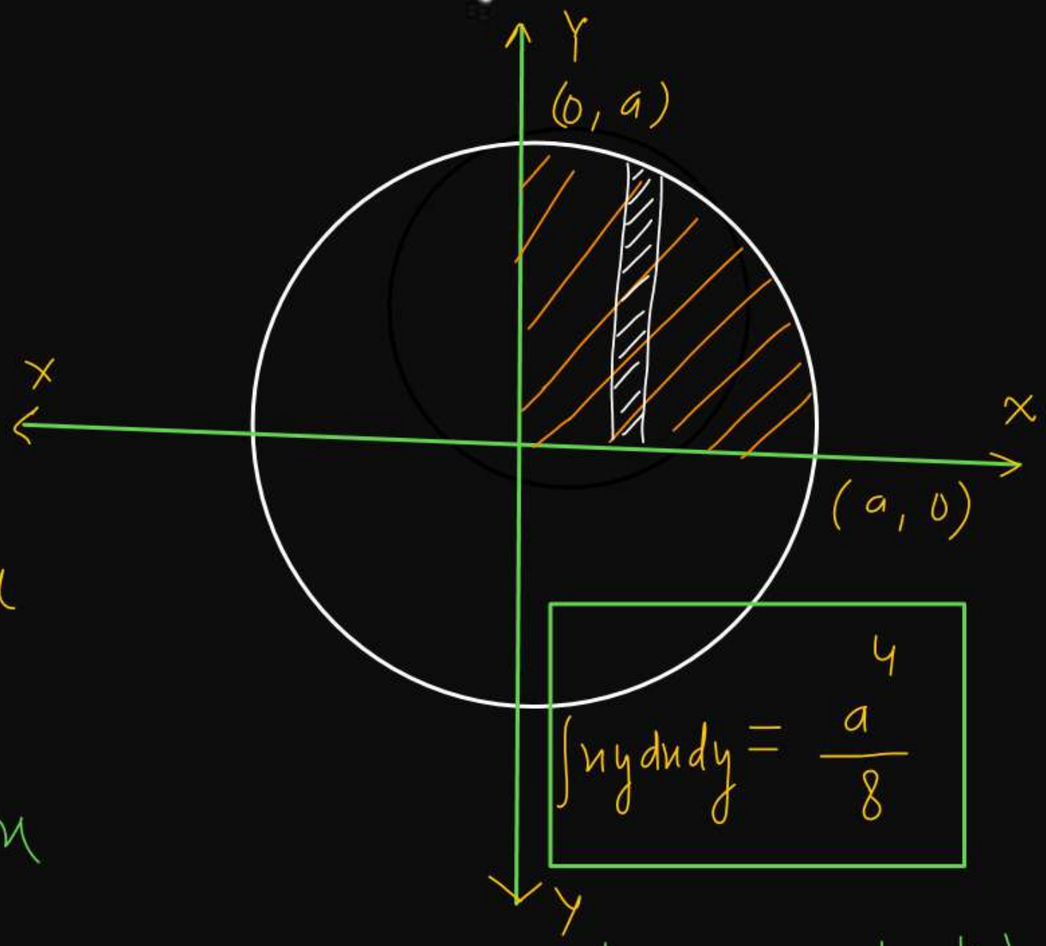
$$= \int_0^a x\left[\int_0^{\sqrt{a^2-x^2}} y\,dy\right]dx$$

$$= \int_0^a x\left(\frac{y^2}{2}\right)_0^{\sqrt{a^2-x^2}} dx$$

$$= \int_0^a x\left(\frac{a^2-x^2}{2}\right)dx$$

$$= \frac{1}{2}\int_0^a (a^2x - x^3)\,dx$$

$$= \frac{1}{2}\left(\frac{a^2x^2}{2} - \frac{x^4}{4}\right)_0^a = \frac{1}{2}\left[\frac{a^4}{2} - \frac{a^4}{4}\right] = \frac{1}{2}\left(\frac{2a^4-a^4}{4}\right)$$



$$\int xy\,dx\,dy = \frac{a^4}{8}$$

Q.5 Evaluate $\int\int x\,y\,dx\,dy$ over the region in the positive quadrant for which $x + y \le 1$.

$x + y \le 1$

$x + y = 1$

x	0	1
y	1	0

$y = 1 - x$

$$\iint xy\,dx\,dy = \int_0^1 \int_0^{1-x} xy\,dy\,dx$$

$$= \int_0^1 x \left[\int_0^{1-x} y\,dy \right] dx$$

$$= \int_0^1 x \left(\frac{y^2}{2} \right)_0^{1-x} dx$$

$$= \int_0^1 x \left[\frac{(1-x)^2}{2} - 0 \right] dx$$

$$= \int_0^1 x \left(\frac{1 + x^2 - 2x}{2} \right) dx$$

$$= \frac{1}{2} \int_0^1 (x + x^3 - 2x^2)\,dx$$

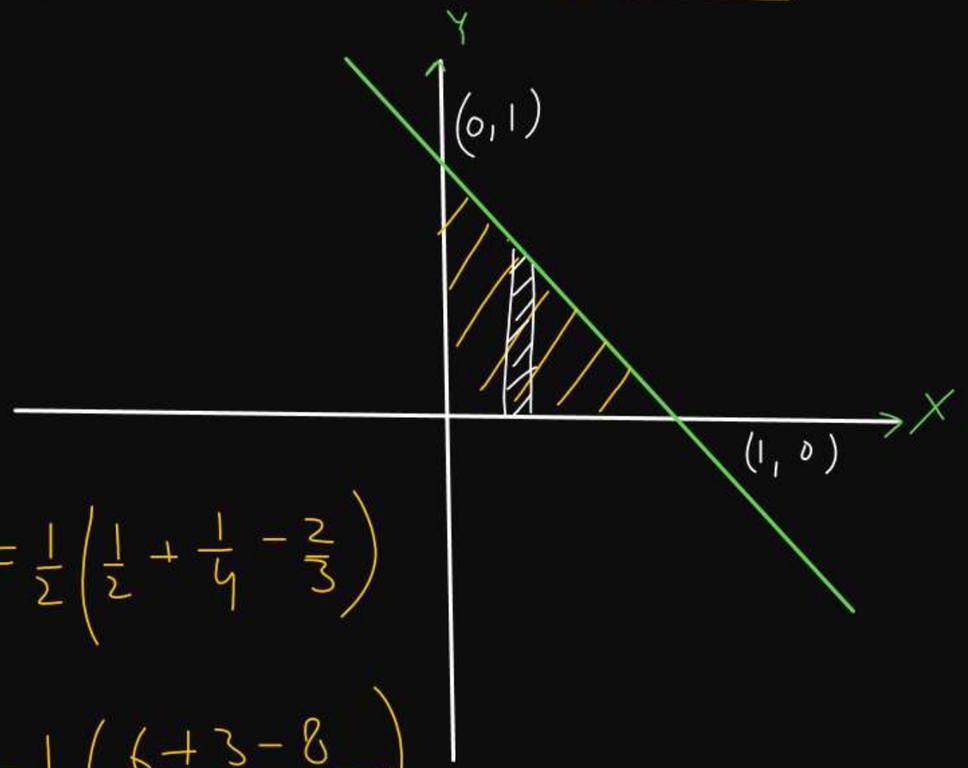
$$= \frac{1}{2} \left(\frac{x^2}{2} + \frac{x^4}{4} - \frac{2x^3}{3} \right)_0^1$$

$$= \frac{1}{2} \left(\frac{1}{2} + \frac{1}{4} - \frac{2}{3} \right)$$

$$= \frac{1}{2} \left(\frac{6 + 3 - 8}{12} \right)$$

$$= \frac{1}{2} \times \frac{1}{12}$$

$$= \frac{1}{24}$$



$$\iint xy\,dx\,dy = \frac{1}{24}$$

Q.6 Evaluate $\iint_A x\,y\,dx\,dy$, where A is the domain bounded by the x-axis, ordinate $x = 2a$ and the curve $x^2 = 4ay$.

X-axis

$x = 2a$ — (1)

$x^2 = 4ay$ — (2)

$$\iint_A xy\,dx\,dy = \int_0^{2a}\int_0^{\frac{x^2}{4a}} xy\,dy\,dx$$

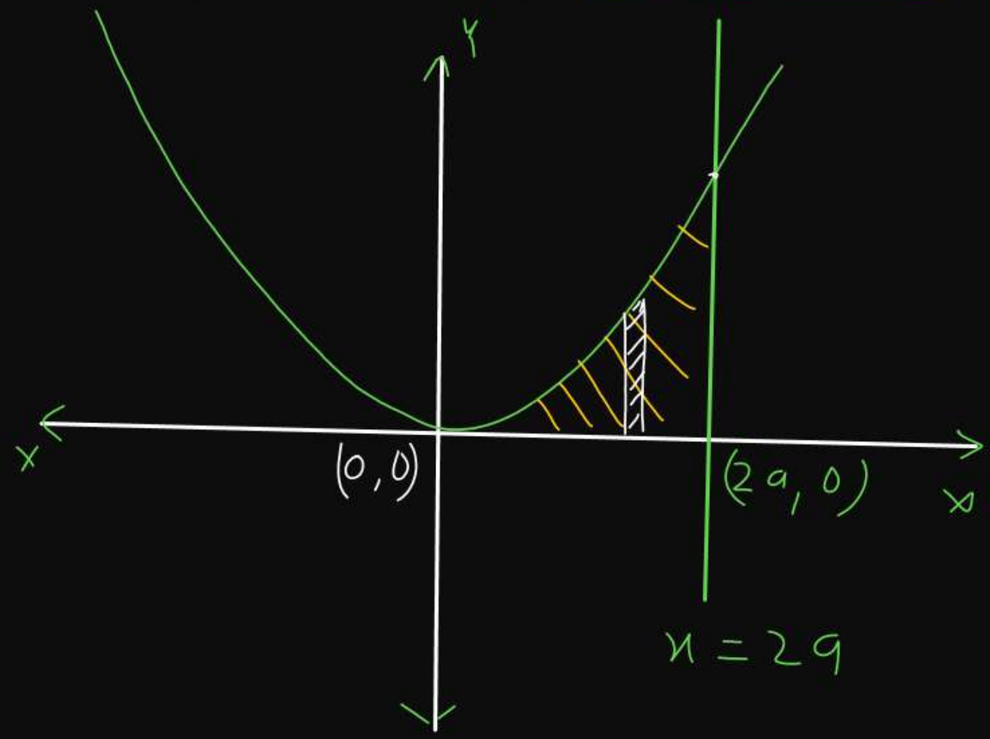
$$= \int_0^{2a} x\left[\int_0^{x^2/4a} y\,dy\right]dx$$

$$= \int_0^{2a} x\left(\frac{y^2}{2}\right)_0^{x^2/4a} dx$$

$$= \frac{1}{2}\int_0^{2a} x\left(y^2\right)_0^{x^2/4a} dx$$

$$= \frac{1}{2}\int_0^{2a} x\left[\left(\frac{x^2}{4a}\right)^2 - 0\right]dx$$

$$= \frac{1}{2}\int_0^{2a} \frac{x^5}{16a^2}\,dx = \frac{1}{32a^2}\int_0^{2a} x^5\,dx$$



$$= \frac{1}{32a^2}\left(\frac{x^6}{6}\right)_0^{2a}$$

$$= \frac{1}{32a^2}\left[\frac{(2a)^6}{6}\right]$$

$$= \frac{1}{32a^2} \times \frac{64\, a^{6-2}}{6_3}$$

$$= \frac{a^4}{3}$$

$$\boxed{\iint_A xy\,dx\,dy = \frac{a^4}{3}}$$

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Multiple Integration : Double Integral

Lec-2

Today's Target

- ✓ Change of order of integration
- ✓ AKTU Questions
- ✓ Practice Questions



By Gulshan sir

Change of order of integration

In a double integral with variable limits, the Change of order of integration changes the limits of integration.

Note - Complicated problems becomes easy after changing the order of integration

Working Rules

Step-1 : Write the given limits of integration

Step-2 : Draw the region for a given limits of integration

Step-3 : Check the strip for variable limits in the given region

Step-4 : If the given strip is horizontal then take a new vertical strip

Step-5 : If the given strip is vertical then take a new horizontal strip

Step-6 : Change the limit of integration , according to new strip

Q.1 ***Evaluate the integrals by changing the order of integration.***

$$\int_0^4 \int_y^4 \frac{x\,dx\,dy}{x^2+y^2}$$

$$\int_0^4 \int_y^4 \frac{x\,dx\,dy}{x^2+y^2}$$

Given limits of integration

$x = y$

$x = 4$

$y = 0$

$y = 4$

Given strip is horizontal

Take a new vertical strip

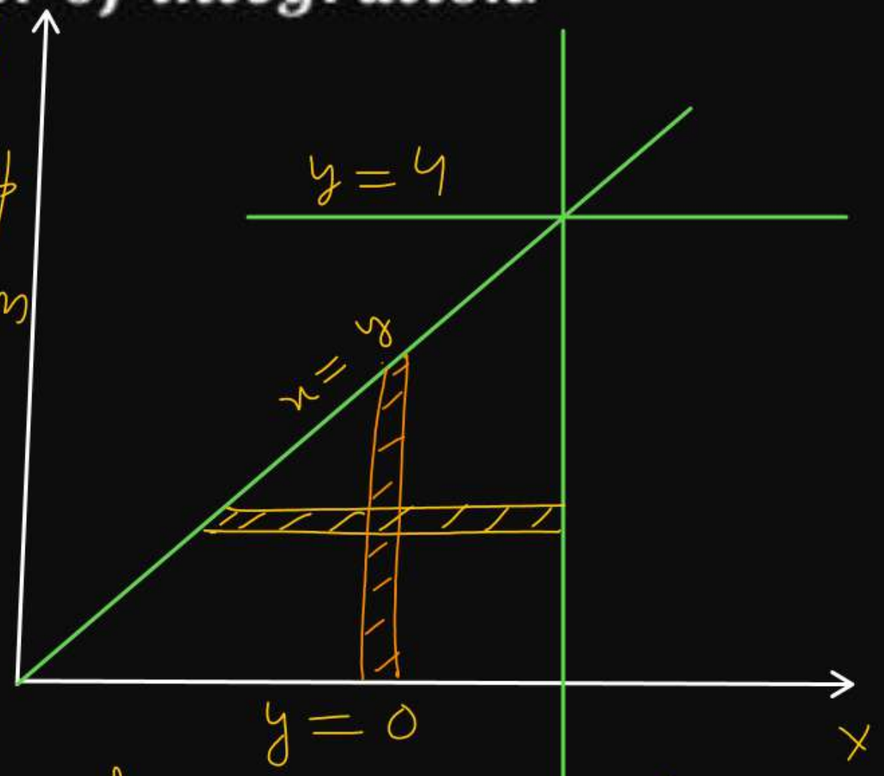
New limit of integration

$y = 0$

$y = x$

$x = 0$

$x = 4$



By changing the order of integration

$$\int_0^4 \int_y^4 \frac{x\,dx\,dy}{x^2+y^2} = \int_0^4 \left[\int_0^x \frac{x\,dy}{x^2+y^2}\right] dx = \int_0^4 x\,dx \int_0^x \frac{dy}{x^2+y^2}$$

$$= \int_0^4 x\,dx \; \frac{1}{x}\left(\tan^{-1}\frac{y}{x}\right)_0^x$$

$$= \int_0^4 dx \times \left(\tan^{-1}1 - \tan^{-1}0\right)$$

$$= \int_0^4 dx \left(\frac{\pi}{4} - 0\right)$$

$$= \int_0^4 \frac{\pi}{4}\,dx$$

$$= \frac{\pi}{4}\int_0^4 dx$$

$$= \frac{\pi}{4}(x)_0^4$$

$$= \frac{\pi}{4}(4-0)$$

$$= \frac{\pi}{4} \times 4$$

$$= \pi \quad \underline{\text{Ans}}$$

Practice Q.1 *Evaluate the integrals by changing the order of integration.*

$$\int_0^a \int_y^a \frac{x\,dx\,dy}{x^2+y^2}$$

Ans. $\pi a/4$

Q.2 ***Evaluate the integrals by changing the order of integration.***

$$\int_0^{\infty}\int_x^{\infty}\frac{e^{-y}}{y}\,dxdy$$

$$\int_0^{\infty}\int_x^{\infty}\frac{e^{-y}}{y}\,dx\,dy$$

Given limits of integration

$y = x$

$y = \infty$

$x = 0$

$x = \infty$

Given strip is vertical

Take a new Horizontal Strip

New limits of integration

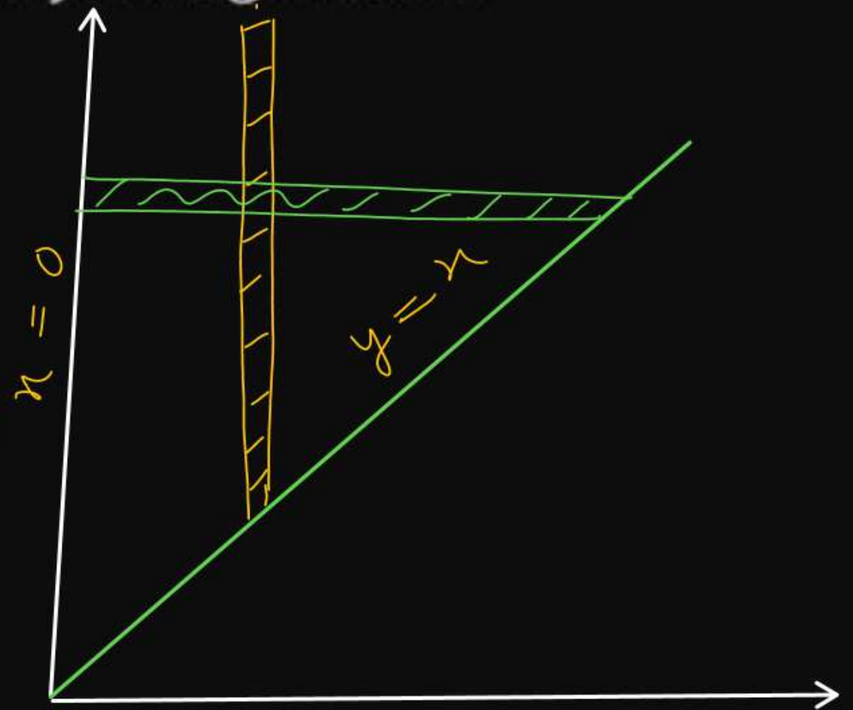
$x = 0$

$x = y$

$y = 0$

$y = \infty$

By changing the order of integration



$$\int_0^\infty \int_x^\infty \frac{e^{-y}}{y}\,dx\,dy = \int_0^\infty \int_0^y \frac{e^{-y}}{y}\,dx\,dy$$

$$= \int_0^\infty \frac{e^{-y}}{y}\,dy \int_0^y dx$$

$$= \int_0^\infty \frac{e^{-y}}{y}\,dy\,(x)_0^y$$

$$= \int_0^\infty \frac{e^{-y}}{y} \times (y-0)\,dy$$

$$= \int_0^\infty \frac{e^{-y}}{y} \times y\,dy$$

$$= \int_0^\infty e^{-y}\,dy$$

$$= \left(\frac{e^{-y}}{-1}\right)_0^\infty$$

$$= -1\left[e^{-\infty} - e^{0}\right]$$

$$= -1\left(\frac{1}{e^\infty} - 1\right)$$

$$= -1\left(\frac{1}{\infty} - 1\right)$$

$$= -1\,(0-1)$$

$$= +1$$

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Thank You

AKTU : Engineering Mathematics-1

Multiple Integration : Double Integral

Lec-3

Today's Target

- ✓ Change of order of integration(Part-2)
- ✓ AKTU Questions
- ✓ Practice Questions



By Gulshan sir

Change of order of integration

In a double integral with variable limits, the Change of order of integration changes the limits of integration.

Note - Some complicated problems become easy after changing the order of integration

Working Rules

Step-1 : Write the given limits of integration

Step-2 : Draw the region for a given limits of integration

Step-3 : Check the strip for variable limits in the given region

Step-4 : If the given strip is horizontal then take a new vertical strip

Step-5 : If the given strip is vertical then take a new horizontal strip

Step-6 : Change the limit of integration , according to new strip

Q.1 ***Change the order of integration in*** $I = \int_0^1 \int_{x^2}^{2-x} xy\,dy\,dx$ **and hence evaluate the same.**

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$$I = \int_0^1 \int_{x^2}^{2-x} xy\,dy\,dx$$

Given limits of integration

$y = x^2$ (Parabola)

$y = 2 - x$ (line)

x	0	2
y	2	0

$x = 0$ (Y-axis)

$x = 1$ (A line parallel to Y-axis)

Solve $y = x^2$ and $y = 2 - x$

$$2 - x = x^2$$

$$x^2 + x - 2 = 0$$

$$x^2 + x - 2 = 0$$

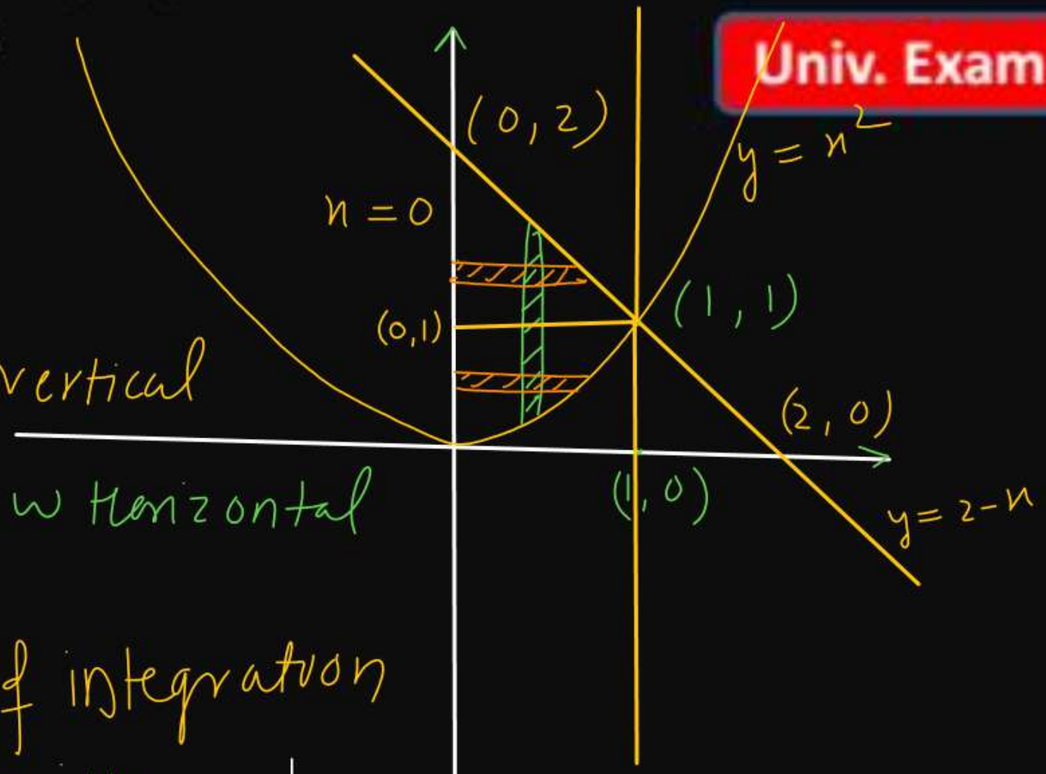
$$\boxed{x = 1}$$

- Given strip is vertical
- Take two new Horizontal Strips
- New limits of integration

For lower region

$$\begin{cases} x = 0 \\ x = \sqrt{y} \end{cases}$$

$$\begin{cases} y = 0 \\ y = 1 \end{cases}$$



For upper region

$$\begin{cases} x = 0 \\ x = 2 - y \end{cases}$$

$$\begin{cases} y = 1 \\ y = 2 \end{cases}$$

$$\int_0^1 \int_{x^2}^{2-x} xy\,dy\,dx = \int_0^1 \int_0^{\sqrt{y}} xy\,dx\,dy + \int_1^2 \int_0^{2-y} xy\,dx\,dy$$

$$= \int_0^1 y\,dy \int_0^{\sqrt{y}} x\,dx + \int_1^2 y\,dy \int_0^{2-y} x\,dx$$

$$= \int_0^1 y\,dy \left(\frac{x^2}{2}\right)_0^{\sqrt{y}} + \int_1^2 y\,dy \left(\frac{x^2}{2}\right)_0^{2-y}$$

$$= \int_0^1 y\,dy \left(\frac{y}{2} - 0\right) + \int_1^2 y\,dy \times \frac{1}{2}(2-y)^2$$

$$= \frac{1}{2}\int_0^1 y^2\,dy + \frac{1}{2}\int_1^2 y(4+y^2-4y)$$

$$= \frac{1}{2}\left(\frac{y^3}{3}\right)_0^1 + \frac{1}{2}\int_1^2 (4y+y^3-4y^2)\,dy$$

$$= \frac{1}{6}(y^3)_0^1 + \frac{1}{2}\left(\frac{4y^2}{2} + \frac{y^4}{4} - \frac{4y^3}{3}\right)_1^2$$

$$= \frac{1}{6}(1-0) + \frac{1}{2}\left[2\times 4 + \frac{16}{4} - \frac{4\times 8}{3} - 2 - \frac{1}{4} + \frac{4}{3}\right]$$

$$= \frac{1}{6} + \frac{1}{2}\left[8 + 4 - \frac{32}{3} - 2 - \frac{1}{4} + \frac{4}{3}\right]$$

$$= \frac{1}{6} + \frac{1}{2}\left[\frac{96+48-128-24-3+16}{12}\right]$$

$$= \frac{1}{6} + \frac{1}{2}\left(\frac{160-155}{12}\right) = \frac{1}{6} + \frac{1}{2}\times\frac{5}{12}$$

$$= \frac{1}{6} + \frac{5}{24} = \frac{9}{24}^{3}_{8}$$

$$= \frac{3}{8}$$

Practice Q.1

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Evaluate the integrals by changing the order of integration

$$\int_0^1 \int_{y^2}^{2-y} xy \, dxdy$$

Ans. 3/8

Practice Q.2

Univ. Exam

Evaluate the integrals by changing the order of integration.

$$\int_0^a \int_{x^2/a}^{2a-x} xy \, dy \, dx$$

Ans. $3a^4/8$

Evaluate the integrals by changing the order of integration.

Practice Q.3

$$\int_{0}^{2a} \int_{x^2/4a}^{3a-x} (x^2 + y^2)\, dy\, dx$$

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Ans. $314a^4/35$

Evaluate the integrals by changing the order of integration.

Practice Q.4

$$\int_0^2 \int_{x^2/4}^{3-x} xy\,dy\,dx$$

Univ. Exam

Ans. 8/3

Evaluate the integrals by changing the order of integration.

Q.2

$$\int_0^1 \int_x^{\sqrt{2-x^2}} \frac{x\,dy\,dx}{\sqrt{x^2+y^2}}.$$

$$\int_0^1 \int_x^{\sqrt{2-x^2}} \frac{x\,dy\,dx}{\sqrt{x^2+y^2}}$$

Given limits of integration

$y = x$ — ① (a line)

$y = \sqrt{2-x^2}$

$y^2 = 2 - x^2$

$x^2 + y^2 = 2$ — ② (a circle)

$x = 0$ (Y-axis)

$x = 1$ (A line ∥ to Y-axis)

Solve ① and ②

$x^2 + x^2 = 2$

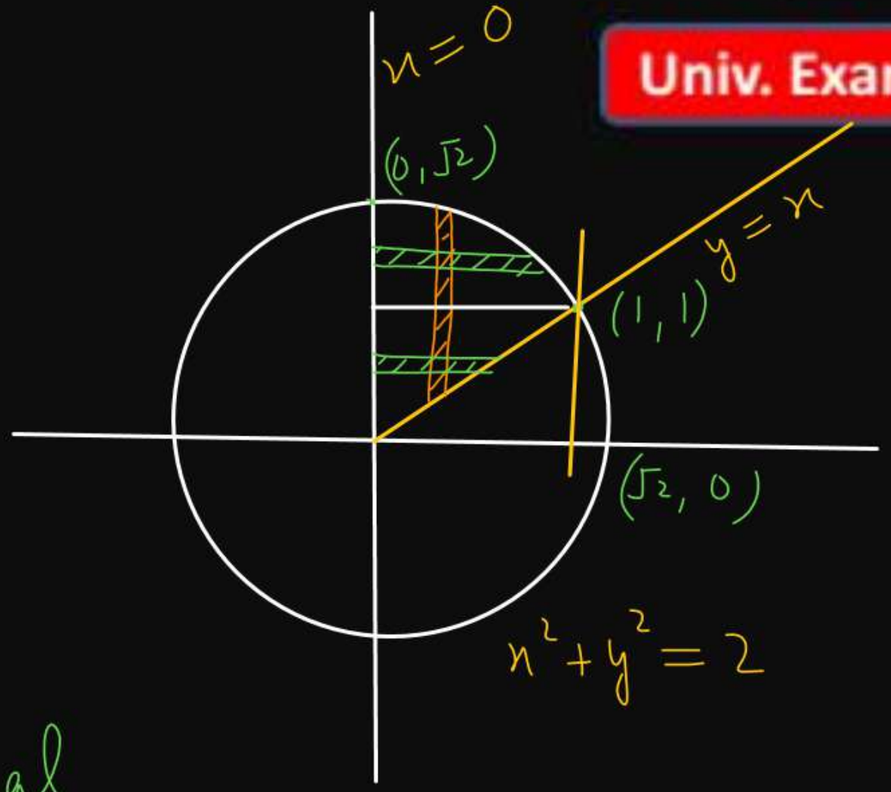
$2x^2 = 2$

$x^2 = 1$

$\boxed{x = 1}$ $\quad x = -1$ ✗

Given strip is vertical

Take two new Horizontal strips



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New limits of integration

For lower region	For upper region
$x = 0$	$x = 0$
$x = y$	$x = \sqrt{2-y^2}$
$y = 0$	$x = 1$
$y = 1$	$x = \sqrt{2}$

change the order of integration

$$\int_0^1 \int_x^{\sqrt{2-x^2}} \frac{x\,dy\,dx}{\sqrt{x^2+y^2}} = \int_0^1 \int_0^y \frac{x\,dy\,dx}{\sqrt{x^2+y^2}} + \int_1^{\sqrt{2}} \int_0^{\sqrt{2-y^2}} \frac{x\,dy\,dx}{\sqrt{x^2+y^2}}$$

Put $x^2+y^2 = t^2$

$\not{2}x\,dx = \not{2}t\,dt$

$x\,dx = t\,dt$

$$= \int_0^1 dy \int_0^y \frac{\not{t}\,dt}{\not{t}} + \int_1^{\sqrt{2}} dy \int_0^{\sqrt{2-y^2}} \frac{\not{t}\,dt}{\not{t}}$$

$$= \int_0^1 dy\,(t)_0^y + \int_1^{\sqrt{2}} dy\,(t)_0^{\sqrt{2-y^2}}$$

$$= \int_0^1 \left(\sqrt{x^2+y^2}\right)_0^y dy + \int_1^{\sqrt{2}} \left(\sqrt{x^2+y^2}\right)_0^{\sqrt{2-y^2}} dy$$

$$= \int_0^1 (\sqrt{2}\,y - y)\,dy + \int_1^{\sqrt{2}} (\sqrt{2} - y)\,dy$$

$$= \int_0^1 (\sqrt{2}y - y)\,dy + \int_1^{\sqrt{2}} (\sqrt{2} - y)\,dy$$

$$= \left(\sqrt{2}\frac{y^2}{2} - \frac{y^2}{2}\right)_0^1 + \left(\sqrt{2}y - \frac{y^2}{2}\right)_1^{\sqrt{2}}$$

$$= \frac{\sqrt{2}}{2} - \frac{1}{2} - 0 + 0 + \sqrt{2}\sqrt{2} - \frac{2}{2} - \sqrt{2} + \frac{1}{2}$$

$$= \frac{\sqrt{2}}{2} + 2 - 1 - \sqrt{2}$$

$$= 1 + \frac{\sqrt{2}}{2} - \sqrt{2}$$

$$= \frac{2 + \sqrt{2} - 2\sqrt{2}}{2}$$

$$= \frac{2 - \sqrt{2}}{2} = 1 - \frac{\sqrt{2}}{2} = 1 - \frac{1}{\sqrt{2}}$$

$$\int_0^1 \int_x^{\sqrt{2-x^2}} \frac{x\,dy\,dx}{\sqrt{x^2+y^2}} = 1 - \frac{1}{\sqrt{2}}$$

Changing the order of integration in the double integral

Q.3 $I = \int_0^8 \int_{x/4}^{2} f(x,y)\,dy\,dx$ leads to the the value $I = \int_r^s \int_p^q f(x,y)\,dx\,dy$. What is the value of q?.

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$$I = \int_0^8 \int_{x/4}^{2} f(x,y)\,dy\,dx$$

Given limits of integration

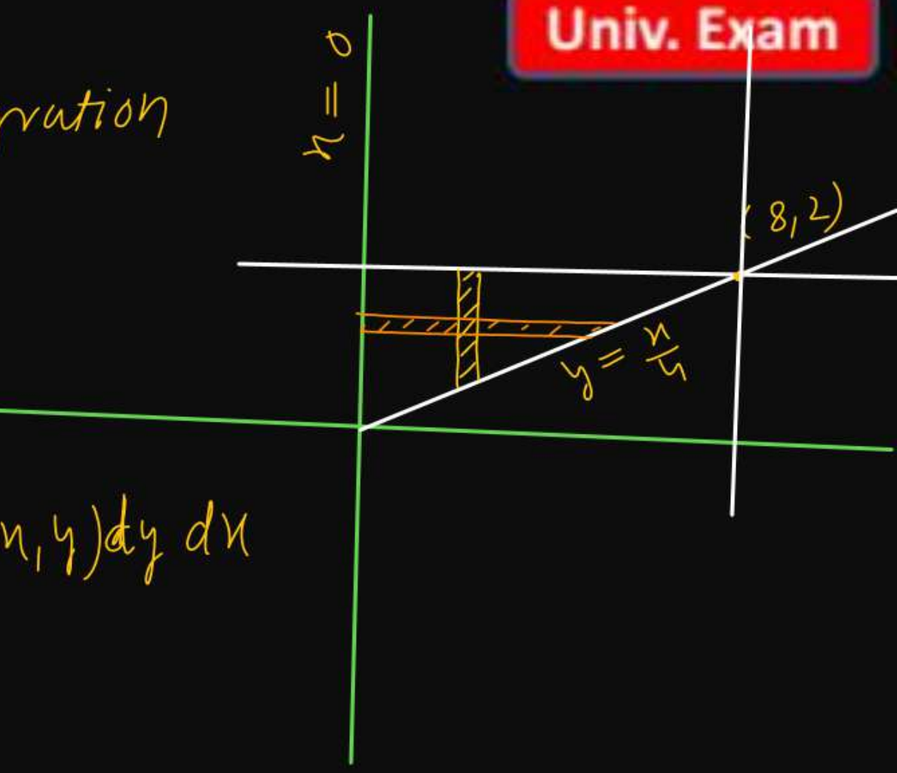
$y = \frac{x}{4}$ (line)

$y = 2$ (line || to X-axis)

$x = 0$ (Y-axis)

$x = 8$ (line || to Y-axis)

Change the order of integration



$$\int_0^8 \int_{x/4}^{2} f(x,y)\,dy\,dx = \int_0^2 \int_0^{4y} f(x,y)\,dx\,dy$$

on compairing

$s = 2$ $\quad P = 0$

$r = 0$

$q = 4y$

Practice Q.5

Changing the order of integration in the double integral $I = \int_{0}^{1} \int_{x^2}^{2-x} f(x, y)\, dy\, dx$

Univ. Exam

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AKTU : Engineering Mathematics-1

Multiple Integration : Double Integral

Lec-4



By Gulshan sir

Today's Target

- ✓ Change of order of integration(Part-3)
- ✓ AKTU Questions
- ✓ Practice Questions

Change of order of integration

In a double integral with variable limits, the Change of order of integration changes the limits of integration.

Working Rules

Step-1 : Write the given limits of integration

Step-2 : Draw the region for a given limits of integration

Step-3 : Check the strip for variable limits in the given region

Step-4 : If the given strip is horizontal then take a new vertical strip

Step-5 : If the given strip is vertical then take a new horizontal strip

Step-6 : Change the limit of integration , according to new strip

Q.1 ***Evaluate the integrals by changing the order of integration.***

$$\int_0^1 \int_{e^x}^{e} \frac{dydx}{\log y}.$$

$$\int_0^1 \int_{e^x}^{e} \frac{dy\,dx}{\log y}$$

Given limits of integration

$y = e^x$ —①

$y = e$ —②

$x = 1$ —③

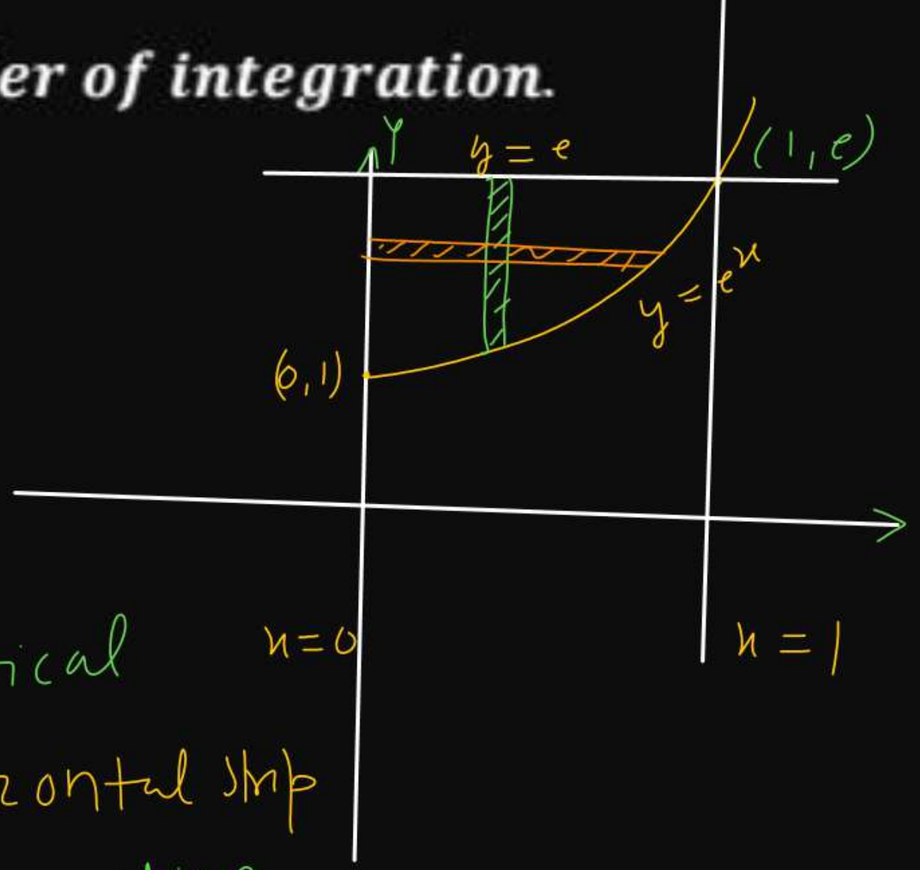
$x = 0$ —④

Solve ① and ②

$e = e^x$

$e^1 = e^x$

$\boxed{x = 1}$ $\boxed{y = e}$



Given strip is vertical

Take a new Horizontal strip

New limits of integration

$x = 0$ —①

$e^x = y$

$\log e^x = \log y$

$x \log e = \log y$

$x = \log y$ —②

$y = 1$ —③

$y = e$ —④

Change the order of integration

$$\int_0^1 \int_{e^x}^{e} \frac{dy\,dx}{\log y} = \int_1^e \int_0^{\log y} \frac{dy\,dx}{\log y}$$

$$= \int_1^e \frac{dy}{\log y} \int_0^{\log y} dx$$

$$= \int_1^e \frac{dy}{\log y} (x)_0^{\log y}$$

$$= \int_1^e \frac{dy}{\log y} \log y$$

$$= \int_1^e dy$$

$$= (y)_1^e$$

$$= e - 1$$

$$\int_0^1 \int_{e^x}^{e} \frac{dy\,dx}{\log y} = e - 1$$

Evaluate the integrals by changing the order of integration.

Practice Q.1

$$\int_0^2 \int_1^{e^x} dydx .$$

Ans. $e^2 - 3$

Q.2 ***Evaluate the integrals by changing the order of integration.***

$$\int_0^{\infty}\int_0^{x} xe^{-\frac{x^2}{y}}\,dy\,dx.$$

$$\int_0^{\infty}\int_0^{x} xe^{-\frac{x^2}{y}}\,dy\,dx$$

Given limits of integration

$y = 0$ —①

$y = x$ —②

$x = 0$ —③

$x = \infty$ —④

Given strip is vertical

Take a new Horizontal strip

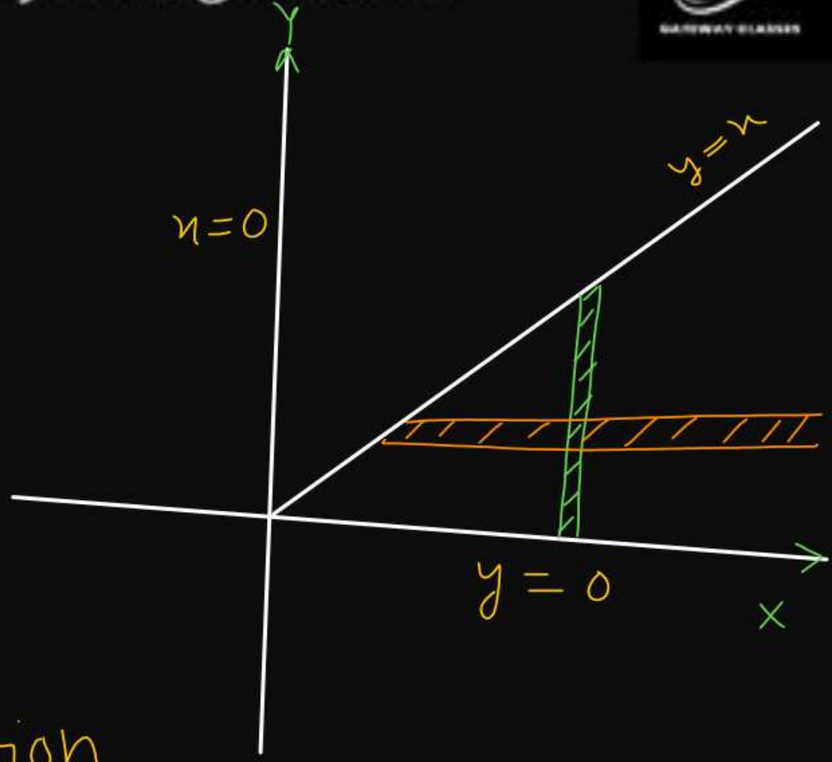
New limits of integration

$x = y$

$x = \infty$

$y = 0$

$y = \infty$



Change the order of integration

$$\int_0^{\infty}\int_0^{x} xe^{-\frac{x^2}{y}}\,dy\,dx = \int_0^{\infty}\int_y^{\infty} xe^{-\frac{x^2}{y}}\,dx\,dy = \int_0^{\infty} dy \int_y^{\infty} e^{-\frac{x^2}{y}}\,x\,dx$$

Put $\frac{x^2}{y} = t$

$$\frac{2x\,dx}{y} = dt$$

$$x\,dx = \frac{y}{2}dt$$

$$= \int_0^\infty dy \int_y^\infty e^{-t} \times \frac{y}{2} dt$$

$$= \int_0^\infty \frac{y}{2} dy \int_y^\infty e^{-t} dt$$

$$= \int_0^\infty \frac{y}{2} dy \left(-e^{-t}\right)_y^\infty$$

$$= -\int_0^\infty \frac{y}{2} dy \left(e^{-\frac{x^2}{y}}\right)_y^\infty$$

$$= -\int_0^\infty \frac{y}{2} dy \left(e^{-\infty} - e^{-y}\right)$$

$$= -\frac{1}{2}\int_0^\infty y\,dy\,(0 - e^{-y})$$

$$= +\frac{1}{2}\int_0^\infty y\,e^{-y}\,dy$$

I II

$$= \frac{1}{2}\left[y(-e^{-y}) - \int 1 \times (-e^{-y})\right]_0^\infty$$

$$= \frac{1}{2}\left[-ye^{-y} + (-e^{-y})\right]_0^\infty$$

$$= \frac{1}{2}\left[-ye^{-y} - e^{-y}\right]_0^\infty$$

$$= -\frac{1}{2}\left[ye^{-y} + e^{-y}\right]_0^\infty$$

$$= -\frac{1}{2}\left[0 + 0 - 0 - 1\right]$$

$$= -\frac{1}{2}(-1) = \frac{1}{2}$$

$$\boxed{\int_0^\infty \int_0^x x\,e^{-\frac{x^2}{y}}\,dy\,dx = \frac{1}{2}}$$

Evaluate the integrals by changing the order of integration.

Practice Q.2

$$\int_0^{\infty}\int_0^{y} xe^{-\frac{y^2}{x}}\,dy\,dx.$$

Ans. 1/2

Evaluate the integrals by changing the order of integration.

Q.3

$$\int_0^1 \int_0^{\sqrt{1-x^2}} \frac{e^y}{(e^y+1)\sqrt{1-x^2-y^2}}\, dy\, dx.$$

$$\int_0^1 \int_0^{\sqrt{1-x^2}} \frac{e^y}{(e^y+1)\sqrt{1-x^2-y^2}}\, dy\, dx$$

Given limits of integration

$y = 0$ —①

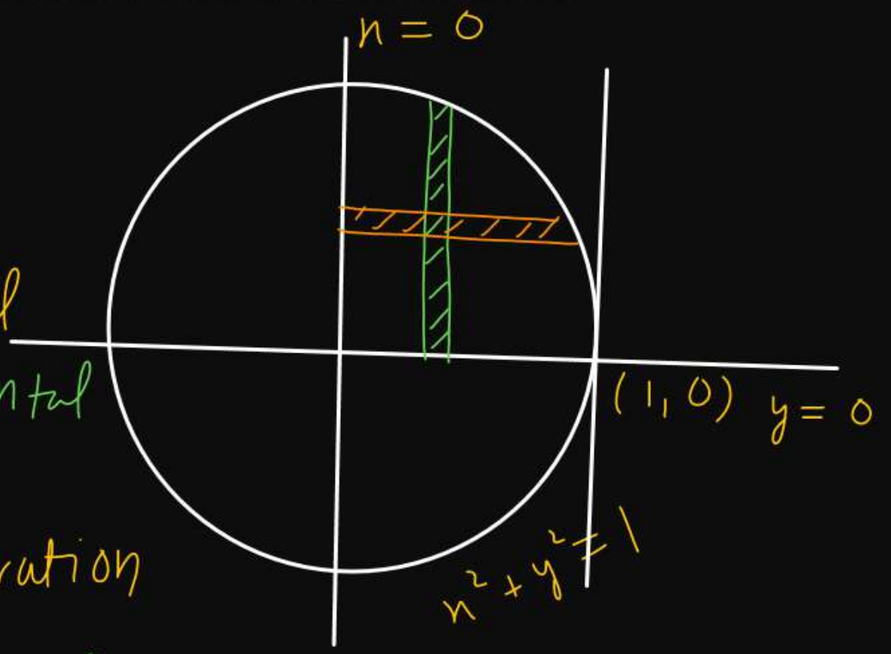
$y = \sqrt{1-x^2}$

$y^2 = 1-x^2$

$x^2+y^2 = 1$ —②

$x = 0$

$x = 1$



Given strip is vertical

Take a new Horizontal strip

New limits of integration

$x = 0$ $\quad y = 0$

$x = \sqrt{1-y^2}$ $\quad y = 1$

Change the order of integration

$$\int_0^1 \int_0^{\sqrt{1-x^2}} \frac{e^y}{(e^y+1)\sqrt{1-x^2-y^2}}\, dy\, dx = \int_0^1 \int_0^{\sqrt{1-x^2}} \frac{e^y\, dy\, dx}{(e^y+1)\sqrt{1-x^2-y^2}}$$

$$= \int_0^1 \frac{e^y}{e^y+1}\,dy \int_0^{\sqrt{1-y^2}} \frac{dx}{\sqrt{1-x^2-y^2}}$$

$$= \int_0^1 \frac{e^y}{e^y+1}\,dy \int_0^{\sqrt{1-y^2}} \frac{dx}{\sqrt{1-y^2-x^2}}$$

$$= \int_0^1 \frac{e^y}{e^y+1}\,dy \int_0^{\sqrt{1-y^2}} \frac{dx}{\sqrt{\left(\sqrt{1-y^2}\right)^2-x^2}}$$

$$= \int_0^1 \frac{e^y}{e^y+1}\,dy \left[\sin^{-1}\left(\frac{x}{\sqrt{1-y^2}}\right)\right]_0^{\sqrt{1-y^2}}$$

$$= \int_0^1 \frac{e^y}{e^y+1}\,dy \left(\sin^{-1}1 - \sin^{-1}0\right)$$

$$= \int_0^1 \frac{e^y}{e^y+1}\,dy \left(\frac{\pi}{2} - 0\right)$$

$$= \frac{\pi}{2}\int_0^1 \frac{e^y}{e^y+1}\,dy$$

Put $e^y+1 = t$

$e^y\,dy = dt$

$$= \frac{\pi}{2}\int_0^1 \frac{dt}{t}$$

$$= \frac{\pi}{2}\left(\log t\right)_0^1$$

$$= \frac{\pi}{2}\left[\log(e^y+1)\right]_0^1$$

$$= \frac{\pi}{2}\left[\log(e+1) - \log 2\right]$$

$$= \frac{\pi}{2}\log\left(\frac{e+1}{2}\right)$$

$$= \frac{\pi}{2}\log\left(\frac{e+1}{2}\right)$$

Evaluate the integrals by changing the order of integration.

Practice Q.3

$$\int_0^3 \int_0^{6/x} x^2 \, dy \, dx$$

Ans. 27

Practice Q.4

Evaluate the integrals by changing the order of integration.

$$\int_0^1 \int_x^1 \sin y^2 \, dydx$$

Ans. ½(1-Cos1)

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AKTU : Engineering Mathematics-1

Multiple Integration : Triple Integral

Lec-5



By Gulshan sir

Today's Target

- ✓ Triple Integral
- ✓ AKTU Questions
- ✓ Practice Questions

Triple Integral

If f(x,y,z) be a continuous function at every point of finite region V of three dimensional space. Then the triple integral is

$$\iiint_V f(x,y,z) = \int_{x=a}^{b} \int_{y=\phi_1(x)}^{\phi_2(x)} \int_{z=f_1(x,y)}^{f_2(x,y)} f(x,y,z)\, dz\, dy\, dx$$

Where V is the region bounded by the curves

$a \leq x \leq b$

$\emptyset_1(x) \leq y \leq \emptyset_2(x)$

$f_1(x,y) \leq z \leq f_2(x,y)$

Note:

(i) Limits involving two variables kept innermost.

(ii) Limits involving one variables kept in middle.

(iii) Limits involving constants kept outermost.

Note: If all the limits are constant then the order of integration is not important.

Note: Start integration from innermost integral.

Q.1 *Evaluate the following inegral* $\int_0^1 \int_0^1 \int_0^1 e^{x+y+z}\,dx\,dy\,dz$

$$\int_0^1 \int_0^1 \int_0^1 e^{x+y+z}\,dx\,dy\,dz$$

$$\int_0^1 \int_0^1 \int_0^1 e^{x}\left(e^{y+z}\right)dx\,dy\,dz$$

$$= \int_0^1 \int_0^1 e^{y+z}\left(e^{x}\right)_0^1 dy\,dz$$

$$= \int_0^1 \int_0^1 e^{y+z}\left(e^{1}-e^{0}\right)dy\,dz$$

$$= (e-1)\int_0^1 \int_0^1 e^{y}\times e^{z}\,dy\,dz$$

$$= (e-1)\int_0^1 e^{z}\left(e^{y}\right)_0^1 dz$$

$$= (e-1)\int_0^1 e^{z}\left(e^{1}-e^{0}\right)dz$$

$$= (e-1)(e-1)\int_0^1 e^{z}\,dz$$

$$= (e-1)^2\left(e^{z}\right)_0^1$$

$$= (e-1)^2\left(e^{1}-e^{0}\right) = (e-1)^2(e-1)$$

$$\int_0^1 \int_0^1 \int_0^1 e^{x+y+z}\,dx\,dy\,dz = (e-1)^3$$

Q.2 Evaluate $\iiint_R (x+y+z)\,dx\,dy\,dz$ where $R: 0 \le x \le 1, 1 \le y \le 2, 2 \le z \le 3$.

Sem.Exam

$$\iiint_R (x+y+z)\,dx\,dy\,dz$$

$$= \int_0^1 \int_1^2 \int_2^3 (x+y+z)\,dz\,dy\,dx$$

$$= \int_0^1 \int_1^2 \left(xz + yz + \frac{z^2}{2}\right)_2^3 dy\,dx$$

$$= \int_0^1 \int_1^2 \left(3x + 3y + \frac{9}{2} - 2x - 2y - 2\right) dy\,dx$$

$$= \int_0^1 \int_1^2 \left(x + y + \frac{5}{2}\right) dy\,dx$$

$$= \int_0^1 \left(xy + \frac{y^2}{2} + \frac{5}{2}y\right)_1^2 dx$$

$$= \int_0^1 \left(2x + 2 + 5 - x - \frac{1}{2} - \frac{5}{2}\right) dx$$

$$= \int_0^1 (x+4)\,dx$$

$$= \left(\frac{x^2}{2} + 4x\right)_0^1$$

$$= \frac{1}{2} + 4 - 0 - 0$$

$$= \frac{1+8}{2}$$

$$= \frac{9}{2}$$

$$\iiint_R (x+y+z)\,dx\,dy\,dz = \frac{9}{2}$$

Q.3 *Find the value of* $\int_0^1 \int_0^x \int_0^{x+y} dx\,dy\,dz$. **Sem.Exam**

$$\int_0^1 \int_0^x \int_0^{x+y} dx\,dy\,dz$$

$x \quad y \quad z$

$$= \int_0^1 \int_0^x \left[\int_0^{x+y} dz\right] dy\,dx$$

$$= \int_0^1 \int_0^x (z)_0^{x+y}\, dy\,dx$$

$$= \int_0^1 \int_0^x (x+y-0)\, dy\,dx$$

$$= \int_0^1 \left[\int_0^x (x+y)\, dy\right] dx$$

$$= \int_0^1 \left(xy + \frac{y^2}{2}\right)_0^x dx$$

$$= \int_0^1 \left(x^2 + \frac{x^2}{2} - 0 - 0\right) dx$$

$$= \int_0^1 \frac{3x^2}{2}\, dx$$

$$= \frac{3}{2}\int x^2\, dx$$

$$= \frac{3}{2}\left(\frac{x^3}{3}\right)_0^1 = \frac{3}{2} \times \frac{1}{3} = \frac{1}{2}$$

$$\int_0^1 \int_0^{x^2} \int_0^{x+y} dx\,dy\,dz = \frac{1}{2}$$

Q.4

Evaluate $\iiint_R (x-2y+z)\,dz\,dy\,dx$ where R is the region determined by $0 \le x \le 1, 0 \le y \le x^2$.

$$\iiint_R (x-2y+z)\,dz\,dy\,dx$$

$$= \int_0^1 \int_0^{x^2} \int_0^{x+y} (x-2y+z)\,dz\,dy\,dx \qquad 0 \le z \le x+y$$

$$= \int_0^1 \int_0^{x^2} \left[x(x+y) - 2y(x+y) + \frac{1}{2}(x+y)^2 \right] dy\,dx$$

$$= \int_0^1 \int_0^{x^2} \left(x^2 + xy - 2xy - 2y^2 + \frac{x^2}{2} + \frac{y^2}{2} + xy \right) dy\,dx$$

$$= \int_0^1 \int_0^{x^2} \left(\frac{3}{2}x^2 - \frac{3}{2}y^2 \right) dy\,dx$$

$$= \frac{3}{2} \int_0^1 \int_0^{x^2} (x^2 - y^2)\,dy\,dx = \frac{3}{2} \int_0^1 \left(x^2 y - \frac{y^3}{3} \right)_0^{x^2} dx$$

$$= \frac{3}{2} \int_0^1 \left(x^2 \cdot x^2 - \frac{x^6}{3} \right) dx$$

$$= \frac{3}{2} \int_0^1 \left(x^4 - \frac{x^6}{3} \right) dx$$

$$= \frac{3}{2} \left(\frac{x^5}{5} - \frac{x^7}{21} \right)_0^1$$

$$= \frac{3}{2} \left(\frac{1}{5} - \frac{1}{21} \right)$$

$$= \frac{3}{2} \left(\frac{21-5}{105} \right)$$

$$= \frac{3}{2} \cdot \frac{16}{105}$$

$$= \frac{8}{35}$$

$$= \frac{3}{2}\int_0^1 \left(x^2y - \frac{y^3}{3}\right)_0^{x^2} dx$$

$$= \frac{3}{2}\int_0^1 \left(x^4 - \frac{x^6}{3}\right) dx$$

$$= \frac{3}{2}\left(\frac{x^5}{5} - \frac{x^7}{3\times 7}\right)_0^1$$

$$= \frac{3}{2}\left[\frac{1}{5} - \frac{1}{21}\right]$$

$$= \frac{3}{2}\left(\frac{21-5}{105}\right)$$

$$= \frac{3}{2} \times \frac{16}{105}^{8}_{35}$$

$$= \frac{8}{35}$$

$$\int\int\int (x-2y+z)\,dz\,dy\,dx = \frac{8}{35}$$

Q.5 Evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \frac{1}{\sqrt{1-x^2-y^2-z^2}} dz\,dy\,dx$

$$\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \frac{1}{\sqrt{1-x^2-y^2-z^2}} dz\,dy\,dx$$

x y z

$$= \int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \frac{1}{\sqrt{\left(\sqrt{1-x^2-y^2}\right)^2 - z^2}} dz\,dy\,dx$$

$$= \int_0^1 \int_0^{\sqrt{1-x^2}} \left(\sin^{-1} \frac{z}{\sqrt{1-x^2-y^2}} \right)_0^{\sqrt{1-x^2-y^2}} dy\,dx$$

$$= \int_0^1 \int_0^{\sqrt{1-x^2}} \left(\sin^{-1} \frac{\sqrt{1-x^2-y^2}}{\sqrt{1-x^2-y^2}} - \sin^{-1} 0 \right) dy\,dx$$

$$= \int_0^1 \int_0^{\sqrt{1-x^2}} \left(\frac{\pi}{2} - 0 \right) dy\,dx$$

$$= \frac{\pi}{2} \int_0^1 \int_0^{\sqrt{1-x^2}} dy\,dx$$

$$= \frac{\pi}{2} \int_0^1 (y)_0^{\sqrt{1-x^2}} dx = \frac{\pi}{2} \int_0^1 \sqrt{1-x^2}\,dx$$

$$= \frac{\pi}{2}\int_0^1 \sqrt{1-x^2}\,dx$$

$$= \frac{\pi}{2}\left[\frac{x}{2}\sqrt{1-x^2} + \frac{1^2}{2}\sin^{-1}\frac{x}{1}\right]_0^1$$

$$= \frac{\pi}{2}\left[0 + \frac{1}{2}\sin^{-1}(1) - 0 - 0\right]$$

$$= \frac{\pi}{2}\left(\frac{1}{2}\times\frac{\pi}{2}\right) = \frac{\pi^2}{8}$$

$$= \frac{\pi^2}{8} \quad \underline{Ans}$$

Q.6 *Evaluate the following inegral* $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} xyz\, dz\, dy\, dx$

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$$\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} xyz\, dz\, dy\, dx$$

$$= \int_0^1 \int_0^{\sqrt{1-x^2}} \left(xy\frac{z^2}{2} \right)_0^{\sqrt{1-x^2-y^2}} dy\, dx$$

$$= \frac{1}{2} \int_0^1 \int_0^{\sqrt{1-x^2}} xy(1-x^2-y^2)\, dy\, dx$$

$$= \frac{1}{2} \int_0^1 \int_0^{\sqrt{1-x^2}} (xy - x^3y - xy^3)\, dy\, dx$$

$$= \frac{1}{2} \int_0^1 \left(\frac{xy^2}{2} - \frac{x^3y^2}{2} - \frac{xy^4}{4} \right)_0^{\sqrt{1-x^2}} dx$$

$$= \frac{1}{2} \int_0^1 \left[\frac{x(1-x^2)}{2} - \frac{x^3(1-x^2)}{2} - \frac{x(1-x^2)^2}{4} \right] dx$$

$$= \frac{1}{8} \int_0^1 \left[2x(1-x^2) - 2x^3(1-x^2) - x(1+x^4-2x^2) \right] dx$$

$$= \frac{1}{8}\int_0^1 \left(x - 2x^3 + x^5\right) dx$$

$$= \frac{1}{8}\left(\frac{x^2}{2} - \frac{x^4}{4_2} + \frac{x^6}{6}\right)_0^1$$

$$= \frac{1}{8}\left(\frac{1}{2} - \frac{1}{2} + \frac{1}{6}\right)$$

$$= \frac{1}{8} \times \frac{1}{6} = \frac{1}{48}$$

$$= \frac{1}{48} \quad \underline{\text{Ans}}$$

Practice Q.1

Evaluate the following inegral $\int_{-1}^{1}\int_{0}^{z}\int_{x-z}^{x+z}(x+y+z)\,dydxdz$

Ans. 0

Practice Q.2

Evaluate the following inegral $\int_0^a \int_0^{a-x} \int_0^{a-x-y} (x+y+z)\, dzdydx$

Ans. $a^4/8$

Practice Q.3 Evaluate the following inegral $\int_0^{log2}\int_0^{x}\int_0^{x+logy} e^{x+y+z}\,dzdydx$

Ans. $\frac{8}{3}log2-\frac{19}{9}$

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AKTU : Engineering Mathematics-1

Multiple Integration : Double Integral in Polar Coordinates

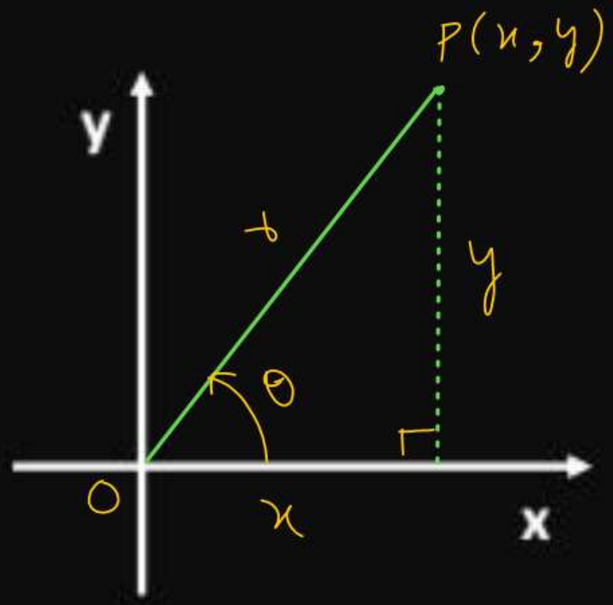
Lec-6

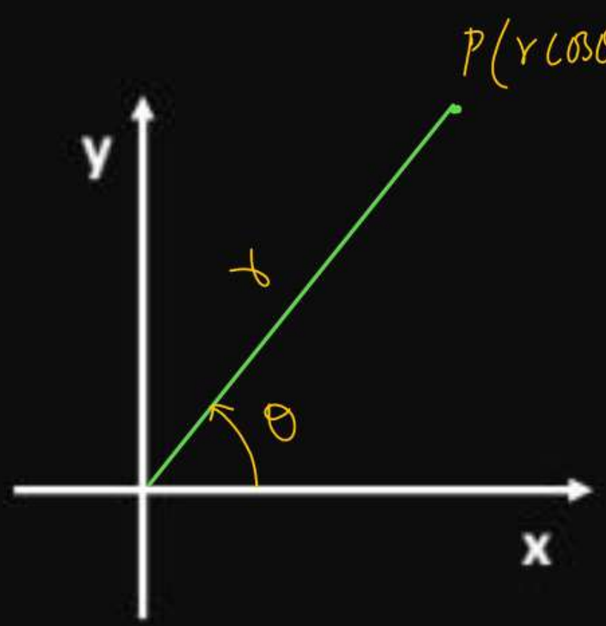


Today's Target

- ✓ Polar Coordinates
- ✓ AKTU Questions
- ✓ Practice Questions

Difference between Cartesian and Polar coordinates

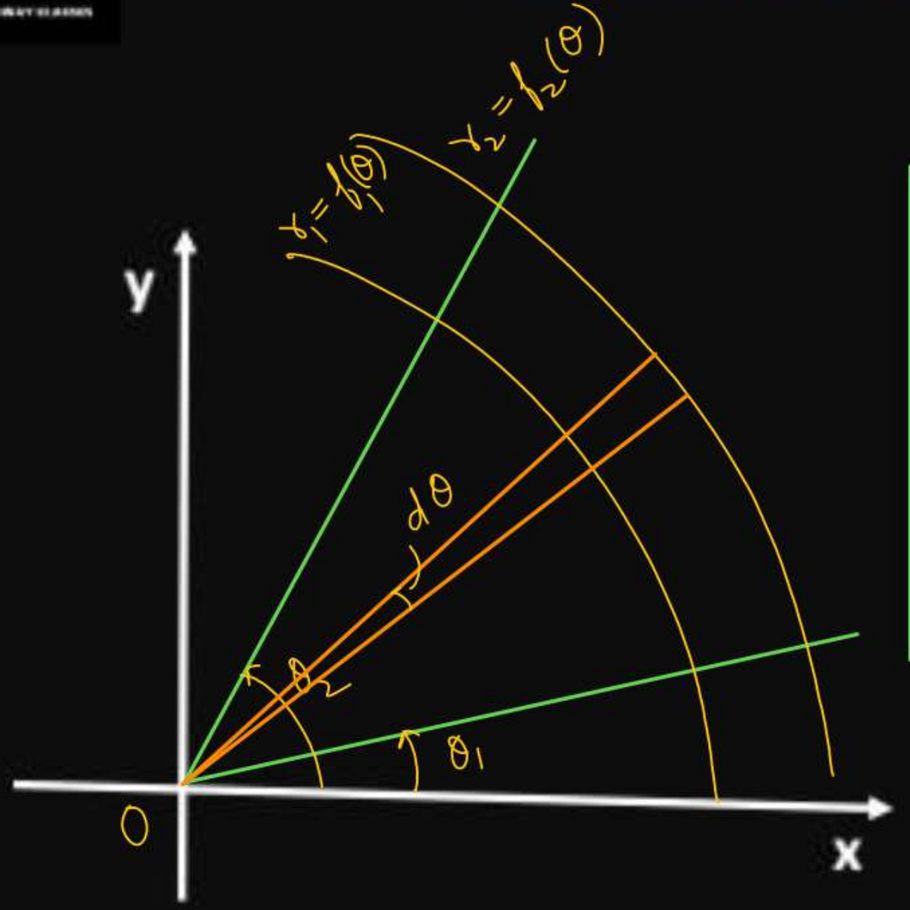




$$x = r\cos\theta$$

$$y = r\sin\theta$$

Double Integral in Polar Form



$$I = \int_{\theta_1}^{\theta_2} \int_{r_1}^{r_2} f(r,\theta)\, dr\, d\theta$$

Where r_1 and r_2 are the function of θ

Q.1 ***Evaluate the following integral***

$$\int_0^{\pi/2}\int_0^{a\cos\theta} r\sin\theta \, dr \, d\theta.$$

$$\int_{\theta=0}^{\pi/2}\int_{r=0}^{a\cos\theta} r\sin\theta \, dr \, d\theta$$

$$= \int_0^{\pi/2} \sin\theta \, d\theta \int_0^{a\cos\theta} r \, dr$$

$$= \int_0^{\pi/2} \sin\theta \, d\theta \left(\frac{r^2}{2}\right)_0^{a\cos\theta}$$

$$= \frac{1}{2}\int_0^{\pi/2} \sin\theta \, d\theta \left(r^2\right)_0^{a\cos\theta}$$

$$= \frac{1}{2}\int_0^{\pi/2} \sin\theta \, d\theta \left\{(a\cos\theta)^2 - 0\right\}$$

$$= \frac{a^2}{2}\int_0^{\pi/2} \sin\theta \cos^2\theta \, d\theta$$

Put $\cos\theta = t$

$-\sin\theta \, d\theta = dt$

$\sin\theta \, d\theta = -dt$

When $\theta = 0$, $t = 1$

When $\theta = \frac{\pi}{2}$, $t = 0$

$$= -\frac{a^2}{2}\int_1^0 t^2 \, dt$$

$$= -\left(-\frac{a^2}{2}\right)\int_0^1 t^2 \, dt$$

$$= \frac{a^2}{2}\left(\frac{t^3}{3}\right)_0^1 = \frac{a^2}{2}\left(\frac{1}{3} - 0\right)$$

$$\boxed{= \frac{a^2}{6}}$$

Q.2 ***Evaluate the following integrals***

$$\int_0^{\pi}\int_0^{a(1-\cos\theta)} r^2\sin\theta\, dr\, d\theta.$$

$$\int_{\theta=0}^{\pi}\int_{r=0}^{a(1-\cos\theta)} r^2\sin\theta\, dr\, d\theta$$

$$= \int_0^{\pi}\sin\theta\, d\theta \int_0^{a(1-\cos\theta)} r^2\, dr$$

$$= \int_0^{\pi}\sin\theta\, d\theta \left(\frac{r^3}{3}\right)_0^{a(1-\cos\theta)}$$

$$= \frac{1}{3}\int_0^{\pi}\sin\theta\, d\theta\left[a^3(1-\cos\theta)^3\right]$$

$$= \frac{a^3}{3}\int_0^{\pi}\sin\theta(1-\cos\theta)^3\, d\theta$$

Put $(1-\cos\theta)=t$

$-(-\sin\theta)d\theta = dt$

$\sin\theta\, d\theta = dt$

When $\theta=0^\circ$, $t=0$

When $\theta=\pi$, $t=2$

$$= \frac{a^3}{3}\int_0^2 t^3\, dt$$

$$= \frac{a^3}{3}\left(\frac{t^4}{4}\right)_0^2$$

$$= \frac{a^3}{3}\times\frac{2^4}{4} = \frac{a^3}{3}\times\frac{16}{4} = \frac{4a^3}{3}$$

$$\int_0^{\pi}\int_0^{a(1-\cos\theta)} r^2\sin\theta\, dr\, d\theta = \frac{4a^3}{3}$$

Q.3 **Evaluate $\int\int r\sin\theta\, dr\, d\theta$ over the area of the cardiod $r = a(1+\cos\theta)$ above the initial line.**

$$\int\int r\sin\theta\, dr\, d\theta$$

$$= \int_{\theta=0}^{\pi}\int_{r=0}^{a(1+\cos\theta)} r\sin\theta\, dr\, d\theta$$

$$= \int_0^{\pi}\sin\theta\, d\theta \left(\frac{r^2}{2}\right)_0^{a(1+\cos\theta)}$$

$$= \frac{1}{2}\int_0^{\pi}\sin\theta\, d\theta \left\{a^2(1+\cos\theta)^2\right\}$$

$$= \frac{a^2}{2}\int_0^{\pi}\sin\theta(1+\cos\theta)^2 d\theta$$

Put $1+\cos\theta = t$

$$\sin\theta\, d\theta = -dt$$

When $\theta = 0$, $t = 2$

When $\theta = \pi$

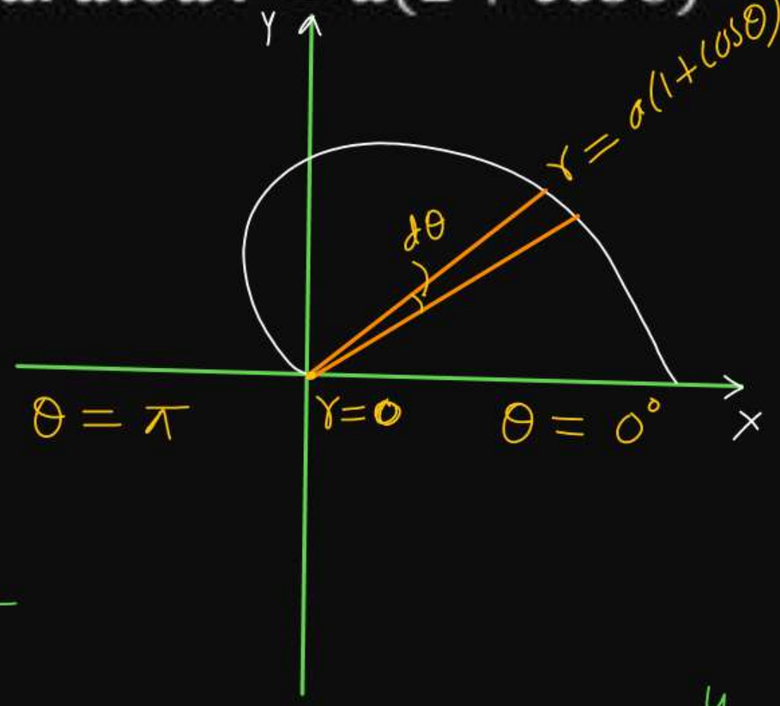
$t = 0$

$$= -\frac{a^2}{2}\int_2^0 t^2\, dt$$

$$= -\frac{a^2}{2}\left(\frac{t^3}{3}\right)_2^0 = -\frac{a^2}{2}\left[0 - \frac{8}{3}\right] = \frac{a^2}{2}\times\frac{8}{3}$$

$$= \frac{4}{3}a^2$$

$$= \int_0^{\pi}\sin\theta\, d\theta \int_0^{a(1+\cos\theta)} r\, dr$$



Practice Q

Evaluate the following integrals

$$\int_0^{\pi} \int_0^{a\sin\theta} r\,dr\,d\theta.$$

$$Ans. \frac{\pi a^2}{4}$$

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AKTU : Engineering Mathematics-1

Multiple Integration : Change of Variables

Lec-7



By Gulshan sir

Today's Target

- ✓ Double Integral using Change of Variables
- ✓ AKTU Questions
- ✓ Practice Questions

Change of Variables

Some times the evaluation of double or triple integral become easy after suitable change of variables.

Double Integral using Change of Variables

If variable x and y in the double integral are change to u and v having relation

$$x = \emptyset_1(u, v), y = \emptyset_2(u, v)$$

Then

$$\int\int_R f(x, y)\, dx\, dy = \int\int_{R'} f[\phi(u, v), \phi_2(u, v)]\, |J|\, du\, dv$$

$$J = \frac{\partial(x, y)}{\partial(u, v)}$$

Q.1 ***Evaluate by changing the variables,*** $\iint_R (x+y)^2 dxdy$ ***where***

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R *is the region bounded by the parallelogram* $x+y=0, x+y=2,$ $3x-2y=0$ ***and*** $3x-2y=3$

$\iint_R (x+y)^2 dx\,dy$

R is the region bounded by

$x+y=0$

$x+y=2$

$3x-2y=0$

$3x-2y=3$

Let $x+y=u$, $3x-2y=v$

$$J' = \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1 \\ 3 & -2 \end{vmatrix} = -2-3$$

$J' = -5$

$$J = \frac{\partial(x,y)}{\partial(u,v)} = \frac{1}{J'}$$

$J = -\frac{1}{5}$

$|J| = \frac{1}{5}$

Limits of integration

$u=0$ to $u=2$

$v=0$ to $v=3$

$dx\,dy = |J|\,du\,dv$

$$\iint_R (x+y)^2\,dx\,dy = \int_0^3 \int_0^2 u^2 |J|\,du\,dv$$

$$= \int_0^3 \int_0^2 u^2 \times \frac{1}{5}\,du\,dv$$

$$= \frac{1}{5}\int_0^3 \left(\frac{u^3}{3}\right)_0^2 dv$$

$$= \frac{1}{5}\int_0^3 \frac{8}{3}\,dv$$

$$= \frac{8}{15}(v)_0^3$$

$$= \frac{8}{15} \times 3$$

$$= \frac{8}{5}$$

Q.2 *Evaluate* $\iint_D (y-x)\,dx\,dy$ *where D is the region in xy − plane bounded by the straight lines* $y = x+1, y = x-3, y = -\frac{1}{3}x + \frac{7}{3}, \quad y = -\frac{1}{3}x + 5$

$$\iint_D (y-x)\,dx\,dy$$

D is the region bounded by

$y = x+1 \implies y - x = 1$

$y = x-3 \implies y - x = -3$

$y = -\frac{1}{3}x + \frac{7}{3} \implies y + \frac{1}{3}x = \frac{7}{3}$

$y = -\frac{1}{3}x + 5 \implies y + \frac{1}{3}x = 5$

Let $y - x = u$ and $y + \frac{1}{3}x = v$

$$J' = \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

$$= \begin{vmatrix} -1 & 1 \\ \frac{1}{3} & 1 \end{vmatrix} = -1 - \frac{1}{3}$$

$$= -\frac{4}{3}$$

$$J = \frac{\partial(x,y)}{\partial(u,v)} = \frac{1}{J'}$$

$J = -\frac{3}{4}$

$|J| = \frac{3}{4}$

Limits of integration

$u = -3$ to $u = 1$

$v = \frac{7}{3}$ to $v = 5$

$dx\,dy = |J|\,du\,dv$

$$\iint_D (y-x)\,dx\,dy = \int_{7/3}^{5}\int_{-3}^{1} u\,|J|\,du\,dv$$

$$= \int_{7/3}^{5}\int_{-3}^{1} u \times \frac{3}{4}\,du\,dv$$

$$= \frac{3}{4}\int_{7/3}^{5}\left(\frac{u^2}{2}\right)_{-3}^{1} dv$$

$$= \frac{3}{4\times 2}\int_{7/3}^{5}(1-9)\,dv$$

$$= \frac{3}{8}\times(-8)\int_{7/3}^{5} dv$$

$$= -3\times(v)_{7/3}^{5}$$

$$= -3\times\left[5-\frac{7}{3}\right]$$

$$= -3\times\left(\frac{15-7}{3}\right)$$

$$= -8\times\frac{8}{8}$$

$$= -8$$

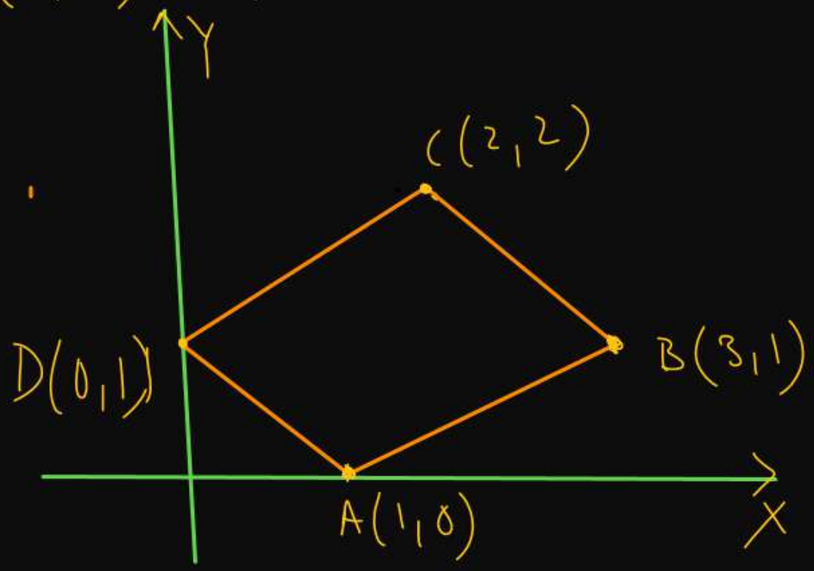
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Q.3 **Evaluate $\iint_R (x+y)^2 dxdy$ where R is the parallelogram in the $xy-$plane with vertices at $(1,0),(3,1),(2,2),(0,1)$ using the transformation $u=x+y$ and $v=x-2y$**

$$\iint_R (x+y)^2 dx\,dy$$

R is a ||gm ABCD where

A(1, 0) B(3, 1) C(2, 2) D(0, 1)



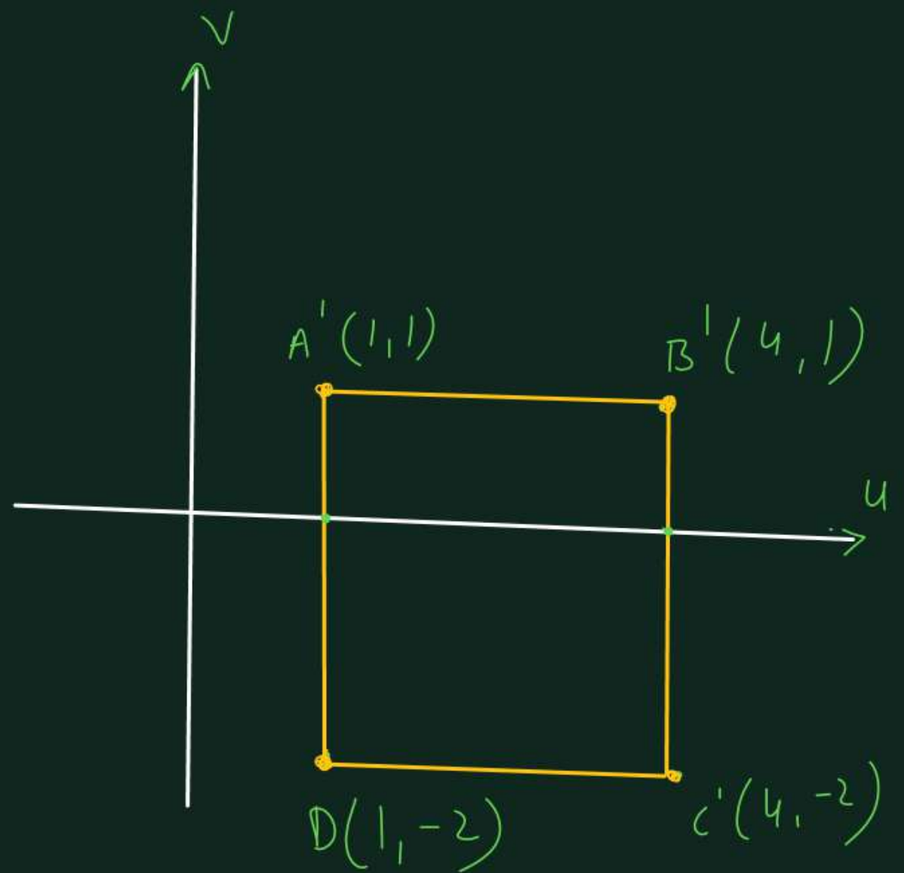
New Region of integration

$u = x+y$

$v = x-2y$

New vertices in uv plane (u, v)

XY Plane	A(1,0)	B(3,1)	C(2,2)	D(0,1)	
u-v plane	A'(1,1)	B'(4,1)	C'(4,-2)	D'(1,-2)	



$$dx\,dy = |J|\,du\,dv$$

$$J' = \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1 \\ 1 & -2 \end{vmatrix} = -2 - 1 = -3$$

$$J = \frac{\partial(x,y)}{\partial(u,v)} = \frac{1}{J'} = \frac{1}{-3}$$

$$|J| = \frac{1}{3}$$

$$\iint_R (x+y)^2\,dx\,dy = \int_{-2}^{1}\int_{1}^{4} u^2\,|J|\,du\,dv$$

$$= \int_{-2}^{1}\int_{1}^{4} u^2 \times \frac{1}{3}\,du\,dv$$

$$= \frac{1}{3}\int_{-2}^{1}\left(\frac{u^3}{3}\right)_1^4 dv$$

$$= \frac{1}{3}\int_{-2}^{1}\frac{1}{3}\left(64-1\right)dv$$

$$= \frac{7}{\cancel{63}}\int_{-2}^{1} dv = 7\,(v)_{-2}^{1}$$

$$= 7(1+2)$$

$$= 7 \times 3$$

$$= 21$$

Practice Q.

Evaluate $\iint_R (x-y)^4 e^{x+y} dxdy$ **where R is the sqare with vertices at** $(1,0), (2,1), (1,2)$ **and** $(0,1)$

Ans. $\frac{e(e^1-1)}{5}$

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AKTU : Engineering Mathematics-1

Multiple Integration : Change of Variables

Lec-8

Today's Target

- ✓ Double Integral using Change of Variables(Part-2)
- ✓ AKTU Questions
- ✓ Practice Questions



By Gulshan sir

Double Integral using Change of Variables

If variable x and y in the double integral are change to u and v having relation

$$x = \emptyset_1(u, v), y = \emptyset_2(u, v)$$

Then

$$\int\int_R f(x,y)\,dx\,dy = \int\int_{R'} f[\phi_1(u,v), \phi_2(u,v)]\,|J|\,du\,dv$$

To change cartesian coordinates (x,y) to polar coordinates (r,Θ)

Working Rules :

Step-1 : Draw the region for the given limits of integration

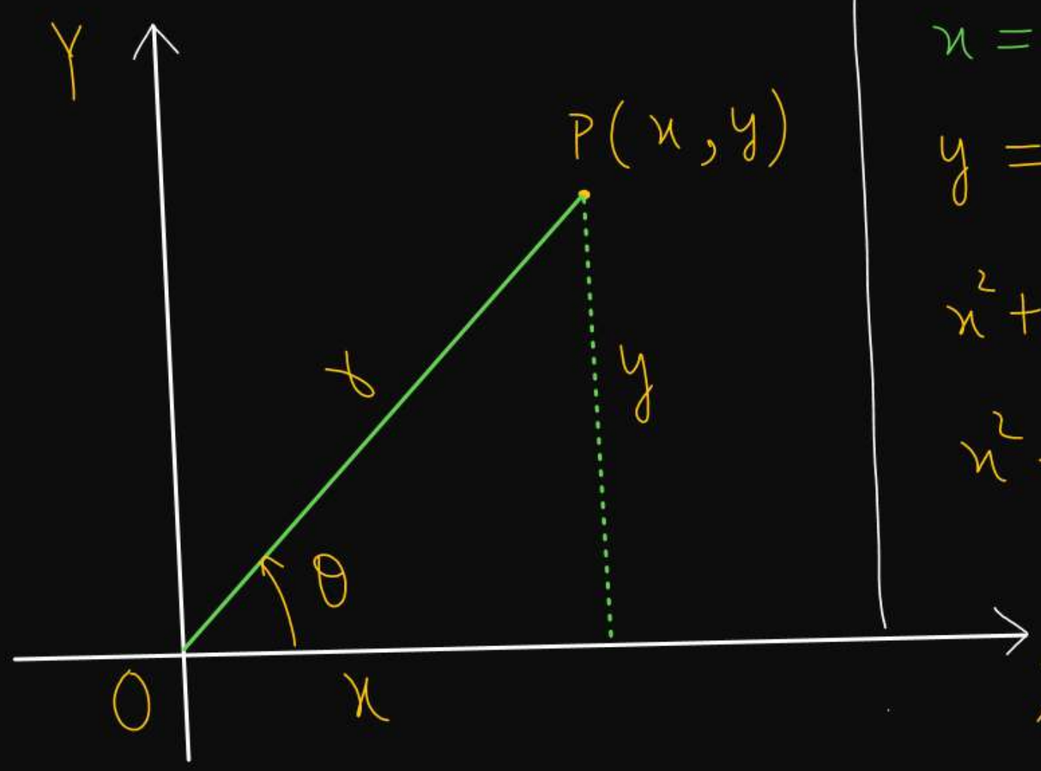
Step-2 : Put $x = r\cos\Theta$, $y = r\sin\Theta$, $dx\,dy = rdr\,d\Theta$ in the given integration

Step-3 : Find the limits of r and Θ from the region

Step-4 : Now solve the polar form

To change cartesian coordinates (x,y) to polar coordinates (r,Θ)

$$\iint_R f(x,y)\,dx\,dy = \iint_R (r\cos\theta, r\sin\theta)\, r\,dr\,d\theta$$



$$x = r\cos\theta \quad -①$$

$$y = r\sin\theta \quad -②$$

$$x^2+y^2 = (r\cos\theta)^2+(r\sin\theta)^2$$

$$x^2+y^2 = r^2(\sin^2\theta+\cos^2\theta)$$

$$\boxed{x^2+y^2 = r^2}$$

$$J = \frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix}$$

$$= \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix}$$

$$= r\cos^2\theta + r\sin^2\theta$$

$$= r(\sin^2\theta+\cos^2\theta)$$

$$\boxed{J = r}$$

Q.1 ***Change into polar coordinates and evaluate*** $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)}\,dy\,dx.$

Hence show that $\int_0^\infty e^{-x^2}\,dx = \frac{\sqrt{\pi}}{2}$

$$\int_0^\infty \int_0^\infty e^{-(x^2+y^2)}\,dy\,dx$$

Given limits of integration

$y = 0$ to $y = \infty$

$x = 0$ to $x = \infty$

To change into polar coordinates

Put $x = r\cos\theta$

$y = r\sin\theta$

$x^2 + y^2 = r^2$

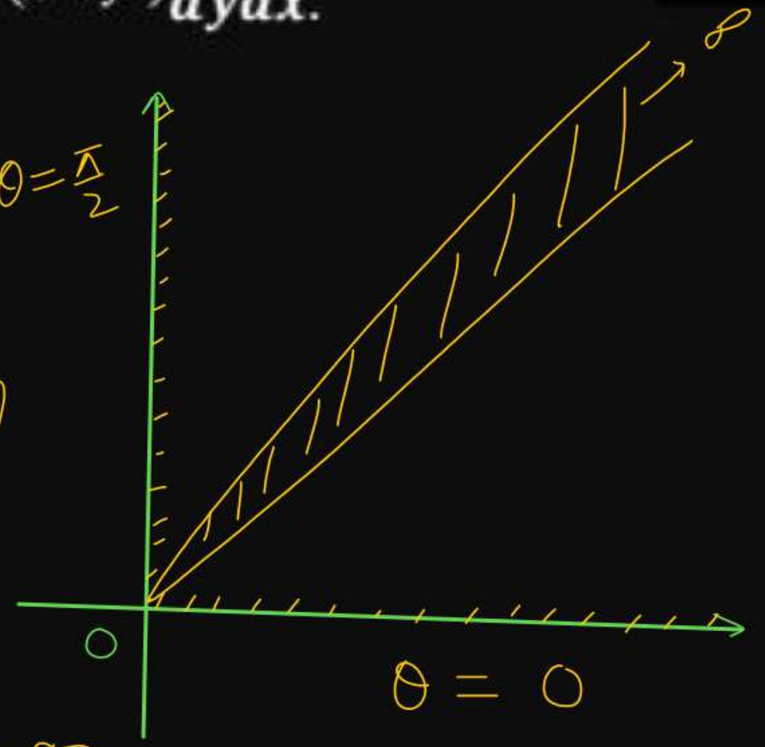
$dx\,dy = r\,dr\,d\theta$

New limits of integration

$\theta = 0$ to $\theta = \frac{\pi}{2}$

$r = 0$ to $r = \infty$

$$\int_0^\infty \int_0^\infty e^{-(x^2+y^2)}\,dy\,dx = \int_0^{\pi/2} \int_0^\infty e^{-r^2}\, r\,dr\,d\theta$$



$$= \int_0^{\pi/2} \int_0^{\infty} e^{-r^2} r\,dr\,d\theta$$

Put $r^2 = t$

$2r\,dr = dt$

$r\,dr = \frac{dt}{2}$

$$= \int_0^{\pi/2} \int_0^{\infty} e^{-t} \frac{dt}{2} d\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} -\left(e^{-t}\right)_0^{\infty} d\theta$$

$$= \frac{-1}{2} \int_0^{\pi/2} (e^{-\infty} - e^{0})\, d\theta$$

$$= \frac{-1}{2} \int_0^{\pi/2} (0-1)\, d\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} d\theta$$

$$= \frac{1}{2} (\theta)_0^{\pi/2}$$

$$= \frac{1}{2} \left(\frac{\pi}{2} - 0\right)$$

$$= \frac{\pi}{4}$$

$$\boxed{\int_0^{\infty} \int_0^{\infty} e^{-(x^2+y^2)}\, dy\, dx = \frac{\pi}{4}}$$

Let $I = \int_0^{\infty} e^{-x^2} dx$ — ①

$$I = \int_0^{\infty} e^{-y^2} dy \quad \text{— ②}$$

Multiply ① and ②

$$I \times I = \int_0^{\infty} \int_0^{\infty} e^{-x^2} \times e^{-y^2}\, dx\, dy$$

$$I^2 = \int_0^{\infty} \int_0^{\infty} e^{-(x^2+y^2)}\, dy\, dx$$

$$I^2 = \frac{\pi}{4}$$

$$\boxed{I = \frac{\sqrt{\pi}}{2}}$$

Q.2 **Evaluate $\int_0^{2a}\int_0^{\sqrt{2ax-x^2}}(x^2+y^2)\,dx\,dy$ by changing into polar coordinates.**

$$\int_0^{2a}\int_0^{\sqrt{2ax-x^2}}(x^2+y^2)\,dx\,dy$$

Given limits of integration

$x=0$ to $x=2a$

$y=0$ to $y=\sqrt{2ax-x^2}$

$$y^2=2ax-x^2$$

$$\boxed{x^2+y^2=2ax}$$

$$x^2-2ax+y^2=0$$

$$\underline{x^2-2ax+a^2}-a^2+y^2=0$$

$$(x-a)^2+y^2=a^2$$

centre $(a,0)$

radius $=a$

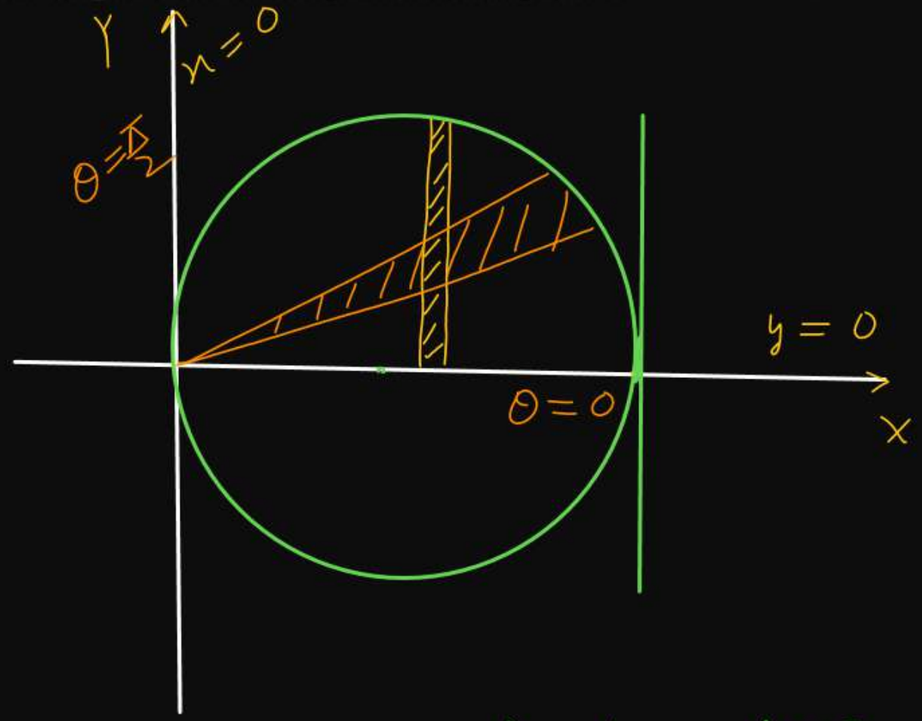
To change in polar coordinates

Put $x=r\cos\theta$

$y=r\sin\theta$

$x^2+y^2=r^2$

$dx\,dy=r\,dr\,d\theta$



New limits of integration

$\theta=0$ to $\theta=\frac{\pi}{2}$

$$x^2+y^2=2ax$$

$$(r\cos\theta)^2+(r\sin\theta)^2=2ar\cos\theta$$

$$r^2\cos^2\theta + r^2\sin^2\theta = 2ar\cos\theta$$

$$r^2(\sin^2\theta + \cos^2\theta) = 2ar\cos\theta$$

$$r^2 = 2ar\cos\theta$$

$$r = 2a\cos\theta$$

$r = 0$ to $r = 2a\cos\theta$

$$\int_0^{2a}\int_0^{\sqrt{2ax-x^2}} (x^2+y^2)\,dy\,dx = \int_0^{\pi/2}\int_0^{2a\cos\theta} r^2 \cdot r\,dr\,d\theta$$

$$= \int_0^{\pi/2}\int_0^{2a\cos\theta} r^3\,dr\,d\theta$$

$$= \int_0^{\pi/2} \left(\frac{r^4}{4}\right)_0^{2a\cos\theta} d\theta$$

$$= \frac{1}{4}\int_0^{\pi/2} \left[(2a\cos\theta)^4 - 0\right] d\theta$$

$$= \frac{1}{4}\int_0^{\pi/2} 16a^4\cos^4\theta\,d\theta$$

$$= \frac{16}{4}\left(\int_0^{\pi/2}\cos^4\theta\,d\theta\right) \times a^4$$

$$= 4 \times a^4 \times \frac{3\times 1}{4\times 2} \times \frac{\pi}{2}$$

$$= \frac{3}{4}\pi a^4$$

Q.3 **Evaluate** $\int_0^2 \int_0^{\sqrt{2x-x^2}} \frac{x\,dy\,dx}{\sqrt{x^2+y^2}}$ **by changing into polar coordinates.**

$$\int_0^2 \int_0^{\sqrt{2x-x^2}} \frac{x\,dy\,dx}{\sqrt{x^2+y^2}}$$

Limits of integration

$x = 0$ to $x = 2$

$y = 0$ to $y = \sqrt{2x-x^2}$

$$y^2 = 2x - x^2$$

$$\boxed{x^2+y^2 = 2x}$$

$$x^2 - 2x + y^2 = 0$$

$$\underline{x^2-2x+1} - 1 + y^2 = 0$$

$$(x-1)^2 + y^2 = 1$$

(centre $(1, 0)$

radius $= 1$

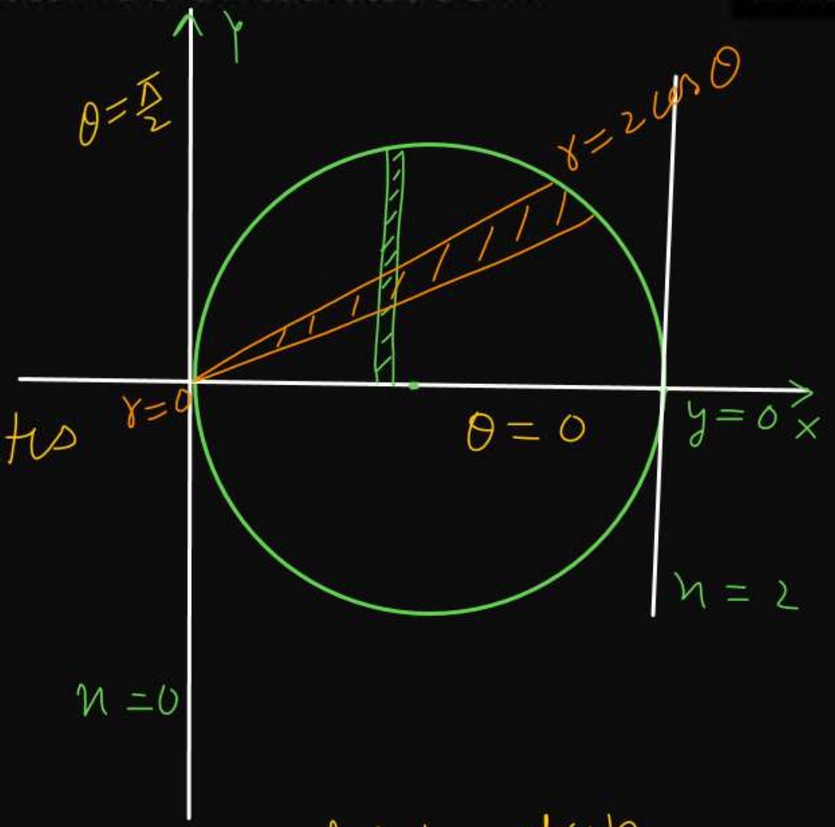
To change in polar coordinates

Put $x = r\cos\theta$

$y = r\sin\theta$

$x^2 + y^2 = r^2$

$dx\,dy = r\,dr\,d\theta$



New limits of integration

$\theta = 0$ to $\theta = \frac{\pi}{2}$

$r^2 = 2x$

$$(r\cos\theta)^2 + (r\sin\theta)^2 = 2r\cos\theta$$

$$r^2\cos^2\theta + r^2\sin^2\theta = 2r\cos\theta$$

$$r^2(\sin^2\theta + \cos^2\theta) = 2r\cos\theta$$

$$r = 2\cos\theta$$

$r = 0$ to $r = 2\cos\theta$

$$\int_0^2 \int_0^{\sqrt{2x-x^2}} \frac{x\,dy\,dx}{\sqrt{x^2+y^2}} = \int_0^{\pi/2} \int_0^{2\cos\theta} \frac{r\cos\theta \times r\,dr\,d\theta}{\sqrt{r^2}}$$

$$= \int_0^{\pi/2} \cos\theta \int_0^{2\cos\theta} r\,dr\,d\theta$$

$$= \int_0^{\pi/2} \cos\theta\,d\theta \left(\frac{r^2}{2}\right)_0^{2\cos\theta}$$

$$= \frac{1}{2}\int_0^{\pi/2} \cos\theta\,d\theta\,(4\cos^2\theta - 0)$$

$$= \frac{1}{2}\int_0^{\pi/2} 4\cos^3\theta\,d\theta$$

$$= \frac{4}{2}\int_0^{\pi/2} \cos^3\theta\,d\theta$$

$$= 2 \times \frac{2}{3} = \frac{4}{3}$$

$$\int_0^2 \int_0^{\sqrt{2x-x^2}} \frac{x\,dy\,dx}{\sqrt{x^2+y^2}} = \frac{4}{3}$$

Q.4 Evaluate the integral $\iint_R \sqrt{x^2+y^2}\,dxdy$ by changing to polar coordinates where R is the region in the $xy-$plane bounded by the circles $x^2+y^2=4$ and $x^2+y^2=9$

$$\iint_R \sqrt{x^2+y^2}\,dx\,dy$$

$x^2+y^2=4$

$x^2+y^2=9$

To change into polar coordinates

Put $x=r\cos\theta$

$y=r\sin\theta$

$x^2+y^2=r^2$

$dx\,dy=r\,dr\,d\theta$

New limits of integration

$\theta=0$ to $\theta=2\pi$

$x^2+y^2=4$

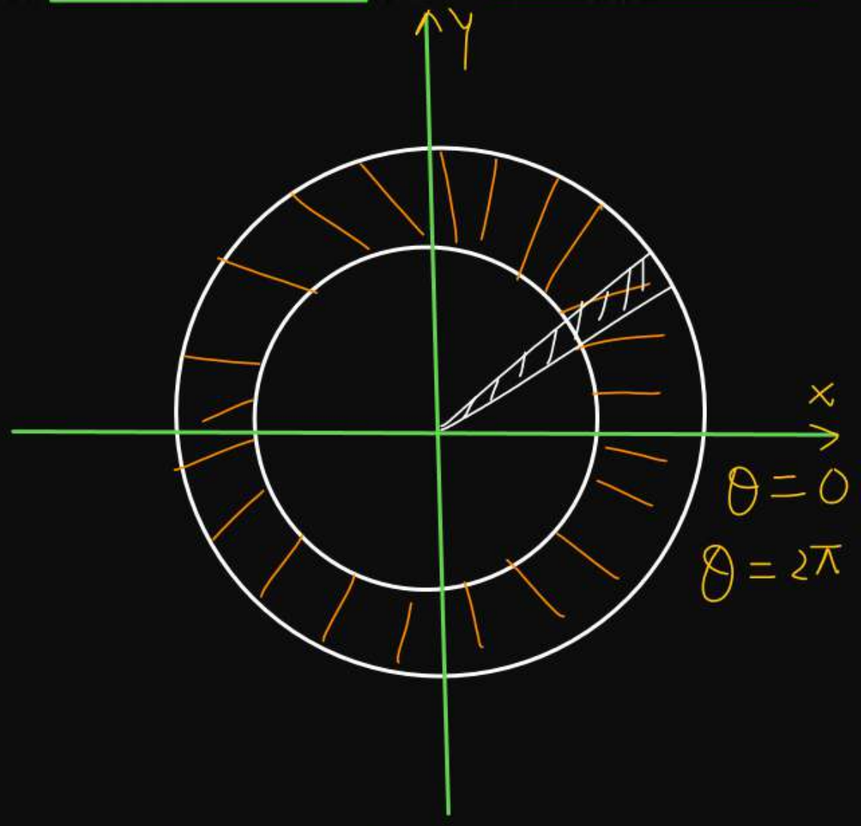
$r^2=4$

$r=2$

$x^2+y^2=9$

$r^2=9$

$r=3$



$r=2$ to $r=3$

$$\iint_R \sqrt{x^2+y^2}\,dx\,dy = \int_0^{2\pi}\int_2^3 \sqrt{r^2}\; r\,dr\,d\theta$$

$$= \int_0^{2\pi} d\theta \int_2^3 r^2\,dr$$

$$= \int_0^{2\pi} d\theta \left(\frac{r^3}{3}\right)_2^3$$

$$= \frac{1}{3}\int_0^{2\pi} d\theta \left(3^3 - 2^3\right)$$

$$= \frac{1}{3}\int_0^{2\pi} (27-8)\,d\theta$$

$$= \frac{19}{3}\int_0^{2\pi} d\theta$$

$$= \frac{19}{3}(\theta)_0^{2\pi}$$

$$= \frac{19}{3}(2\pi - 0) = \frac{38}{3}\pi$$

$$\boxed{\iint_R \sqrt{x^2+y}\,dx\,dy = \frac{38}{3}\pi}$$

Practice Q.1

Evaluate $\int_0^a \int_0^{\sqrt{a^2-y^2}} (x^2+y^2)dxdy$ by changing into polar coordinates.

Practice Q.2

Evaluate the following by changing into polar coordinates

$$\int_0^a \int_0^a \frac{x\,dx\,dy}{x^2+y^2}.$$

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AKTU : Engineering Mathematics-1

Multiple Integration : Change of Variables

Lec-9



By Gulshan sir

Today's Target

- ✓ Triple Integral using Change of Variables
- ✓ AKTU Questions
- ✓ Practice Questions

Triple Integral using Change of Variables

Case – 1. To change cartesian coordinates (x, y, z) to cylindrical coordinates (r, θ, z)

Put $x = r\cos\theta$

$y = r\sin\theta$

$z = z$

$x^2 + y^2 = r^2$

$dx\,dy\,dz = J\,dr\,d\theta\,dz$

$$J = \frac{\partial(x, y, z)}{\partial(r, \theta, z)} = r$$

$dx\,dy\,dz = r\,dr\,d\theta\,dz$

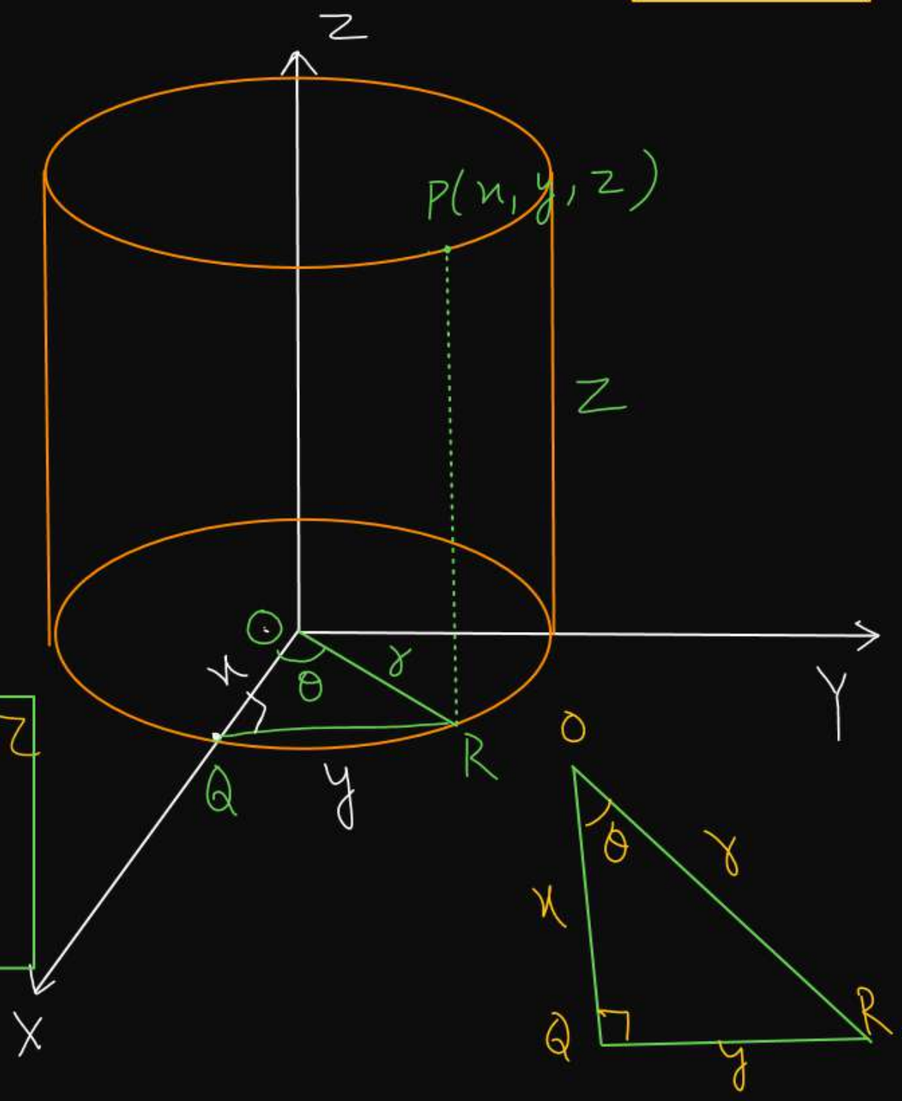
New limits of integrations

$\theta = 0$ to $\theta = 2\pi$

$r = 0$ to $r = ---$

$z = ---$ to $z = ---$

$$\iiint_V f(x, y, z)\,dx\,dy\,dz = \int\int\int_{V'} f(r\cos\theta, r\sin\theta, z)\,r\,dr\,d\theta\,dz$$



Q.1 ***By transforming into cylindrical coordidates , evaluate the integral***

$$\iiint (x^2+y^2+z^2)\,dx\,dy\,dz \text{ taken over the region } 0 \le z \le x^2+y^2 \le 1$$

$\iiint (x^2+y^2+z^2)\,dx\,dy\,dz$

Put $x = r\cos\theta$

$y = r\sin\theta$

$z = z$

$x^2+y^2 = r^2$

$dx\,dy\,dz = r\,dr\,d\theta\,dz$

New limits of integration

$\theta = 0$ to $\theta = 2\pi$

$r = 0$ to $r = 1$

$z = 0$ to $z = 1$

$$\iiint (x^2+y^2+z^2)\,dx\,dy\,dz = \int_{z=0}^{1}\int_{\theta=0}^{2\pi}\int_{r=0}^{1} (r^2+z^2)\,r\,dr\,d\theta\,dz$$

$$= \int_0^1 \int_0^{2\pi} \int_0^1 (r^3 + z^2 r)\, dr\, d\theta\, dz$$

$$= \int_0^1 \int_0^{2\pi} \left(\frac{r^4}{4} + z^2 \frac{r^2}{2} \right)_0^1 d\theta\, dz$$

$$= \int_0^1 \int_0^{2\pi} \left(\frac{1}{4} + \frac{z^2}{2} \right) d\theta\, dz$$

$$= \int_0^1 \left(\frac{1}{4} + \frac{z^2}{2} \right) dz \int_0^{2\pi} d\theta$$

$$= \int_0^1 \left(\frac{1}{4} + \frac{z^2}{2} \right) dz \, (\theta)_0^{2\pi}$$

$$= \int_0^1 \left(\frac{1}{4} + \frac{z^2}{2} \right) 2\pi$$

$$= 2\pi \int_0^1 \left(\frac{1}{4} + \frac{z^2}{2} \right) dz$$

$$= 2\pi \left(\frac{1}{4} z + \frac{z^3}{2 \times 3} \right)_0^1$$

$$= 2\pi \left(\frac{1}{4} + \frac{1}{3 \times 2} \right)$$

$$= 2\pi \left(\frac{3+2}{12} \right)$$

$$= 2\pi \times \frac{5}{12}$$

$$= \frac{5\pi}{6}$$

Practice Q.1

Evaluate $\int\int\int z(x^2+y^2)dxdydz$ *over the volume of the cylinder*

$x^2+y^2=1$ *intercepted by the plane* $z=2$ *and* $z=3$.

Triple Integral using Change of Variables

Case-2: To change cartesian coordinates (x, y, z) to spherical polar coordinates (r, θ, ϕ)

Put $x = r\sin\theta\cos\phi$

$y = r\sin\theta\sin\phi$

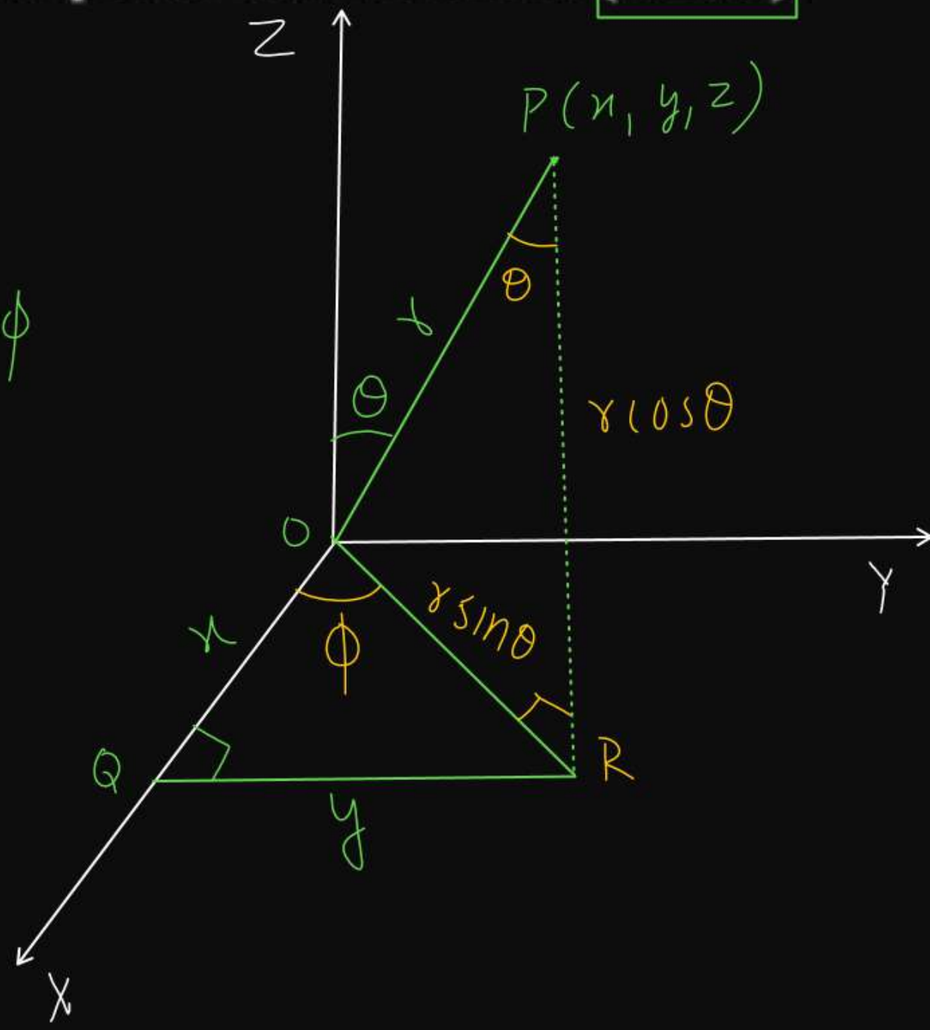
$z = r\cos\theta$

$r^2 = x^2 + y^2 + z^2$

$dx\,dy\,dz = J\,dr\,d\theta\,d\phi$

$J = \dfrac{\partial(x, y, z)}{\partial(r, \theta, \phi)} = r^2\sin\theta$

$dx\,dy\,dz = r^2\sin\theta\,dr\,d\theta\,d\phi$



Limits of integration

① Sphere

$r = 0$ to $r = a$

$\theta = 0$ to $\theta = \pi$

$\phi = 0$ to $\theta = 2\pi$

$$\iiint_V f(x,y,z)\,dx\,dy\,dz = \iiint_V f(r\sin\theta\cos\phi,\ r\sin\theta\sin\phi,\ r\cos\theta)\ r^2\sin\theta\,dr\,d\theta\,d\phi$$

(2) For Hemisphere

$r = 0$ to $r = a$

$\theta = 0$ to $\theta = \frac{\pi}{2}$

$\phi = 0$ to $\phi = 2\pi$

(3) For octant

$r = 0$ to $r = a$

$\theta = 0$ to $\theta = \frac{\pi}{2}$

$\phi = 0$ to $\phi = \frac{\pi}{2}$

Q.2 **Evaluate** $\iiint \frac{dxdydz}{(x^2+y^2+z^2)}$ **taken throughout the volume of the sphere** $x^2+y^2+z^2=a^2$

$$\iiint \frac{dx\,dy\,dz}{(x^2+y^2+z^2)}$$

Put

$x = r\sin\theta\cos\phi$

$y = r\sin\theta\sin\phi$

$z = r\cos\theta$

$x^2+y^2+z^2 = r^2$

$dx\,dy\,dz = r^2\sin\theta\,dr\,d\theta\,d\phi$

$x^2+y^2+z^2 = a^2$

$r^2 = a^2$

$r = a$

Limits of integration

$r = 0$ to $r = a$

$\theta = 0$ to $\theta = \pi$

$\phi = 0$ to $\phi = 2\pi$

$$\iiint \frac{dx\,dy\,dz}{x^2+y^2+z^2} = \int_0^{2\pi}\int_0^{\pi}\int_0^{a} \frac{r^2\sin\theta\,dr\,d\theta\,d\phi}{r^2}$$

$$= \int_0^{2\pi}\int_0^{\pi}\int_0^{a} \sin\theta\,d\theta\,dr\,d\phi$$

$$= \int_0^{2\pi}\int_0^{\pi} \sin\theta\,d\theta\,d\phi\,(r)_0^a$$

$$a\int_0^{2\pi}\int_0^{\pi} \sin\theta\,d\theta\,d\phi$$

$$= a\int_0^{2\pi} (-\cos\theta)_0^{\pi} \, d\phi$$

$$= -a\int_0^{2\pi} (\cos\pi - \cos 0)\, d\phi$$

$$= -a\int_0^{2\pi} (-1-1)\, d\phi$$

$$= 2a\int_0^{2\pi} d\phi$$

$$= 2a\left(\phi\right)_0^{2\pi}$$

$$= 2a(2\pi - 0)$$

$$\boxed{= 4\pi a}$$

Practice Q.2

Evaluate $\int\int\int xyz\, dxdydz$ over the positive octant of the sphere $x^2 + y^2 + z^2 = b^2$

by transforming to spherical polar coordinate

Practice Q.3

Evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \frac{dzdydx}{\sqrt{1-x^2-y^2-z^2}}$, by changing to spherical polar coordidates.

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AKTU : Engineering Mathematics-1

Beta Gamma Function

Lec-10

Today's Target

- ✓ Gamma function
- ✓ AKTU Questions
- ✓ Practice Questions



By Gulshan sir

Gamma function

If n is positive, then the definite inegral $\int_0^\infty e^{-x}x^{n-1}dx$ *is called gamma function of n.*

It is denoted by Γn

$$\Gamma n = \int_0^\infty e^{-x}\, x^{n-1}\, dx \qquad n > 0$$

Note: ① $\Gamma n+1 = \int_0^\infty e^{-x}\, x^{n}\, dx$

② $\Gamma 1 = 1$

$$\Gamma n = \int_0^\infty e^{-x}\, x^{n-1}\, dx$$

Put $n = 1$

$$\Gamma 1 = \int_0^\infty e^{-x}\, x^{0}\, dx$$

$$\Gamma 1 = \int_0^\infty e^{-x}\, dx$$

$$\Gamma 1 = -\left(e^{-x}\right)_0^\infty$$

$$\Gamma 1 = -\left[e^{-\infty} - e^{0}\right]$$

$$\Gamma 1 = -[0-1]$$

$$\Gamma 1 = \Gamma 1$$

③ $\Gamma\frac{1}{2} = \sqrt{\pi}$

④ $\Gamma 0 = \infty$

Properties of gamma function

1. Prove that $\boxed{\Gamma n+1 = n\Gamma n}$

We know that

$$\Gamma n = \int_0^\infty e^{-x} x^{n-1} dx$$

Put $n = n+1$

$$\Gamma n+1 = \int_0^\infty \underset{②}{e^{-x}} \underset{①}{x^n} dx$$

$$\Gamma n+1 = \left(-x^n e^{-x}\right)_0^\infty - \int_0^\infty n x^{n-1} (-e^{-x}) dx$$

$$\Gamma n+1 = -(0-0) + n\int_0^\infty e^{-x} x^{n-1} dx$$

$$\boxed{\Gamma n+1 = n\int_0^\infty e^{-x} x^{n-1} dx}$$

$$\boxed{\Gamma n+1 = n\Gamma n}$$ Hence proved

$\Gamma n = (n-1)\Gamma n-1$

$\Gamma n-1 = (n-2)\Gamma n-2$

$\Gamma n+3 = (n+2)\Gamma n+2$

$\Gamma 10 = 9\Gamma 9$

Properties of gamma function

2. Prove that $\boxed{\Gamma n+1 = n!}$

We know that

$\Gamma n+1 = n\,\Gamma n$ — ①

Replace n by (n-1) in ①

$\Gamma n = (n-1)\,\Gamma n-1$ — ②

Replace n by (n-1) in ②

$\Gamma n-1 = (n-2)\,\Gamma n-2$ — ③

Replace n by (n-1) in ③

$\Gamma n-2 = (n-3)\,\Gamma n-3$ — ④

.... - ... and so on

Put all the values in ①

$\Gamma n+1 = n\,\Gamma n$

$= n(n-1)\,\Gamma n-1$

$= n(n-1)(n-2)\,\Gamma n-2$

$= n(n-1)(n-2)(n-3)\,\Gamma n-3$

$\Gamma n+1 = n(n-1)(n-2)(n-3) - - - - 3.2.1\,\Gamma$

$\Gamma n+1 = n! \times \Gamma$

$\boxed{\Gamma n+1 = n!}$

Hence proved

$$\boxed{\int_0^\infty e^{-x} x^n dx = n!}$$

Note

① $\Gamma n = (n-1)!$

② If n is a positive integer

$$\Gamma n = (n-1)!$$

EX-

$$\Gamma 6 = (6-1)! = 5!$$

$$= 5 \times 4 \times 3 \times 2 \times 1$$

$$= 120$$

③ If n is a positive fraction

$$\Gamma n = (n-1)\Gamma n-1$$

EX-

$$\Gamma \frac{3}{2} = \left(\frac{3}{2}-1\right)\Gamma\left(\frac{3}{2}-1\right)$$

$$= \frac{1}{2}\Gamma\frac{1}{2}$$

$$= \frac{1}{2}\sqrt{\pi}$$

③ If n is a negative fraction

$$\Gamma n = \frac{\Gamma n+1}{n}$$

EX

$$\Gamma\frac{-5}{3} = \frac{\Gamma\frac{-5}{3}+1}{\frac{-5}{3}} = \frac{\Gamma\frac{-2}{3}}{\frac{-5}{3}}$$

$$= \frac{\Gamma\frac{-2}{3}+1}{\left(\frac{-5}{3}\right)\left(\frac{-2}{3}\right)}$$

$$\Gamma\frac{-5}{3} = \frac{\Gamma\frac{1}{3}}{\frac{10}{9}}$$

$$\Gamma\frac{-5}{3} = \frac{9}{10}\Gamma\frac{1}{3}$$

④ If n is a negative integer

$$\Gamma n = \frac{\Gamma n+1}{n}$$

EX-

$$\Gamma -4 = \frac{\Gamma -4+1}{-4} = \frac{\Gamma -3}{-4} = \frac{\Gamma -3+1}{(-4)(-3)}$$

$$= \frac{\Gamma -2}{(-4)(-3)} = \frac{\Gamma -2+1}{(-4)(-3)(-2)}$$

$$= \frac{\Gamma -1}{-24} = \frac{\Gamma -1+1}{-24(-1)} = \frac{\Gamma 0}{24}$$

$$= \infty$$

Q.1 Evaluate $\Gamma\left(\frac{7}{2}\right)$

$\Gamma\left(\frac{7}{2}\right)$

We know that

$\Gamma n = (n-1)\Gamma(n-1)$

$\Gamma\left(\frac{7}{2}\right) = \left(\frac{7}{2}-1\right)\Gamma\left(\frac{7}{2}-1\right)$

$= \frac{5}{2}\Gamma\left(\frac{5}{2}\right)$

$= \frac{5}{2}\left(\frac{5}{2}-1\right)\Gamma\left(\frac{5}{2}-1\right)$

$= \frac{5}{2}\times\frac{3}{2}\Gamma\left(\frac{3}{2}\right)$

$\Gamma\left(\frac{7}{2}\right) = \frac{5}{2}\times\frac{3}{2}\Gamma\left(\frac{3}{2}\right)$

$= \frac{5}{2}\times\frac{3}{2}\times\left(\frac{3}{2}-1\right)\Gamma\left(\frac{3}{2}-1\right)$

$= \frac{5}{2}\times\frac{3}{2}\times\frac{1}{2}\Gamma\left(\frac{1}{2}\right)$

$= \frac{15}{8}\Gamma\left(\frac{1}{2}\right)$

$$\boxed{\Gamma\left(\frac{7}{2}\right) = \frac{15}{8}\sqrt{\pi}}$$

Q.2 Evaluate $\Gamma\left(\frac{-5}{2}\right)$

We know that

$$\Gamma n = \frac{\Gamma n+1}{n}$$

$$\Gamma\left(\frac{-5}{2}\right) = \frac{\Gamma\left(\frac{-5}{2}+1\right)}{\frac{-5}{2}} = \frac{\Gamma\left(\frac{-3}{2}\right)}{\frac{-5}{2}}$$

$$= \frac{\Gamma\left(\frac{-3}{2}+1\right)}{\left(\frac{-5}{2}\right)\times\left(\frac{-3}{2}\right)}$$

$$= \frac{\Gamma\left(\frac{-1}{2}\right)}{\left(\frac{-5}{2}\right)\times\left(\frac{-3}{2}\right)}$$

$$\Gamma\left(\frac{-5}{2}\right) = \frac{\Gamma\left(\frac{-1}{2}\right)}{\left(\frac{-5}{2}\right)\times\left(\frac{-3}{2}\right)} = \frac{\Gamma\left(\frac{-1}{2}+1\right)}{\left(\frac{-5}{2}\right)\times\left(\frac{-3}{2}\right)\times\left(\frac{-1}{2}\right)}$$

$$= \frac{\Gamma\left(\frac{1}{2}\right)}{\frac{-15}{8}}$$

$$\Gamma\left(\frac{-5}{2}\right) = \frac{-8}{15}\Gamma\left(\frac{1}{2}\right)$$

$$\boxed{\Gamma\left(\frac{-5}{2}\right) = \frac{-8}{15}\sqrt{\pi}}$$

Q.3 *Evaluate* $\int_0^\infty e^{-x^2}dx$.

$$\int_0^\infty e^{-n^2}dn$$

Put $n^2 = t$

$$2n\,dn = dt$$

$$dn = \frac{dt}{2n}$$

$$dn = \frac{dt}{2\sqrt{t}}$$

$$dn = \frac{1}{2}t^{-1/2}dt$$

When $n = 0$, $t = 0$

When $n = \infty$, $t = \infty$

$$= \int_0^\infty e^{-t} \times \frac{1}{2} t^{-1/2}dt$$

$$= \frac{1}{2}\int_0^\infty e^{-t} t^{-1/2}dt$$

$$= \frac{1}{2}\int_0^\infty e^{-t} t^{\frac{1}{2}-1}dt$$

$$\int_0^\infty e^{-n^2}dn = \frac{1}{2}\int_0^\infty e^{-n} n^{\frac{1}{2}-1}dn$$

$$= \frac{1}{2}\Gamma\frac{1}{2} = \frac{1}{2}\sqrt{\pi}$$

$$\int_0^\infty e^{-n^2}dn = \frac{1}{2}\sqrt{\pi}$$

Q.4 *Evaluate* $\int_0^\infty \sqrt[4]{x}\, e^{-\sqrt{x}}\, dx.$

$$\int_0^\infty \sqrt[4]{x}\, e^{-\sqrt{x}}\, dx$$

Put $\sqrt{x} = t$

$x = t^2$

$dx = 2t\, dt$

When $x = 0$

$t = 0$

When $x = \infty$

$t = \infty$

$$= \int_0^\infty (t^2)^{\frac{1}{4}} e^{-t}\, 2t\, dt$$

$$= 2\int_0^\infty t^{1/2} e^{-t} \times t\, dt$$

$$= 2\int_0^\infty e^{-t}\, t^{3/2}\, dt$$

$$= 2\int_0^\infty e^{-x}\, x^{3/2}\, dx$$

$$\int_a^b f(x)\, dx = \int_a^b f(x)\, dy$$

$$\int_0^\infty \sqrt[4]{x}\, e^{-\sqrt{x}}\, dx = 2\int_0^\infty e^{-x}\, x^{\frac{5}{2}-1}\, dx$$

$$= 2\lceil \tfrac{5}{2} = 2 \times \left(\tfrac{5}{2}-1\right)\lceil \tfrac{5}{2}-1$$

$$= \cancel{2} \times \tfrac{3}{\cancel{2}} \lceil \tfrac{3}{2} = 3 \times \left(\tfrac{3}{2}-1\right)\lceil \tfrac{3}{2}-1$$

$$= 3 \times \tfrac{1}{2} \lceil \tfrac{1}{2}$$

$$= \frac{3}{2}\sqrt{\pi}$$

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AKTU : Engineering Mathematics-1

Beta Gamma Function

Lec-02



By Gulshan sir

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- ✓ Gamma function
- ✓ AKTU Questions
- ✓ Practice Questions

Transformation of gamma function

1. Prove that $\Gamma n = k^n \int_0^\infty e^{-kx} x^{n-1} dx$

We know that

$$\Gamma n = \int_0^\infty e^{-x} x^{n-1} dx$$

Put $x = ky$

$dx = k\,dy$

When $x = 0$, $y = 0$

When $x = \infty$, $y = \infty$

$$\Gamma n = \int_0^\infty e^{-ky} (ky)^{n-1} k\,dy$$

$$\Gamma n = \int_0^\infty e^{-ky} k^{n-1} y^{n-1} k\,dy$$

$$\Gamma n = k^{n-1+1} \int_0^\infty e^{-ky} y^{n-1} dy$$

$$\Gamma n = k^n \int_0^\infty e^{-ky} y^{n-1} dy$$

$$\Gamma n = k^n \int_0^\infty e^{-kx} x^{n-1} dx$$

$$\int_a^b f(x)\,dx = \int_a^b f(y)\,dy$$

$$\int_0^\infty e^{-kx} x^{n-1} dx = \frac{\Gamma n}{k^n}$$

Transformation of gamma function

2. Prove that

$$\Gamma n = \int_0^1 \left(\log \frac{1}{y}\right)^{n-1} dy$$

$$= \int_0^1 \left(\log \frac{1}{y}\right)^{n-1} dy$$

Put $\log \frac{1}{y} = x$

$$\frac{1}{y} = e^x$$

$$y = \frac{1}{e^x}$$

$$y = e^{-x}$$

$$y = e^{-x}$$

$$dy = -e^{-x} dx$$

When $y = 0$, $x = \infty$

When $y = 1$, $x = 0$

$$= -\int_{\infty}^{0} x^{n-1} e^{-x} dx$$

$$\int_a^b f(x)\, dx = -\int_b^a f(x)\, dx$$

$$\Gamma n = \int_0^{\infty} e^{-x} x^{n-1} dx$$

$\Rightarrow$

$$\Gamma n = \int_0^1 \left(\log \frac{1}{y}\right)^{n-1} dy$$

Q.5 *Evaluate* $\int_0^1 (x \log x)^3 dx$.

$$\int_0^1 (x \log x)^3 dx$$

$$= \int_0^1 x^3 (\log x)^3 dx$$

Put $\log x = -t$

$$x = e^{-t}$$

$$dx = -e^{-t} dt$$

When $x = 0$, $t = \infty$

When $x = 1$, $t = 0$

$$= \int_\infty^0 (e^{-t})^3 (-t)^3 (-e^{-t}) dt$$

$$= +\int_\infty^0 e^{-3t} t^3 e^{-t} dt$$

$$= -\int_0^\infty e^{-4t} t^3 dt$$

$$= -\int_0^\infty e^{-4t} t^{4-1} dt$$

$$\int_0^\infty e^{-kx} x^{n-1} dx = \frac{\Gamma n}{k^n}$$

$$= -\frac{\Gamma 4}{(4)^4}$$

$$= -\frac{3!}{4 \times 4 \times 4 \times 4} \quad (\Gamma n = (n-1)!)$$

$$= \frac{-3 \times \cancel{2} \times 1}{4 \times 4 \times 4 \times \cancel{4} \times 2} = \frac{-3}{128}$$

$$= -\frac{3}{128}$$

Practice Q. $Evaluate\ \int_0^1 (x\ log x)^4 dx.$

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AKTU : Engineering Mathematics-1

Beta Gamma Function

Lec-3

Today's Target

- ✓ Beta Function
- ✓ AKTU Questions
- ✓ Practice Questions



By Gulshan sir

Beta Function

If m , n are positive , then the definite inegral

$\int_0^1 x^{m-1}(1-x)^{n-1}dx$ *which a function of* m *and* n *is called beta function .*

It is denoted by

$$\beta(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx$$

$$\overline{n} = \int_0^\infty e^{-x}\, x^{n-1}\, dx$$

Properties of Beta Function

1. Symmetry of Beta Function

$$\beta(m,n) = \beta(n,m)$$

$$\beta(m,n) = \int_0^1 x^{n-1}(1-x)^{m-1}\,dx$$

$$\beta(m,n) = \beta(n,m)$$

Hence property

We know that

$$\beta(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx$$

$$\beta(m,n) = \int_0^1 (1-x)^{m-1}\left[1-(1-x)\right]^{n-1}dx \quad \left\{\because \int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx\right\}$$

$$\beta(m,n) = \int_0^1 (1-x)^{m-1}(x-x+x)^{n-1}\,dx$$

Properties of Beta Function

2. Transformation of Beta Function

(a) $$\beta(m,n) = \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}}\,dx = \int_0^\infty \frac{x^{n-1}}{(1+x)^{m+n}}\,dx$$

We know that

$$\beta(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx$$

Put $x = \frac{1}{1+y}$ when $x = 0$, $y = \infty$

$dx = -\frac{1}{(1+y)^2}\,dy$ when $x = 1$, $y = 0$

$$\beta(m,n) = \int_\infty^0 \left(\frac{1}{1+y}\right)^{m-1}\left(1-\frac{1}{1+y}\right)^{n-1}\left(-\frac{1}{(1+y)^2}\right)dy$$

$$= -\int_\infty^0 \left(\frac{1}{1+y}\right)^{m-1}\left(\frac{1+y-1}{1+y}\right)^{n-1}\frac{1}{(1+y)^2}\,dy$$

$$= -(-)\int_0^\infty \frac{1}{(1+y)^{m-1}} \times \frac{y^{n-1}}{(1+y)^{n-1}} \times \frac{1}{(1+y)^2}\,dy$$

$$= \int_0^\infty \frac{y^{n-1}}{(1+y)^{m-1+n-1+2}}\,dy$$

$$B(m,n) = \int_0^\infty \frac{y^{n-1}}{(1+y)^{m+n-z+z}}\,dy$$

$$B(m,n) = \int_0^\infty \frac{y^{n-1}}{(1+y)^{m+n}}\,dy$$

$$\boxed{B(m,n) = \int_0^\infty \frac{x^{n-1}}{(1+x)^{m+n}}\,dx}$$

Hence proved

$$B(n,m) = \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}}$$

$$\therefore \boxed{B(m,n) = \int_0^\infty \frac{x^{n-1}}{(1+x)^{m+n}}\,dx = \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}}\,dx}$$

Properties of Beta Function

2. Transformation of Beta Function

(b) $\beta(m,n) = 2\int_0^{\pi/2} \sin^{2m-1}\theta \cos^{2n-1}\theta \, d\theta$

We know that

$$\beta(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx$$

Put $x = \sin^2\theta$

$dx = 2\sin\theta\cos\theta\, d\theta$

When $x = 0$, $\theta = 0$

When $x = 1$, $\theta = \frac{\pi}{2}$

$$\beta(m,n) = \int_0^{\pi/2} (\sin^2\theta)^{m-1}(1-\sin^2\theta)^{n-1} 2\sin\theta\cos\theta\, d\theta$$

$$= \int_0^{\pi/2} (\sin\theta)^{2m-2}(\cos^2\theta)^{n-1} \times 2\sin\theta\cos\theta\, d\theta$$

$$= \int_0^{\pi/2} (\sin\theta)^{2m-2} \times (\cos\theta)^{2n-2} \times 2\sin\theta\cos\theta\, d\theta$$

$$= 2\int_0^{\pi/2} (\sin\theta)^{2m-2+1} \times (\cos\theta)^{2n-2+1}\, d\theta$$

$$\beta(m,n) = 2\int_0^{\pi/2} (\sin\theta)^{2m-1} \times (\cos\theta)^{2n-1} d\theta$$

$$\beta(m,n) = 2\int_0^{\pi/2} \sin^{2m-1}\theta \times \cos^{2n-1}\theta \, d\theta$$

Hence proved

Q.1 *Evaluate* $\int_0^1 (1-x^3)^{-1/2}dx$ **Univ. Exam**

$$\int_0^1 (1-x^3)^{-1/2}\,dx$$

Put $x^3 = t$

$3x^2\,dx = dt$

When $x = 0$, $t = 0$

When $x = 1$, $t = 1$

$$= \int_0^1 (1-t)^{-1/2}\frac{dt}{3x^2}$$

$$= \int_0^1 (1-t)^{-1/2} \times \frac{1 \times dt}{3t^{2/3}}$$

$$= \int_0^1 (1-t)^{-1/2} \times \frac{t^{-2/3}}{3}\,dt$$

$$= \frac{1}{3}\int_0^1 t^{-2/3}(1-t)^{-1/2}\,dt$$

$$= \frac{1}{3}\int_0^1 t^{\frac{1}{3}-1}(1-t)^{\frac{1}{2}-1}\,dt$$

$$= \frac{1}{3}B\left(\frac{1}{3}, \frac{1}{2}\right)$$

$$\int_0^1 (1-x^3)^{-1/2}\,dx = \frac{1}{3}B\left(\frac{1}{3}, \frac{1}{2}\right)$$

Q.2 Evaluate $\int_0^1 \frac{x^{m-1}+x^{n-1}}{(1+x)^{m+n}}dx$

Univ. Exam

$$\int_0^1 \frac{x^{m-1}+x^{n-1}}{(1+x)^{m+n}}dx$$

Using transformation of Beta function

$$B(m,n)=\int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}}dx$$

$$B(m,n)=\int_0^1 \frac{x^{m-1}}{(1+x)^{m+n}}dx+\int_1^\infty \frac{x^{m-1}}{(1+x)^{m+n}}dx \quad \text{---(1)}$$

Consider second integral

$$\int_1^\infty \frac{x^{m-1}}{(1+x)^{m+n}}dx$$

Put $x=\frac{1}{t}$

$$dx=-\frac{1}{t^2}dt$$

When $x=1$

$t=1$

When $x=\infty$

$t=0$

$$=\int_1^0 \frac{\left(\frac{1}{t}\right)^{m-1}}{\left(1+\frac{1}{t}\right)^{m+n}}\times\left(\frac{-1}{t^2}\right)dt$$

$$=-\int_1^0 \frac{dt}{\left(\frac{1+t}{t}\right)^{m+n}\times t^2\times t^{m-1}}$$

$$=\int_0^1 \frac{dt}{\frac{(1+t)^{m+n}}{(t)^{m+n}}\times t^2\times t^{m-1}}$$

$$= \int_0^1 \frac{t^{m+n}}{(1+t)^{m+n} t^2 t^{m-1}} dt$$

$$= \int_0^1 \frac{t^{m+n} \times t^{-2} \times t^{-m+1}}{(1+t)^{m+n}}$$

$$= \int_0^1 \frac{t^{m+n-2-m+1}}{(1+t)^{m+n}} dt$$

$$= \int_0^1 \frac{t^{n-1}}{(1+t)^{m+n}} dt$$

$$\int_1^\infty \frac{x^{m-1}}{(1+x)^{m+n}} dx = \int_0^1 \frac{x^{n-1}}{(1+x)^{m+n}} dx$$

Put in (1)

$$B(m,n) = \int_0^1 \frac{x^{m-1}}{(1+x)^{m+n}} dx + \int_0^1 \frac{x^{n-1}}{(1+x)^{m+n}} dx$$

$$\boxed{B(m,n) = \int_0^1 \frac{x^{m-1} + x^{n-1}}{(1+x)^{m+n}} dx}$$

Q.3 Prove that $\beta(m,n) = \int_0^\infty \frac{x^{m-1}}{(ax+b)^{m+n}}\,dx$ $a^m\ b^n$

Univ. Exam

RHS

$$\int_0^\infty \frac{x^{m-1}}{(ax+b)^{m+n}}\,dx$$

$ax = bt$

$a\,dx = b\,dt$

$dx = \frac{b}{a}dt$

When $x=0$, $t=0$

When $x=\infty$, $t=\infty$

$$= \int_0^\infty \frac{\left(\frac{b}{a}t\right)^{m-1}}{(bt+b)^{m+n}} \times \frac{b}{a}\,dt$$

$$= \int_0^\infty \frac{\left(\frac{b}{a}\right)^{m-1} \times t^{m-1} \times \left(\frac{b}{a}\right)}{b^{m+n}\,(1+t)^{m+n}}\,dt$$

$$= \frac{\left(\frac{b}{a}\right)^{m-1} \times \frac{b}{a}}{b^{m+n}} \int_0^\infty \frac{t^{m-1}}{(1+t)^{m+n}}\,dt$$

$$= \frac{b^{m-1} \times b}{a^{m-1} \times a \times b^{m+n}} \int_0^\infty \frac{t^{m-1}}{(1+t)^{m+n}}\,dt$$

$$= \frac{b^{m-1+1}}{a^{m-1+1} * b^{m+n}} \int_0^\infty \frac{t^{m-1}}{(1+t)^{m+n}}\,dt$$

$$= \frac{b^m}{a^m \times b^{m+n}} \int_0^\infty \frac{t^{m-1}}{(1+t)^{m+n}}\,dt$$

$$= \frac{b^m \times b^{-m-n}}{a^m} \int_0^\infty \frac{t^{m-1}}{(1+t)^{m+n}}\,dt$$

$$= b^{-n} a^{-m} \int_0^\infty \frac{t^{m-1}}{(1+t)^{m+n}} dt$$

$$= a^{-m} b^{-n} \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}} dx$$

$$= a^{-m} b^{-n} \times a^m b^n \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}} dx$$

$$= a^{-m+m} b^{-n+n} \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}} dx$$

$$= a^0 b^0 \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}} dx$$

$$= \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}} dx$$

$$= B(m,n)$$

LHS = RHS

Practice Q. $\int_0^1 x^4(1-\sqrt{x})^5 dx$

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AKTU : Engineering Mathematics-1

Beta Gamma Function

Lec-4

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By Gulshan sir

Beta and Gamma function

1. Relation between Beta and Gamma function

$$\beta(m,n) = \frac{\Gamma m \, \Gamma n}{\Gamma m+n}$$

We know that

$$\Gamma n = \int_0^\infty e^{-x} x^{n-1} dx \quad -(1)$$

$$\Gamma n = k^n \int_0^\infty e^{-kx} x^{n-1} dx \quad -(2)$$

Replace k by z

$$\Gamma n = z^n \int_0^\infty e^{-zx} x^{n-1} dx \quad -(3)$$

Multiply (3) by e^{-z} and z^{m-1}

$$\Gamma n \, e^{-z} z^{m-1} = z^n e^{-z} z^{m-1} \int_0^\infty e^{-zx} x^{n-1} dx$$

$$\Gamma n \, e^{-z} z^{m-1} = \int_0^\infty e^{-z} e^{-zx} z^n z^{m-1} x^{n-1} dx$$

$$\Gamma n \, e^{-z} z^{m-1} = \int_0^\infty e^{-(1+x)z} \times z^{m+n-1} x^{n-1} dx$$

Integrate above equation wrt z from 0 to ∞

$$\Gamma n \int_0^\infty e^{-z} z^{m-1} dz = \int_0^\infty \int_0^\infty e^{-(1+x)z} \times z^{m+n-1} \times x^{n-1} dx$$

$$\Gamma n\ \Gamma m = \int_0^\infty x^{n-1}\,dx \int_0^\infty e^{-(1+x)z}\, z^{m+n-1}\,dz$$

$$\Gamma n\ \Gamma m = \int_0^\infty x^{n-1}\,dx\ \frac{\Gamma(m+n)}{(1+x)^{m+n}}$$

$$\Gamma n\ \Gamma m = \Gamma(m+n) \int_0^\infty \frac{x^{n-1}}{(1+x)^{m+n}}\,dx$$

$$\Gamma n\ \Gamma m = \Gamma(m+n)\ \beta(m,n)$$

$$\beta(m,n) = \frac{\Gamma m\ \Gamma n}{\Gamma(m+n)}$$

Beta and Gamma function

2. Prove that

$$\int_0^{\frac{\pi}{2}} \sin^m\theta \cos^n\theta \, d\theta = \frac{\lfloor\frac{m+1}{2} \lfloor\frac{n+1}{2}}{2\lfloor\frac{m+n+2}{2}}$$

We know that

$$\beta(m,n) = 2\int_0^{\pi/2} \sin^{2m-1}\theta \cos^{2n-1}\theta \, d\theta$$

$$\int_0^{\pi/2} \sin^{2m-1}\theta \cos^{2n-1}\theta \, d\theta = \frac{1}{2}\beta(m,n)$$

$$\int_0^{\pi/2} \sin^{2m-1}\theta \cos^{2n-1}\theta \, d\theta = \frac{\lfloor m \lfloor n}{2\lfloor m+n}$$

Put $m = \frac{m+1}{2}$ and $n = \frac{n+1}{2}$

$$\int_0^{\pi/2} \sin^m\theta \cos^n\theta \, d\theta = \frac{\lfloor\frac{m+1}{2} \lfloor\frac{n+1}{2}}{2\lfloor\frac{m+1}{2}+\frac{n+1}{2}}$$

$$\int_0^{\pi/2} \sin^m\theta \cos^n\theta \, d\theta = \frac{\lfloor\frac{m+1}{2} \lfloor\frac{n+1}{2}}{2\lfloor\frac{m+n+2}{2}}$$

$$\int_0^{\pi/2} \sin^m\theta \cos^n\theta \, d\theta = \frac{\Gamma\frac{m+1}{2}\,\Gamma\frac{n+1}{2}}{2\,\Gamma\frac{m+n+2}{2}}$$

special cases

(i) Put $n = 0, m = 0$

$$\int_0^{\pi/2} d\theta = \frac{\Gamma\frac{1}{2}\,\Gamma\frac{1}{2}}{2\,\Gamma 1}$$

$$(\theta)_0^{\pi/2} = \frac{\left(\Gamma\frac{1}{2}\right)^2}{2}$$

$$\frac{\pi}{2} = \frac{\left(\Gamma\frac{1}{2}\right)^2}{2}$$

$$\Gamma\frac{1}{2} = \sqrt{\pi}$$

(ii) Put $n = 0$

$$\int_0^{\pi/2} \sin^m\theta \, d\theta = \frac{\Gamma\frac{m+1}{2}\,\Gamma\frac{1}{2}}{2\,\Gamma\frac{m+2}{2}}$$

$$\int_0^{\pi} \sin^m\theta \, d\theta = \frac{\sqrt{\pi}}{2}\,\frac{\Gamma\frac{m+1}{2}}{\Gamma\frac{m+2}{2}}$$

(iii) Put $m = 0$

$$\int_0^{\pi/2} \cos^n\theta \, d\theta = \frac{\sqrt{\pi}}{2}\,\frac{\Gamma\frac{n+1}{2}}{\Gamma\frac{n+2}{2}}$$

Beta and Gamma function

An important deduction using $\int_0^\infty \frac{x^{n-1}}{1+x} = \frac{\pi}{\sin n\pi}$

3. Prove that $\boxed{\Gamma n \ \Gamma 1-n = \frac{\pi}{\sin n\pi}}$

We know that

$$\beta(m,n) = \frac{\Gamma m \ \Gamma n}{\Gamma m+n}$$

$$\int_0^\infty \frac{x^{n-1}}{(1+x)^{m+n}} = \frac{\Gamma m \ \Gamma n}{\Gamma m+n}$$

Put $m+n = 1$

$m = 1-n$

$$\int_0^\infty \frac{x^{n-1}}{1+x} = \frac{\Gamma 1-n \ \Gamma n}{\Gamma 1-n+n}$$

$$\int_0^\infty \frac{x^{n-1}}{1+x} = \frac{\Gamma 1-n \ \Gamma n}{\Gamma 1}$$

$$\frac{\pi}{\sin n\pi} = \Gamma 1-n \ \Gamma n$$

$$\boxed{\Gamma n \ \Gamma 1-n = \frac{\pi}{\sin n\pi}}$$

Note : ① Put $n = \frac{1}{2}$

$$\Gamma\frac{1}{2} \ \Gamma 1-\frac{1}{2} = \frac{\pi}{\sin\frac{\pi}{2}}$$

$$\Gamma\frac{1}{2} \ \Gamma\frac{1}{2} = \pi$$

$$\boxed{\Gamma\frac{1}{2} = \sqrt{\pi}}$$

② Put $n = \frac{1}{3}$

$$\Gamma\frac{1}{3} \ \Gamma 1-\frac{1}{3} = \frac{\pi}{\sin\frac{\pi}{3}}$$

$$\boxed{\Gamma\frac{1}{3} \ \Gamma\frac{2}{3} = \frac{2\pi}{\sqrt{3}}}$$

Beta and Gamma function

4. Duplication formula

$$\Gamma m \ \Gamma\left(m+\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2^{2m-1}} \Gamma 2m$$

We know that

$$\beta(m,n) = 2\int_0^{\pi/2} \sin^{2m-1}\theta \cos^{2n-1}\theta \, d\theta$$

$$\int_0^{\pi/2} \sin^{2m-1}\theta \cos^{2n-1}\theta \, d\theta = \frac{1}{2}\beta(m,n)$$

$$\int_0^{\pi/2} \sin^{2m-1}\theta \cos^{2n-1}\theta \, d\theta = \frac{1}{2}\frac{\Gamma m \Gamma n}{\Gamma m+n}$$

Put $2n-1 = 0$

$n = \frac{1}{2}$

$$\int_0^{\pi/2} \sin^{2m-1}\theta \, d\theta = \frac{\Gamma m \Gamma \frac{1}{2}}{2\Gamma m+\frac{1}{2}} \quad -(1)$$

Put $m = n$

$$\int_0^{\pi/2} \sin^{2m-1}\theta \cos^{2m-1}\theta \, d\theta = \frac{\Gamma m \Gamma m}{2\Gamma 2m}$$

$$\int_0^{\pi/2} (\sin\theta\cos\theta)^{2m-1} \, d\theta = \frac{\Gamma m \Gamma m}{2\Gamma 2m}$$

$\sin 2\theta = 2\sin\theta\cos\theta$

$$\int_0^{\pi/2} \left(\frac{2\sin\theta\cos\theta}{2}\right)^{2m-1} d\theta = \frac{\lfloor m \lfloor m}{2 \lfloor 2m}$$

$$\int_0^{\pi/2} \left(\frac{\sin 2\theta}{2}\right)^{2m-1} d\theta = \frac{\lfloor m \lfloor m}{2 \lfloor 2m}$$

$$\frac{1}{2^{2m-1}} \int_0^{\pi/2} (\sin 2\theta)^{2m-1} d\theta = \frac{\lfloor m \lfloor m}{2 \lfloor 2m}$$

$$\int_0^{\pi/2} \sin^{2m-1} 2\theta \, d\theta = \frac{\lfloor m \lfloor m}{2 \lfloor 2m} 2^{2m-1}$$

Put $2\theta = t$ When $\theta = 0$ $t = 0$

$2 d\theta = dt$

$d\theta = \frac{dt}{2}$ When $\theta = \frac{\pi}{2}$, $t = \pi$

$$\int_0^{\pi} \sin^{2m-1} t \frac{dt}{2} = \frac{\lfloor m \lfloor m}{2 \lfloor 2m} 2^{2m-1}$$

$$\frac{1}{2} \int_0^{\pi/2} \sin^{2m-1} t \, dt = \frac{\lfloor m \lfloor m}{2 \lfloor 2m} 2^{2m-1}$$

$$\left[\int_0^{2a} f(x)dx = 2\int_0^{a} f(x)dx\right]$$

$$\int_0^{\pi/2} \sin^{2m-1}\theta \, d\theta = \frac{\lfloor m \lfloor m}{2 \lfloor 2m} 2^{2m-1} \quad \text{— (2)}$$

From (1) and (2)

$$\frac{\cancel{\Gamma m}\,\Gamma m}{\cancel{2}\,\Gamma 2m}\,2^{2m-1} = \frac{\cancel{\Gamma m}\,\Gamma\frac{1}{2}}{\cancel{2}\,\Gamma m+\frac{1}{2}}$$

$$\frac{\Gamma m\; 2^{2m-1}}{\Gamma 2m} = \frac{\sqrt{\pi}}{\Gamma m+\frac{1}{2}}$$

$$\boxed{\Gamma m\;\Gamma m+\tfrac{1}{2} = \frac{\sqrt{\pi}}{2^{2m-1}}\;\Gamma 2m}$$

Q.1 *Prove that* $\frac{B(p,q+1)}{q} = \frac{B(p+1,q)}{p} = \frac{B(p,q)}{p+q}, (p>0, q>0)$ Univ. Exam

We know that

$$B(m,n) = \frac{\Gamma m \, \Gamma n}{\Gamma m+n}$$

$$\frac{B(p,q+1)}{q} = \frac{\Gamma p \, \Gamma q+1}{q \Gamma p+q+1}$$

$$= \frac{\Gamma p \, q \, \Gamma q}{q (p+q) \Gamma p+q}$$

$$= \frac{\Gamma p \, \Gamma q}{(p+q) \Gamma p+q}$$

$$\frac{B(p,q+1)}{q} = \frac{B(p,q)}{p+q} \quad \text{— (1)}$$

NOW

$$\frac{B(p+1,q)}{p} = \frac{\Gamma p+1 \, \Gamma q}{p \Gamma p+q+1}$$

$$= \frac{p \Gamma p \, \Gamma q}{p (p+q) \Gamma p+q}$$

$$= \frac{\Gamma p \, \Gamma q}{(p+q) \Gamma p+q}$$

$$\frac{B(p+1,q)}{p} = \frac{B(p,q)}{(p+q)} \quad \text{— (2)}$$

From (1) and (2)

$$\frac{B(p,q+1)}{q} = \frac{B(p+1,q)}{p} = \frac{B(p,q)}{p+q}$$

Hence prove

Q.2 *Using beta gamma functions, evaluate* $\int_0^1 \left(\frac{x^3}{1-x^3}\right)^{\frac{1}{2}} dx$

Univ. Exam

$$\int_0^1 \left(\frac{x^3}{1-x^3}\right)^{\frac{1}{2}} dx$$

Put $x^3 = t$

$x = t^{1/3}$

$dx = \frac{1}{3} t^{-2/3} dt$

When $x = 0$

$t = 0$

When $x = 1$

$t = 1$

$$= \int_0^1 \left(\frac{t}{1-t}\right)^{\frac{1}{2}} \frac{1}{3} t^{-2/3} dt$$

$$= \frac{1}{3} \int_0^1 \frac{t^{1/2} \times t^{-2/3}}{(1-t)^{1/2}} dt$$

$$= \frac{1}{3} \int_0^1 \frac{t^{-1/6}}{(1-t)^{1/2}} dt$$

$$= \frac{1}{3} \int_0^1 t^{-1/6} (1-t)^{-1/2} dt$$

$$\beta(m,n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

$$= \frac{1}{3} \int_0^1 x^{-1/6} (1-x)^{-1/2} dx$$

$$= \frac{1}{3} \int_0^1 x^{\frac{5}{6}-1} (1-x)^{\frac{1}{2}-1} dx$$

$$= \frac{1}{3} \beta\left(\frac{5}{6}, \frac{1}{2}\right)$$

$$\beta(m,n) = \frac{\Gamma m \, \Gamma n}{\Gamma m+n}$$

$$\Gamma n = n-1 \, \Gamma n-1$$

$$= \frac{1}{3} \frac{\Gamma\frac{5}{6} \, \Gamma\frac{1}{2}}{\Gamma\frac{5}{6}+\frac{1}{2}}$$

$$= \frac{\sqrt{\pi}}{3} \frac{\Gamma\frac{5}{6}}{\Gamma\frac{4}{3}}$$

$$= \frac{\sqrt{\pi}}{3} \frac{\Gamma\frac{5}{6}}{\left(\frac{4}{3}-1\right)\Gamma\frac{4}{3}-1}$$

$$= \frac{\sqrt{\pi}}{3} \frac{\Gamma\frac{5}{6}}{\frac{1}{3}\Gamma\frac{1}{3}} = \frac{\Gamma\frac{5}{6}\sqrt{\pi}}{\Gamma\frac{1}{3}}$$

Q.3 Evaluate $\int_0^\infty \frac{dx}{1+x^4}$

Univ. Exam

$$\int_0^\infty \frac{dx}{1+x^4}$$

Put $x^4 = t$

$x = t^{1/4}$

$dx = \frac{1}{4} t^{-3/4}$

When $x = 0$

$t = 0$

When $x = \infty$

$t = \infty$

$$= \frac{1}{4}\int_0^\infty \frac{t^{-3/4}}{1+t}\,dt$$

$$= \frac{1}{4}\int_0^\infty \frac{t^{\frac{1}{4}-1}}{1+t}$$

$$\left[\because \int_0^\infty \frac{x^{n-1}}{1+x}\,dx = \frac{\pi}{\sin n\pi}\right]$$

$$= \frac{1}{4} \times \frac{\pi}{\sin\frac{\pi}{4}}$$

$$= \frac{1}{4} \times \frac{\pi}{\frac{1}{\sqrt{2}}}$$

$$\int_0^\infty \frac{dx}{1+x^4} = \frac{\sqrt{2}\,\pi}{4}$$

Q. 4 Prove the following result $\int_0^{\pi/2} \sqrt{\tan\theta}\, d\theta = \int_0^{\pi/2} \sqrt{\cot\theta}\, d\theta = \frac{\pi}{\sqrt{2}}$

Univ. Exam

$$\int_0^{\pi/2} \sqrt{\tan\theta}\, d\theta = \int_0^{\pi/2} \sqrt{\cot\theta}\, d\theta = \frac{\pi}{\sqrt{2}}$$

$$\int_0^{\pi/2} \sqrt{\tan\theta}\, d\theta = \int_0^{\pi/2} \sqrt{\tan\left(\frac{\pi}{2}-\theta\right)}\, d\theta$$

$$= \int_0^{\pi/2} \sqrt{\cot\theta}\, d\theta$$

$$= \int_0^{\pi/2} \sqrt{\frac{\cos\theta}{\sin\theta}}\, d\theta$$

$$= \int_0^{\pi/2} \left(\frac{\cos\theta}{\sin\theta}\right)^{1/2} d\theta$$

$$= \int_0^{\pi/2} \sin^{-\frac{1}{2}}\theta \cos^{\frac{1}{2}}\theta\, d\theta$$

$$= \frac{\left|\frac{-\frac{1}{2}+1}{2}\right| \left|\frac{\frac{1}{2}+1}{2}\right|}{2\left|\frac{-\frac{1}{2}+\frac{1}{2}+2}{2}\right|}$$

$$= \frac{\left|\frac{1}{4}\right| \left|\frac{3}{4}\right|}{2\Gamma}$$

$$= \frac{\left|\frac{1}{4}\right| \left|\frac{3}{4}\right|}{2}$$

$$= \frac{\left|\frac{1}{4}\right| \left|1-\frac{1}{4}\right|}{2}$$

$$= \frac{1}{2} \times \frac{\pi}{\sin\frac{\pi}{4}}$$

$$= \frac{1}{2} \times \sqrt{2}\,\pi$$

$$= \frac{\pi}{\sqrt{2}}$$

Hence proved

Q.5 Prove that $\dfrac{\Gamma\left(\frac{1}{3}\right)\Gamma\left(\frac{5}{6}\right)}{\Gamma\left(\frac{2}{3}\right)} = (2)^{1/3}\sqrt{\pi}$

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$$\frac{\Gamma\frac{1}{3}\,\Gamma\frac{5}{6}}{\Gamma\frac{2}{3}} = 2^{\frac{1}{3}}\sqrt{\pi}$$

Using Duplication Formula

$$\Gamma m\,\Gamma\left(m+\tfrac{1}{2}\right) = \frac{\sqrt{\pi}}{2^{2m-1}}\Gamma 2m$$

Put $m = \frac{1}{3}$

$$\Gamma\tfrac{1}{3}\,\Gamma\left(\tfrac{1}{3}+\tfrac{1}{2}\right) = \frac{\sqrt{\pi}}{2^{2\times\frac{1}{3}-1}}\Gamma 2\times\tfrac{1}{3}$$

$$\Gamma\tfrac{1}{3}\,\Gamma\tfrac{5}{6} = \frac{\sqrt{\pi}}{2^{-\frac{1}{3}}}\Gamma\tfrac{2}{3}$$

$$\boxed{\frac{\Gamma\frac{1}{3}\,\Gamma\frac{5}{6}}{\Gamma\frac{2}{3}} = 2^{\frac{1}{3}}\sqrt{\pi}}$$

Practice Q. 1 *Evaluate* $\int_0^1 \frac{dx}{\sqrt{1+x^4}}$

Univ. Exam

Practice Q. 2 *Evaluate* $\int_0^\infty \frac{x^8(1-x^6)}{(1+x)^{24}}\,dx$

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Practice Q. 3 ***Prove the following result*** $\int_0^{\pi/2} \frac{d\theta}{\sqrt{\sin\theta}} X \int_0^{\pi/2} \sqrt{\sin\theta}\, d\theta = \pi$

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Practice Q.4 *Using beta gamma functions, evaluate* $\int_0^1 x^5(1-x^3)^{10}dx$

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Practice Q. 5 Prove that $\Gamma\left(-\frac{3}{2}\right) = 4/3\sqrt{\pi}$

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$$\Gamma n = \frac{\Gamma n+1}{n}$$

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AKTU : Engineering Mathematics-1

Beta Gamma Function

Lec-5

Today's Target

- ✓ Dirichlet's Inegral and Liouville's extension of Dirichlet's integral
- ✓ AKTU Questions
- ✓ Practice Questions



By Gulshan sir

Dirichlet's Integral or Dirichlet's Theorem

If l, m, n are all positive then the

$$\iiint_V x^{l-1} y^{m-1} z^{n-1} dxdydz = \frac{\lceil l \ \lceil m \ \lceil n}{\lceil l+m+n+1}$$

where the integrals are extended for $x \geq 0, y \geq 0, z \geq 0$ ***subject to the condition*** $x + y + z \leq 1$

Note: Volume = $\iiint dxdydz$

Mass = $\iiint \rho dxdydz$

Q.1 ***Evaluate*** $\iiint dxdydz$ ***, where*** $x + y + z \leq 1$

$$\iiint dx\,dy\,dz = \iiint x^{1-1}\, y^{1-1}\, z^{1-1}\, dx\, dy\, dz$$

$$= \frac{\Gamma\ \Gamma\ \Gamma}{\Gamma 1+1+1+1}$$

$$= \frac{1 \times 1 \times 1}{\Gamma 4}$$

$$= \frac{1}{3!}$$

$$= \frac{1}{6}$$

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Q.2 *The plane* $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$, *meets the axes in A, B and C. Apply Dirichlet's integral to find the volume of the tetrahedron OABC. Also find its mass if the density at any point is kxyz.*

$$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$$

Let $\frac{x}{a}=u$ $\quad \frac{y}{b}=v$ $\quad \frac{z}{c}=w$

$x=au$ $\quad y=bv$ $\quad z=cw$

$dx=a\,du$ $\quad dy=b\,dv$ $\quad dz=c\,dw$

such that $u+v+w\le 1$

Volume

$$V=\iiint dx\,dy\,dz$$

$$=\iiint a\,du\times b\,dv\times c\,dw$$

$$=abc\iiint du\,dv\,dw$$

$$=abc\iiint u^{l-1}v^{l-1}w^{l-1}\,du\,dv\,dw$$

$$=abc\left[\frac{\Gamma l\ \Gamma l\ \Gamma l}{\Gamma l+l+l+1}\right]$$

$$= abc\left[\frac{1 \times 1 \times 1}{\lfloor 4}\right]$$

$$= \frac{abc}{3!}$$

$$\boxed{V = \frac{abc}{6}}$$

$$\text{Mass} = \iiint \rho \, dx \, dy \, dz$$

$$= \iiint k x y z \, dx \, dy \, dz$$

$$= \iiint k au \times bv \times cw \times a \, du \times b \, dv \times c \, dw$$

$$= k a^2 b^2 c^2 \iiint u v w \, du \, dv \, dw$$

$$= k a^2 b^2 c^2 \iiint u^{2-1} v^{2-1} w^{2-1} \, du \, dv \, dw$$

$$= k a^2 b^2 c^2 \frac{\lfloor 2 \; \lfloor 2 \; \lfloor 2}{\lfloor 2+2+2+1}$$

$$= \frac{k a^2 b^2 c^2 \times 1 \times 1 \times 1}{\lfloor 7} = \frac{k a^2 b^2 c^2}{6!}$$

$$\boxed{m = \frac{k a^2 b^2 c^2}{720}}$$

Q.3 ***Apply Dirichlet's integral to find the volume and the mass contain in the solid region in the first octant of the ellipsoid*** $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$ ***if density at any point is*** $kxyz$.

Univ. Exam

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$$

Let $\frac{x^2}{a^2}=u$ | $\frac{y^2}{b^2}=v$ | $\frac{z^2}{c^2}=w$

$x^2=a^2u$ | $y^2=b^2v$ | $z^2=c^2w$

$x=a\sqrt{u}$ | $y=b\sqrt{v}$ | $z=c\sqrt{w}$

$dx=\frac{a\,du}{2\sqrt{u}}$ | $dy=\frac{b\,dv}{2\sqrt{v}}$ | $dz=\frac{c\,dw}{2\sqrt{w}}$

such that $u+v+w\le 1$

Volume

$$V=\iiint dx\,dy\,dz$$

$$=\iiint \frac{a}{2\sqrt{u}}\times\frac{b}{2\sqrt{v}}\times\frac{c}{2\sqrt{w}}\,du\,dv\,dw$$

$$=\frac{abc}{8}\iiint u^{-\frac{1}{2}}v^{-\frac{1}{2}}w^{-\frac{1}{2}}\,du\,dv\,dw$$

$$= \frac{abc}{8} \iiint u^{\frac{1}{2}-1} v^{\frac{1}{2}-1} w^{\frac{1}{2}-1} \, du\, dv\, dw$$

$$= \frac{abc}{8} \left[\frac{\Gamma\frac{1}{2}\,\Gamma\frac{1}{2}\,\Gamma\frac{1}{2}}{\Gamma\frac{1}{2}+\frac{1}{2}+\frac{1}{2}+1} \right]$$

$$= \frac{abc}{8} \left[\frac{\sqrt{\pi}\,\sqrt{\pi}\,\sqrt{\pi}}{\Gamma\frac{5}{2}} \right]$$

$$= \frac{abc}{8} \left[\frac{\sqrt{\pi}\,\sqrt{\pi}\,\sqrt{\pi}}{\frac{3}{2}\times\frac{1}{2}\times\sqrt{\pi}} \right]$$

$$= \frac{abc}{2\cancel{8}} \left[\frac{\sqrt{\pi}\,\sqrt{\pi}\,\cancel{\sqrt{\pi}}}{\frac{3}{\cancel{2}}\times\frac{1}{\cancel{2}}\times\cancel{\sqrt{\pi}}} \right]$$

$$\boxed{V = \frac{\pi abc}{6}}$$

$$\text{Mass} = \iiint \rho \, dx\, dy\, dz$$

$$= \iiint kxyz \, dx\, dy\, dz$$

$$= \iiint k\, a\cancel{\sqrt{u}}\, b\cancel{\sqrt{v}}\, c\cancel{\sqrt{w}} \times \frac{a}{2\cancel{\sqrt{u}}} \times \frac{b}{2\cancel{\sqrt{v}}} \times \frac{c}{2\cancel{\sqrt{w}}} \, du\, dv\, dw$$

$$= \frac{ka^2b^2c^2}{8} \iiint du\, dv\, dw$$

$$= \frac{ka^2b^2c^2}{8} \iiint u^{1-1} v^{1-1} w^{1-1} \, du\, dv\, dw$$

$$= \frac{ka^2b^2c^2}{8} \left[\frac{\Gamma 1\,\Gamma 1\,\Gamma 1}{\Gamma 1+1+1+1} \right] = \frac{ka^2b^2c^2 \times 1 \times 1 \times 1}{8 \times 3!}$$

$$\boxed{m = \frac{ka^2b^2c^2}{48}}$$

Q.4 Compute $\iiint_V x^2 dx dy dz$ over the volume tetrahedron bounded by $x = 0, y = 0, z = 0$ and $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.

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$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

Let $\frac{x}{a} = u$ | $\frac{y}{b} = v$ | $\frac{z}{c} = w$

$x = au$ | $y = bv$ | $z = cw$

$dx = a\,du$ | $dy = b\,dv$ | $dz = c\,dw$

such that

$u + v + w \le 1$

$$V = \iiint_V x^2 \, dx\, dy\, dz$$

$$V = \iiint_V a^2u^2 \times a\,du \times b\,dv \times c\,dw$$

$$= a^3bc \iiint u^2 \, du\, dv\, dw$$

$$= a^3bc \iiint u^{3-1} v^{1-1} w^{1-1} \, du\, dv\, dw$$

$$= a^3bc \left[\frac{\Gamma 3 \; \Gamma 1 \; \Gamma 1}{\Gamma 3+1+1+1}\right]$$

$$= \frac{a^3bc \times 2 \times 1}{\Gamma 6}$$

$$= \frac{a^3bc \times 2}{5 \times 4 \times 3 \times 2 \times 1}$$

$$V = \frac{a^3bc}{60}$$

Practice Q.1 *Apply Dirichlet's integral to find the volume of an octant of sphere*

$$x^2 + y^2 + z^2 \leq 25$$

$$\frac{x^2}{25} + \frac{y^2}{25} + \frac{z^2}{25} \leq 1$$

$$\frac{x^2}{5^2} + \frac{y^2}{5^2} + \frac{z^2}{5^2} \leq 1$$

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Practice Q.2 *Find the mass of solid* $\left(\frac{x}{a}\right)^p + \left(\frac{y}{b}\right)^q + \left(\frac{z}{c}\right)^r = 1$, the density at any point being

$\rho = kx^{l-1}y^{m-1}z^{n-1}$, *where* x, y, z *all are positive* .

Practice Q. 3

Evaluate $\iiint (ax^2 + by^2 + cz^2)dxdydz$, where v is the region bounded by $x^2 + y^2 + z^2 \leq 1$

$$\int\int\int ax^2\,dx\,dy\,dx + \int\int\int by^2\,dx\,dy\,dz + \int cz^2\,dx\,dy\,dz$$

Univ. Exam

Liouville's extension of Dirichlet Theorem

If x, y, z are all positive such that $h_1 < (x+y+z) < h_2$ *then*

$$\iiint f(x+y+z)\, x^{l-1} y^{m-1} z^{n-1} dx dy dz = \frac{\lfloor l \; \lfloor m \; \lfloor n}{\lfloor l+m+n} \int_{h_1}^{h_2} f(u)\, u^{l+m+n-1} du$$

Q.5 *Evaluate* $\iiint e^{x+y+z}\,dxdydz$, *taken over the positive octant such that* $x+y+z \le 1$.

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For positive octant

$0 \le x+y+z \le 1$

$0 \le u \le 1$

$$\int\int\int e^{x+y+z}\,dx\,dy\,dz$$

$$= \int\int\int e^{x+y+z}\, x^{1-1}\, y^{1-1}\, z^{1-1}\,dx\,dy\,dz$$

$$= \frac{\Gamma 1\ \Gamma 1\ \Gamma 1}{\Gamma 1+1+1} \int_0^1 e^u \times u^{1+1+1-1}\,du$$

$$= \frac{1}{\Gamma 3}\int_0^1 u^2 e^u\,du$$

(① u^2, ② e^u)

$$= \frac{1}{2!}\left[(u^2 e^u)_0^1 - \int_0^1 2u\,e^u\,du\right]$$

(① $2u$, ② e^u)

$$= \frac{1}{2}\left[e - 2\{(ue^u)_0^1 - (e^u)_0^1\}\right]$$

$$= \frac{1}{2}\left[e - 2\{e-(e-1)\}\right]$$

$$= \frac{1}{2}\left[e - 2\{e - e + 1\}\right]$$

$$= \frac{1}{2}\left[e - 2\right]$$

$$= \frac{1}{2}e - 1$$

Univ. Exam

Q.6 *Prove that* $\iiint \frac{dxdydz}{\sqrt{1-x^2-y^2-z^2}} = \frac{\pi^2}{8}$, *the integral being extended to all positive values of the variables for which the expression is real.*

$$\iiint \frac{dx\,dy\,dz}{\sqrt{1-x^2-y^2-z^2}} = \frac{\pi^2}{8}$$

$$\iiint \frac{dx\,dy\,dz}{\sqrt{1-(x^2+y^2+z^2)}} = \frac{\pi^2}{8}$$

The expression is real if

$$0 \le x^2+y^2+z^2 \le 1$$

Put $x^2 = u$, $y^2 = v$, $z^2 = w$

$x = \sqrt{u}$, $y = \sqrt{v}$, $z = \sqrt{w}$

$dx = \frac{du}{2\sqrt{u}}$, $dy = \frac{dv}{2\sqrt{v}}$, $dz = \frac{dw}{2\sqrt{w}}$

such that

$$0 \le u+v+w \le 1$$

$$\iiint \frac{1}{\sqrt{1-(u+v+w)}} \times \frac{du}{2\sqrt{u}} \times \frac{dv}{2\sqrt{v}} \times \frac{dw}{2\sqrt{w}}$$

$$= \frac{1}{8} \iiint \frac{u^{-\frac{1}{2}} v^{-\frac{1}{2}} w^{-\frac{1}{2}}}{\sqrt{1-(u+v+w)}} \, du\, dv\, dw$$

$$= \frac{1}{8} \iiint \frac{u^{\frac{1}{2}-1} v^{\frac{1}{2}-1} w^{\frac{1}{2}-1}}{\sqrt{1-(u+v+w)}} \, du\, dv\, dw$$

$$= \frac{1}{8} \left[\frac{\Gamma\frac{1}{2}\,\Gamma\frac{1}{2}\,\Gamma\frac{1}{2}}{\Gamma\frac{1}{2}+\frac{1}{2}+\frac{1}{2}} \int_0^1 f(h)\, h^{\frac{1}{2}+\frac{1}{2}+\frac{1}{2}-1} \, dh \right.$$

$$= \frac{1}{8} \left[\frac{\sqrt{\pi}\,\sqrt{\pi}\,\sqrt{\pi}}{\Gamma\frac{3}{2}} \right] \int_0^1 \frac{1}{\sqrt{1-h}}\, h^{\frac{1}{2}} \, dh$$

$$= \frac{1}{48} \left[\frac{\sqrt{\pi}\,\sqrt{\pi}\,\sqrt{\pi}}{\frac{1}{2}\sqrt{\pi}} \right] \int_0^1 h^{\frac{1}{2}} (1-h)^{-\frac{1}{2}} \, dh$$

$$= \frac{\pi}{4} \int_0^1 h^{\frac{3}{2}-1} (1-h)^{\frac{1}{2}-1}$$

$$= \frac{\pi}{4} B\left(\frac{3}{2}, \frac{1}{2}\right) \qquad B(m,n) = \frac{\Gamma m\, \Gamma n}{\Gamma m+n}$$

$$= \frac{\pi}{4} \frac{\Gamma\frac{3}{2}\,\Gamma\frac{1}{2}}{\Gamma\frac{3}{2}+\frac{1}{2}} = \frac{\pi}{4} \frac{\frac{1}{2}\Gamma\frac{1}{2}\,\Gamma\frac{1}{2}}{\Gamma 2}$$

$$= \frac{\pi}{4} \times \frac{1}{2} \times \sqrt{\pi} \times \sqrt{\pi}$$

$$= \frac{\pi^2}{8} = \text{RHS}$$

Hence proved

Practice Q. 3 *Evaluate* $\iiint \frac{dxdydz}{\sqrt{a^2 - x^2 - y^2 - z^2}}$ *the integral being extended to* Univ. Exam

all positive values of the variables for which the expression is real.

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